

Energex

Pricing Proposal

1 July 2013 to 30 June 2014



positive energy

Version Control

Version	Date	Description
Draft	19/04/13	Draft sent to AER for early review.
1	30/04/13	Document submitted to the AER for approval.



Energex Limited T/A Energex
ABN 40 078 849 055
26 Reddacliff Street
NEWSTEAD QLD 4006

<http://www.energex.com.au>

Requests and inquiries concerning this document should be addressed to:

Pricing Manager
Energex
GPO Box 1461
BRISBANE QLD 4001

Table of contents

LIST OF TABLES	V
LIST OF FIGURES	VI
LIST OF EQUATIONS	VI
1. INTRODUCTION AND OUTLINE.....	1
2. PRICING FRAMEWORK.....	3
2.1 Pricing principles and objectives	3
2.2 Setting the 2013/14 tariffs	4
2.2.1 Maximum allowed revenue and revenue cap	4
2.2.2 Modelling outputs	6
2.2.3 DUOS overs and unders account	9
2.2.4 Demand, energy and customer numbers forecasts	10
2.2.5 Jurisdictional schemes	11
3. NETWORK TARIFF CLASSES AND TARIFF CLASS ASSIGNMENT	12
3.1 Tariff classes.....	12
3.1.1 Tariff class membership	13
3.2 Assignment and re-assignment of customers to tariff classes.....	14
3.2.1 Tariff class assignment process.....	14
3.2.2 Customers with micro-generation facilities	15
3.2.3 Customer notification process for tariff class changes	18
3.2.4 Tariff class assignment review process	19
4. SCS TARIFFS AND CHARGES	20
4.1 Description of tariffs	20
4.2 Proposed charges	20
5. CHARGING PARAMETERS	24
5.1 Application of charging parameters.....	24
5.2 Recovery of DUOS.....	24
5.2.1 Fixed charges.....	26
5.2.2 Demand and capacity charges	26
5.2.3 Flat volume charges.....	26
5.2.4 ToU volume charges	26
5.3 Recovery of DPPC	27
6. WEIGHTED AVERAGE REVENUE	29
7. APPLICATION OF PRICING PRINCIPLES.....	30
7.1 Estimating avoidable and stand alone costs.....	30
7.1.1 The DCOS model.....	31
7.1.2 Lower bound test (avoidable cost)	32
7.1.3 Upper bound test (stand alone cost).....	32
7.1.4 Cost estimates	32
7.2 Long run marginal cost.....	33

7.3	Transaction costs.....	34
7.4	Response to price signals	35
7.5	Tariff adjustment to address revenue shortfalls	35
8.	SIDE CONSTRAINTS FOR SCS TARIFFS	36
9.	TRANSMISSION COST RECOVERY	39
9.1	Expenses	40
9.1.1	DPPC paid to TNSPs	40
9.1.2	Avoided customer TUOS charges	40
9.1.3	Payments to other DNSPs	41
9.2	Revenue	41
9.2.1	Recovery of DPPC through tariffs	41
9.2.2	Payments received from other DNSPs	41
9.3	DPPC overs and unders accounts	41
10.	CHANGES FROM PREVIOUS REGULATORY YEAR	43
10.1	Summary of annual adjustments	43
10.1.1	Adjustments required by the Final Determination	44
10.1.2	Other adjustments	44
10.2	Adjustments detailed in Energex's 2013/14 SCI	44
10.3	Changes to tariffs	45
10.3.1	Straight kVA pricing for large customers	45
10.3.2	PeakSmart ToU tariff for residential customers	46
10.4	Changes to pricing approach	47
10.4.1	Smoothing transition between tariff classes	47
10.4.2	Allocation of Solar PV FiT across all tariff classes	47
10.5	Proposed removal of tariffs	47
10.6	Changes within the 2013/14 regulatory year	48
11.	PRICING STRATEGY AND TARIFF TRIALS	49
11.1	Strategic outlook	49
11.1.1	ToU demand charges for large customers	49
11.1.2	Shift to straight fixed charges for unmetered supply	49
11.1.3	Capacity and/or demand charges for small customers	49
11.1.4	Widening the application of kVA pricing for DUOS charges	50
11.1.5	Move towards fixed charges comprising greater percentage of tariffs	50
11.1.6	Australian Energy Market Commission power of choice review	50
11.2	Tariff trials and demand management programs	50
11.2.1	Controlled load tariffs	50
11.2.2	Reward Based Tariff Trial	51
12.	ALTERNATIVE CONTROL SERVICES	52
12.1	ACS tariff classes	52
12.2	Street lighting	54
12.2.1	Overview	54
12.2.2	Framework	54
12.2.3	Proposed charges	55
12.2.4	Charging parameters	55
12.3	Fee-based and quoted services	56

12.3.1	Overview	56
12.3.2	Framework	56
12.3.3	Proposed charges	57
12.3.4	Charging parameters	58
12.3.5	Long run marginal costs.....	58
12.3.6	Estimating avoidable and stand alone costs	59
12.4	Large customer connections	59
12.4.1	Overview	59
12.4.2	Framework	59
12.4.3	Proposed charges	60
12.4.4	Charging parameters	60
13.	CUSTOMER IMPACTS	63
13.1	SCS	63
13.1.1	Customer numbers and consumption	63
13.1.2	2013/14 Price impacts.....	64
13.2	ACS	67
14.	PUBLICATION OF INFORMATION ABOUT TARIFFS AND TARIFF CLASSES.....	69
15.	CONFIDENTIAL INFORMATION.....	70
16.	APPENDICES.....	71
16.1	Table of contents (Appendices).....	71
16.2	List of tables (Appendices).....	72
16.3	List of figures and equations (Appendices)	72
APPENDIX 1.	2013/14 TARIFFS – STANDARD CONTROL SERVICES	1
Appendix 1.1	– ICC site-specific tariffs	2
Appendix 1.2	– CAC and EG site-specific tariffs	5
Appendix 1.3	– kVA calculation methodology.....	28
APPENDIX 2.	2013/14 TARIFFS – ALTERNATIVE CONTROL SERVICES	29
Appendix 2.1	– Street lighting services.....	30
Appendix 2.2	– Fee-based services	31
Appendix 2.3	– Quoted services	33
Appendix 2.4	– Price development for ACS.....	34
Appendix 2.5	– Additional reporting for ACS	39
APPENDIX 3.	TARIFF COST ALLOCATION PROCESS.....	42
Appendix 3	– Tariff cost allocation process	43
APPENDIX 4.	SUMMARY OF COMPLIANCE	55
Appendix 4.1	– Compliance with the National Electricity Rules.....	56
Appendix 4.2	– Compliance with the Final Determination.....	60
APPENDIX 5.	GLOSSARY	62
Appendix 5.1	– Acronyms and abbreviations.....	63
Appendix 5.2	– Measurements	65
Appendix 5.3	– Definitions.....	66

List of tables

Table 1.1 - Pricing Proposal structure	2
Table 2.1 - Energex's pricing objectives.....	4
Table 2.2 - 2013/14 Revenue cap calculations	5
Table 2.3 - Redistribution of smoothed revenue cap to cost components	6
Table 2.4 - DUOS overs and unders account	10
Table 2.5 - 2013/14 Demand, energy and customer number forecasts by SCS tariff class	11
Table 3.1 - 2013/14 SCS tariff classes.....	13
Table 3.2 - Customer notification process for tariff class changes.....	18
Table 4.1 - Descriptions of SCS tariffs	21
Table 4.2 - 2013/14 SCS tariff charges (proposed DUOS, DPPC and NUOS charges)	23
Table 5.1 - Charging parameters relating to each element of service	24
Table 5.2 - Tariff charging parameters for DUOS charges	25
Table 5.3 - ToU charging timeframes.....	27
Table 5.4 - Tariff charging parameters for DPPC.....	28
Table 6.1 - Expected Weighted Average Revenue by tariff class	29
Table 7.1 - DCOS categories used in price boundary calculations.....	31
Table 7.2 - 2013/14 stand alone and avoidable cost boundaries	33
Table 8.1 - Compliance with side constraint formula	38
Table 9.1 - DPPC overs and unders account.....	42
Table 10.1 - Summary of annual adjustments.....	43
Table 12.1 - 2013/14 ACS tariff classes.....	52
Table 12.2 - LCC pricing framework.....	62
Table 13.1 - Outline of estimated average percentage price change by tariff class from 2012/13 to 2013/14.....	65
Table 13.2 - Indicative NUOS price change from 2012/13 to 2013/14 for varying usage profiles for SAC Demand and SAC Non-Demand tariffs.....	66
Table 13.3 - Street light services percentage increase from 2012/13 to 2013/14.....	67
Table 13.4 - Fee-based services percentage increase from 2012/13 to 2013/14.....	68
Table 15.1 - Confidential information	70

List of figures

Figure 2.1 - 2013/14 Tariff class cost allocation flowchart	8
Figure 3.1 - Assignment of customers to SCS tariff classes (Flowchart A)	16
Figure 3.2 - Assignment of customers to SCS tariff classes (Flowchart B)	17
Figure 12.1 - Assignment of customers to ACS tariff classes	53
Figure 12.2 - Service classification for LCC	61
Figure 13.1 - 2011/12 - Customer numbers (count) by tariff class.....	64
Figure 13.2 - 2011/12 Consumption by tariff class.....	64

List of equations

Equation 7.1 - LRMC: average incremental cost method	34
Equation 8.1 - Side constraint formula	37
Equation 12.1 - Formula for pricing fee-based and quoted services	58

1. Introduction and outline

RULE REQUIREMENT

Clause 6.18.2 Pricing proposals

- (a) A Distribution Network Service Provider must:
- (2) submit to the AER, at least 2 months before the commencement of the second and each subsequent regulatory year of the Regulatory Control Period, a further pricing proposal (an annual pricing proposal) for the relevant regulatory year.

This document is Energex's Annual Pricing Proposal for 2013/14. It has been prepared for the fourth year of Energex's 2010–2015 Regulatory Control Period and is submitted for review and approval by the Australian Energy Regulator (AER), in accordance with clause 6.18.2(a)(2) of the National Electricity Rules (the *Rules*) and additional requirements as specified by the AER in Energex's Final Determination 2010–2015 (the Final Determination).

This document is submitted in accordance, and complies, with the requirements of:

- The National Electricity Law (NEL)¹
- The National Electricity Rules (the *Rules*), Version 55 (7 March 2013)
- Final Determination, Queensland, 2010/11 to 2014/15 (AER, May 2010)²
- Australian Competition Tribunal order (19 May 2011)^{3,4}
- Final decision, Framework and Approach Paper, Classification of services and control mechanisms, Energex and Ergon Energy 2010–15 (AER, August 2008)⁵.

Additionally, in accordance with clause 6.18.2(b)(7) of the *Rules*, and as required by the AER in the Final Determination, the following items are included in this Pricing Proposal:

- a record of the amount of revenues recovered from designated pricing proposal charges (DPPC⁶) and associated payments
- for street lighting, the proposed price for each service in the regulatory year and the revenues collected from the provision of each service in the preceding regulatory year
- for fee-based and quoted services, the volume of services provided and revenues collected from the provision of each service in the preceding regulatory year, in addition to the proposed price for each service in the regulatory year
- information on a distribution use of system (DUOS) overs and unders account

¹ The NEL is established by the *National Electricity (South Australia) Act 1996*. The current version came into force on 1 February 2013.

² AER, 2010. Energex - Distribution determination 2010-11 to 2014-15, 4 May 2010. <http://www.aer.gov.au/sites/default/files/Queensland%20determination%20-%20Energex.pdf> (Date accessed: 24/04/13).

³ Australian Competition Tribunal, 2011. Review of a Distribution Determination by Energex Limited, 19 May 2011. <http://www.aer.gov.au/sites/default/files/Australian%20Competition%20Tribunal%20order%20Energex%20-%2019%20May%202011.pdf> (Date accessed: 24/04/13).

⁴ More information about the Australian Competition Tribunal order is included in Section 2.2.

⁵ AER, 2008. Final decision, Framework and approach paper, Classification of services and control mechanisms, Energex and Ergon Energy 2010-2015, August 2008. <http://www.aer.gov.au/sites/default/files/AER%20-%20Final%20Decision%20-%20Framework%20and%20approach%20paper%20-%20classification%20of%20service%20and%20control%20mechanisms%20-%20August%202008.pdf> (Date accessed: 24/04/2013).

⁶ DPPC (Designated Pricing Proposal Charges) is new terminology for what was previously known as Transmission Use of System (TUOS) Charges.

- information on a DPPC overs and unders account.

Specifically, this Pricing Proposal describes the methodology and principles Energex has followed during tariff development to recover its allowed revenue. It outlines the tariff classes, proposed network tariffs and charging parameters for Standard Control Services (SCS) and Alternative Control Services (ACS), and expected revenue for the year commencing 1 July 2013 and ending 30 June 2014. The structure of this Pricing Proposal is outlined in Table 1.1.

Table 1.1 - Pricing Proposal structure

Chapter	Title	Overview
2	Pricing framework	Outlines Energex's framework and methodology for setting tariffs. It provides details of the modelling inputs and outputs used to develop the network tariffs to recover regulated revenue.
3	Network tariff classes and tariff class assignment	Sets out the tariff classes for SCS, the basis for the proposed tariff classes and the procedures for the assignment and re-assignment of customers to tariff classes.
4	SCS tariffs and charges	Sets out the proposed tariffs and charges for each SCS tariff.
5	Charging parameters	Sets out the charging parameters for each SCS tariff.
6	Weighted average revenue	Sets out the weighted average revenue by tariff class and compares it to the efficient revenue boundary defined by the standalone cost and avoidable cost.
7	Application of pricing principles	Provides information about how Energex applies the Pricing Principles stipulated in the <i>Rules</i> .
8	Side constraints for SCS tariffs	Sets out the formula for calculating side constraints and, by tariff class, demonstrates adherence to the formula.
9	Transmission cost recovery	Outlines how adjustments to charges are calculated for DPPC resulting from over or under recovery.
10	Changes from previous regulatory year	Sets out annual adjustments to revenue cap components, and changes to tariffs and Energex's approach to pricing setting between 2012/13 and 2013/14.
11	Pricing strategy and tariff trials	Provides a brief overview of Energex's pricing strategy and tariff trials currently underway.
12	Alternative control services	For each ACS tariff, sets out the proposed tariffs, charges and charging parameters. Also sets out the large customer connection framework Energex applies for large connection services.
13	Customer impacts	Discusses the customer impacts from the tariffs that will be implemented in 2013/14.
14	Publication of information about tariffs and tariff classes	Sets out the documents relating to tariffs and pricing that will be published on the Energex website.
15	Confidential information	Details which parts of this document are confidential and reasons supporting a claim of confidentiality.
16	Appendices	Provides additional supporting information.

2. Pricing framework

2.1 Pricing principles and objectives

RULE REQUIREMENT

Clause 6.18.5 Pricing principles

- (a) For each tariff class, the revenue expected to be recovered should lie on or between:
 - (1) an upper bound representing the stand alone cost of serving the retail customers who belong to that class; and
 - (2) a lower bound representing the avoidable cost of not serving those retail customers.
- (b) A tariff, and if it consists of 2 or more charging parameters, each charging parameter for a tariff class:
 - (1) must take into account the long run marginal cost for the service or, in the case of a charging parameter, for the element of the service to which the charging parameter relates; and
 - (2) must be determined having regard to:
 - (i) transaction costs associated with the tariff or each charging parameter;
 - (ii) whether retail customers of the relevant tariff class are able or likely to respond to price signals.

When setting SCS tariffs, Energex's objective is to ensure its regulated revenue, as set by the AER, is recovered from customers in a manner consistent with the Pricing Principles, as outlined in clause 6.18.5 of the *Rules*. Detailed information about Energex's application of and compliance with the Pricing Principles is set out in Chapter 7.

For ACS (subject to a price cap), the objective is to ensure that the price charged is cost-reflective and is consistent with the Pricing Principles. More information about ACS is included in Chapter 12.

In addition to the Pricing Principles established under the *Rules*, Energex applies a number of Pricing Objectives in the formulation of tariffs which are described in Table 2.1. These Pricing Objectives are intended to complement the Pricing Principles and provide clarity when formulating tariffs. Additionally, where practical, Energex works to develop network tariffs that preserve the economic signals present in the structure of DPPC.

Table 2.1 - Energex's pricing objectives

Pricing Objective	Description
No cross-subsidisation	For a network user, or group of users, there should be no cross subsidies between each SCS tariff class, or between SCS and ACS tariffs.
Network efficiency	Tariffs should incorporate appropriate signals to inform network users of their impact on existing and future network capacity and costs, and to encourage demand management.
Equity	Tariffs should be equitable for customers and should reflect the users' utilisation of the existing network and the use of specific dedicated assets.
Price stability	Tariffs should not widely fluctuate over time to permit customers to make informed investment decisions.
Cost-reflectivity	As far as possible, tariffs should reflect the actual cost of service provision to customers.
Simplicity	Tariffs should be simple and straightforward to apply, based on a well-defined and clearly explained methodology and be readily understood by customers.

2.2 Setting the 2013/14 tariffs

This section provides a high level overview of how the regulated revenue is recovered through tariffs for SCS. More information about tariff classes and tariffs is available in Chapters 3 and 4, respectively.

2.2.1 Maximum allowed revenue and revenue cap

In May 2011, the Australian Competition Tribunal made a revised determination based on its decision that the value of gamma is 0.25. The annual revenue requirement is set out in the Tribunal's Order dated 19 May 2011.⁷

On 11 February 2012, Energex received a direction notice from the Queensland Government under Section 115 of the *Government Owned Corporations Act 1993* (Queensland). This direction requires Energex to exclude revenue related to the expected reduction in its capital expenditure (Capex) program, arising from the 2011 Electricity Network Capital Program (ENCAP) Review, from its Pricing Proposals for 2012/13 and remaining years of the Regulatory Control Period. As such, the 2013/14 MAR includes an adjustment for this direction, as shown in Table 2.2.

On 10 January 2013, Energex received approval from the AER of its pass through application for the Queensland Solar Bonus Scheme 2011/12. This allows for a positive Feed-in-Tariff (FiT) direct payment pass through to be incorporated into SCS tariffs in 2013/14 in accordance with the SCS control mechanism set out in the Final Determination.⁸

⁷ Refer to Footnote 3 for full citation.

⁸ AER, 2013. Determination - Energex - Pass through application for Queensland solar bonus scheme 2011-12, January 2013. <http://www.aer.gov.au/sites/default/files/AER%20Determination%20-%20Energex%20Limited%20-%20pass%20through%20for%20the%20Queensland%20Solar%20Bonus%20Scheme%202011-12%20-%20For%20Web%20Upload.pdf> (Date accessed: 24/04/13).

Table 2.2 details the revenue cap calculation for 2013/14. The revenue cap is based on a building block approach, which includes each of the regulated cost components:

- Regulatory depreciation
- Return on capital
- Operating expenditure (Opex)
- Tax allowance.

Table 2.2 - 2013/14 Revenue cap calculations

Component	Amount (\$m)	Comments/Reference
Annual Revenue Requirement - 2012/13	1,468.9	As approved by the AER in 2012/13 Pricing Proposal
• Consumer Price Index (CPI) ¹	2.50%	CPI factors as published on the Australian Bureau of Statistics (ABS) website
• X Factor	-11.04%	
Impact on Revenue	202.9	
Annual Revenue Requirement - 2013/14	1,671.9	As per Section 4.5.1 in Final Determination ⁴
Other Adjustments:		
• Capital Contributions	9.3	Adjustment for under recovery in 2011/12
• Service Target Performance Incentive Scheme (STPIS) Factor ²	0.3	Adjustment consistent with S-banking approval from AER received 22 April 2013
• Solar Photovoltaic (PV) FiT pass through	78.6	Approved pass through for Solar PV FiT based on under recovery in 2011/12 ⁵
Maximum Allowed Revenue (MAR)	1,760.1	
Revenue Reduction (ENCAP) ²	-59.9	Adjustment to Capex due to the 2011 Electricity Network Capital Program (ENCAP) Review.
Adjusted MAR	1,700.2	
Unders/Overs Adjustments:		
DUOS ^{2,3}	0.2	Adjustment for 2011/12 DUOS under recovery.
Revenue Cap⁶	1,700.4	MAR plus DUOS over/under adjustment

Notes:

1. The CPI is the annual percentage change from March 2012 to March 2013.
2. Adjustment as detailed in Energex's 2013/14 SCI. For more information, refer to Section 10.2.
3. Energex's DUOS overs and unders account is included in Table 2.4 (Section 2.2.3).
4. Refer to Footnote 2 for full citation.
5. Refer to Footnote 8 for full citation.
6. Due to rounding, individual components may not sum to the total.

2.2.2 Modelling outputs

The total revenue cap for 2013/14 is \$1,700.4 million, as detailed in Table 2.2 and Table 2.3. The allowances for each of the regulatory cost components are adjusted to account for the smoothed revenue and the annual adjustments as per Table 2.3.

Table 2.3 - Redistribution of smoothed revenue cap to cost components

Component	Sub-Total (\$m)	Total (\$m)
Regulatory Depreciation		
Original building block revenue requirement	110.3	121.4
Smoothing/CPI allowance	3.8	
Tax allowance	7.4	
Solar PV FiT pass through	5.5	
Revenue Reduction (ENCAP) ¹	-5.5	
Return on Capital		
Original building block revenue requirement	1,101.2	1,148.4
Smoothing/CPI allowance	43.7	
Tax allowance	73.6	
Solar PV FiT pass through	54.6	
Revenue Reduction (ENCAP) ¹	-54.5	
Capital Contributions	-74.2	
Revenue from Shared Assets	-6.1	
Overs/Unders Adjustments:		
• DUOS ¹	0.2	
• Capital Contributions	9.3	
• STPIS ¹	0.3	
Operating Expenditure		
Original building block revenue requirement	372.5	430.6
Smoothing/CPI Allowance	14.7	
Tax Allowance	24.9	
Solar PV FiT pass through	18.5	
Revenue Reduction (ENCAP) ¹	0	
Revenue Cap ²	1,700.4	1,700.4

Notes:

- Adjustments as detailed in Energex's 2013/14 SCI. For more information, refer to Section 10.2.
- Due to rounding, individual components may not sum to the total.

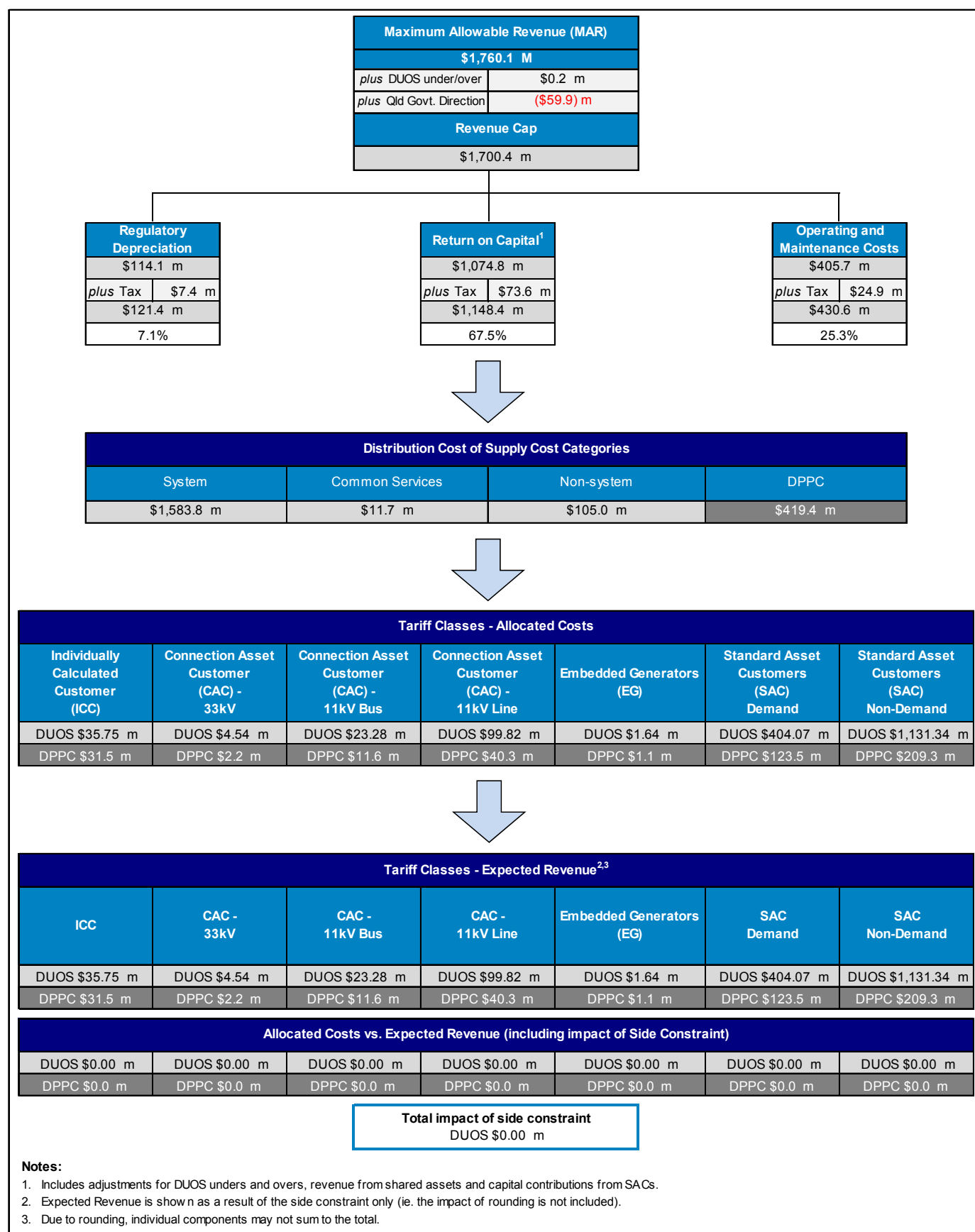
The first stage of the tariff setting process is to allocate or assign network costs to the tariff classes in the most cost-reflective way. Energex's tariff classes are:

- Individually Calculated Customers (ICC)
- Connection Asset Customers (CAC)
- Embedded Generators (EG)
- Standard Asset Customer (SAC) Demand
- SAC Non-Demand.

A description of these tariff classes, including customer eligibility, is included in Section 3.1.

Energex allocates costs to its tariff classes using a Distribution Cost of Supply (DCOS) model. This modelling process is explained in Appendix 3. The allocation of costs to recover the revenue cap is illustrated in Figure 2.1.

Figure 2.1 - 2013/14 Tariff class cost allocation flowchart



2.2.3 DUOS overs and unders account

FINAL DETERMINATION REQUIREMENT

Energex is to keep and providing information on a distribution use of system (DUOS) overs and unders account, in accordance with Clause 6.18.2(b)(7) of the *Rules*.

As part of the requirements of the Final Determination, the AER requires Energex to provide entries in its DUOS overs and unders account for the most recently completed regulatory year (t-2) and the next regulatory year (t). For this 2013/14 Pricing Proposal, year t-2 is 2011/12 and year t is 2013/14.

The overs and unders account is detailed in Table 2.4. In 2011/12 the closing balance on the account is \$81.6 million, representing 4.6% of the MAR.

In accordance with Section 4 of the Final Determination, where Energex has an over or under recovery balance of between two and five percent of the MAR, the adjustment to revenue may span two years. Where the over or under recovery is not cleared in one year, additional interest is charged on the remaining balance to index it to the year in which it is adjusted.

To moderate price variations for customers, Energex will not be recovering the entire 2011/12 DUOS under recovery in 2013/14, instead spreading the recovery across two years. More information about this is included in Section 10.2.

The AER requires the amounts used in the table above for the most recently completed regulatory year (t-2) (i.e. 2011/12) to be audited. Energex believes this requirement has been met as the amounts are either:

- (a) audited as part of the statutory or regulatory account audits undertaken by Queensland Audit Office (QAO)
- (b) provided by the AER in the Final Determination
- (c) calculated based on underlying figures which are included under points (a) and/or (b) above.

Amounts for the next regulatory year (t) are forecast amounts.

Table 2.4 - DUOS overs and unders account

Over/Under account element	2011/12 actual (\$'000)	2013/14 forecast (\$'000)
Revenue from DUOS charges	1,152,313	1,700,440
Over/Under adjustment approved for year t-2 (from t-4)	-258	0
Less foregone revenues	-52,300	-59,942
Less MAR for the relevant year	1,272,677	1,760,133
Allowed revenues (AR_t)	1,302,319	1,671,893
STPIS reward/penalty (S_t)	0	334
Capital Contribution overs/unders adjustment (C_t)	-5,833	9,344
Transitional adjustments ($Transitional_t$)	-23,809	0
Approved pass throughs ($Passthrough_t$)	0	78,561
Over/Under recovery for regulatory year	-67,806	249
DUOS overs and unders account		
Nominal WACC 2011/12	9.72%	N/A
Opening balance from year t-2 (2011/12)	0	-81,628
Interest on opening balance	0	N/A
Over/Under recovery for regulatory year	-67,806	249
Interest on over/under recovery for regulatory year	-13,822	N/A
Closing balance¹	-81,628	-81,378
Note 1. Due to rounding, individual components may not sum to total.		

2.2.4 Demand, energy and customer numbers forecasts

As part of the Regulatory Information Notice for the Final Determination, Energex provided the AER with details of the demand and energy forecasts, and expected customer numbers throughout the Regulatory Control Period. A set of forecasts were approved in the Final Determination; these approved forecast figures have now been revised and are listed in Table 2.5. The revised energy forecast for 2013/14 follows conservative energy purchases during the current year (i.e. 2012/13), primarily resulting from reduced energy sales following mild temperatures, slowed economic resurgence, increased energy conservation and the impact of the growth in residential Solar PV.

Energy sales for 2012/13 are forecast to reach 21,100 gigawatt hours (GWh). Reflecting the current conservative usage of electricity and the growth of Solar PV, the energy sales for 2013/14 are forecast

to vary between 20,819 and 21,481⁹ as per Energex's 2013/14 Long Term Energy Forecasts¹⁰. 21,180 GWh has been used for development of 2013/14 prices.

The estimated customer numbers are based on actual customer numbers at 30 November 2012 obtained from network billing systems, with a small allowance for population growth. Forecasts of demand, energy and customer numbers are used to allocate different costs to tariff classes. This is described further in the tariff cost allocation process outlined in Appendix 3.

Table 2.5 - 2013/14 Demand, energy and customer number forecasts by SCS tariff class

Tariff Class	Maximum Demand (MW)	Forecast Customer Numbers	Forecast Energy Consumption (GWh)
2013/14 Original Forecast¹	5,555	1,448,548	24,795
2013/14 Revised Forecast	5,081	1,379,387	21,180
Split of revised forecast into tariff classes:			
ICC	262	33	796
CAC – 33 kV	38	11	136
CAC – 11 kV Bus	79	54	806
CAC – 11 kV Line	412	393	2,920
EG	22	23	73
SAC Demand	1,243	11,033	5,550
SAC Non-Demand	3,025	1,367,840	9,901
Total	5,081	1,379,387	21,180
Note 1. Approved in the Final Determination.			

2.2.5 Jurisdictional schemes

In 2013/14, Energex is not subject to any jurisdictional schemes that are to be passed on to customers.

⁹ 95% confidence interval.

¹⁰ In accordance with Section 5.13.2 of the *Rules*, Energex is required to provide the AER with Long Term Energy Forecasts in its Distribution Annual Planning Report (DAPR) by its DAPR date of 30 September.

3. Network tariff classes and tariff class assignment

RULE REQUIREMENT

Clause 6.18.2 Pricing proposals

- (b) A pricing proposal must:
 - (1) set out the tariff classes that are to apply for the relevant regulatory year.

Under Chapter 10 of the *Rules*, tariff classes are defined as representing ‘a class of customers for one or more direct control services who are subject to a particular tariff or particular tariffs.’

3.1 Tariff classes

RULE REQUIREMENT

Clause 6.18.3 Tariff classes

- (a) A pricing proposal must define the tariff classes into which retail customers for direct control services are divided.
- (b) Each customer for direct control services must be a member of 1 or more tariff classes.
- (c) Separate tariff classes must be constituted for retail customers to whom standard control services are supplied and retail customers to whom alternative control services are supplied (but a customer for both standard control services and alternative control services may be a member of 2 or more tariff classes).
- (d) A tariff class must be constituted with regard to:
 - (1) the need to group retail customers together on an economically efficient basis; and
 - (2) the need to avoid unnecessary transaction costs.

In line with the *Rules* and to ensure economic efficiency, tariff classes have been designed to group similar customers together according to voltage level, customer size and usage profiles, and connection characteristics. The underpinning characteristics of the existing tariff classes broadly reflect the costs associated with provision of service to those customers within the tariff class.

Each tariff class consists of a grouping of individual tariffs that are established on the same basis as the tariff classes. This ensures there are not an excessive number of tariffs and available tariffs are clear and easily understood. Ultimately, this minimises transaction costs that may be incurred by customers from switching between tariffs and by Energex in managing the provision of an excessive number tariffs.

All customers who take supply from Energex for direct control services are a member of at least one tariff class. Direct control services comprise SCS and ACS. For the 2010–15 Regulatory Control Period, the AER has classified network services, connection services and metering services as SCS; and, fee-based services, quoted services and street lighting, are classified as ACS. Where a customer

has both SCS and ACS supplied, they may be a member of two or more tariff classes. More information about tariff class membership is included in Section 3.1.1. Tariff classes for SCS are outlined in Table 3.1, and detailed descriptions of each tariff class are included in Table 4.1. More information about ACS tariff classes, charging parameters and customer assignment to tariffs is available in Chapter 12.

Table 3.1 - 2013/14 SCS tariff classes

Tariff Class	Applicable Customer
Individually Calculated Customers (ICC)	Customers: - with electricity consumption greater than 40 GWh per year at a single connection point; and/or - where the customers demand is greater than or equal to 10 megavolt amperes (MVA); or - where a customer's circumstances mean that the average shared network charge becomes meaningless or distorted.
Connection Asset Customers (CAC)	Customers: - with electricity consumption greater than 4 GWh, but less than 40 GWh per year at a single connection point; and/or - where demand is greater than or equal to 1 MVA at a single connection point; and/or - where a customer has a dedicated supply system with connection assets; or - where the customer has contributed to their dedicated connection assets; or - where the uniqueness of the connection assets would result in distortion of the SAC pricing.
Embedded Generator (EG)	Generators with an installed capacity greater than 1 MVA in accordance with the Energy Networks Association definition as follows: - Medium: 1 – 5 MVA (Low Voltage - LV or High Voltage - HV) or less than 1 MVA (HV); and - Large: greater than 5 MVA.
SAC Demand	Customers with consumption less than 4 GWh per year, but greater than 100 megawatt hours (MWh) per year. Customers must have a meter installed that is capable of measuring energy consumption (kilowatt hours - kWh) and demand (kilowatts - kW), and is capable of measuring demand over 30 minute periods.
SAC Non-Demand	Customers with consumption below 100 MWh per year. A Non-Demand tariff may also apply when the customer's connection point has a meter installed that is capable of measuring energy consumption (kWh) only.

3.1.1 Tariff class membership

To comply with the *Rules*, Energex's process for tariff assignment, as illustrated in Figure 3.1, ensures no direct control services customers can take supply without being a member of at least one tariff

class. Prior to supply being provided to a customer, they must be assigned to a relevant network tariff. As explained in Table 3.2, where a new customer connection request is received and no tariff is nominated, using the tariff assignment process, the customer will be allocated first to a tariff class and then to a tariff. In these instances, Energex will assign the customer to the relevant Default Tariff for the tariff class. The relevant Default Tariff is determined considering the guidelines outlined in Section 3.2.

3.2 Assignment and re-assignment of customers to tariff classes

RULE REQUIREMENT

Clause 6.18.4 Principles governing assignment or re-assignment of retail customers to tariff classes and assessment and review of basis of charging

- (a) [When assigning customers to tariff classes] the AER must have regard to the following principles:
 - (1) retail customers should be assigned to tariff classes on the basis of one or more of the following factors:
 - (i) the nature and extent of their usage;
 - (ii) the nature of their connection to the network;
 - (iii) whether remotely-read interval metering or other similar metering technology has been installed at the retail customer's premises as a result of a regulatory obligation or requirement;
 - (2) retail customers with a similar connection and usage profile should be treated on an equal basis;
 - (3) however, retail customers with micro-generation facilities should be treated no less favourably than retail customers without such facilities but with a similar load profile;
 - (4) a Distribution Network Service Provider's decision to assign a customer to a particular tariff class, or to re-assign a customer from one tariff class to another should be subject to an effective system of assessment and review.
- (b) If the charging parameters for a particular tariff result in a basis of charge that varies according to the usage or load profile of the customer, a distribution determination must contain provisions for an effective system of assessment and review of the basis on which a customer is charged.

3.2.1 Tariff class assignment process

The following customer characteristics are taken into account by Energex when determining the applicable tariff class for a customer:

- energy consumption and/or demand
- voltage level
- nature of the connection.

In addition to the above, the following guidelines apply:

- Allocation of a customer with micro-generation facilities to a tariff will be made on the same basis as other connections. Details of Energex's policy is detailed in Section 3.2.2

- Where a new tariff is applied to a customer, it is standard practice to apply the tariff from the next billing period
- For new connections with no previous load history, the Default Tariff is based on their expected energy usage, supply voltage and meter type
- Instead of the Default Tariff, a customer will be assigned to a specific tariff for which they are eligible if requested by their electricity retailer or electrical contractor
- In accordance with clauses 6.18.4(a)(4) and 6.18.4(b), assignment of customers to tariff classes is reviewed periodically to assess if the tariff assignment is still applicable, given potential changes in annual usage. A change in connection voltage means that the connection is treated as if it is a new connection and the process in Figure 3.1 will be followed to assign the customer to a suitable tariff class.

The procedure for assigning and re-assigning customers to tariff classes relates specifically to the application of mandated tariffs. Customers who have chosen to participate in a tariff trial, such as the Reward Based Tariff trial (refer to Section 11.2.2. for more information), will not normally be subject to this review process. Energex believes this approach satisfies the requirements of the *Rules* by:

- balancing the grouping of customers on an economically efficient basis with the need to avoid unnecessary transaction costs
- ensuring customers with a similar connection and usage profile are treated on an equal basis and customers with micro-generation facilities are not treated less favourably.

The process for assignment to tariff class (and applicable network tariff codes) for SCS is outlined in Figure 3.1 and Figure 3.2. As depicted, within each tariff class there are a number of tariffs available; typically, a specific tariff will be applied to customers within the same tariff class. To limit transaction costs and ensure pricing signals are not distorted by constant changes in customer tariff assignment, SAC customers are generally only allowed one requested tariff change per 12 month period. For customers with demand levels that fluctuate annually, Energex may apply a reasonable tolerance limit up to 20 per cent on tariff thresholds to mitigate frequent tariff re-assignment, and subsequently limit customer impact.

3.2.2 Customers with micro-generation facilities

It is Energex's policy to treat customers with micro-generation facilities no less favourably than customers without these facilities but with a similar consumption profile. Allocation of a micro-generation customer to a tariff class will be made on the same basis as other customers; this being the extent and nature of usage and the nature of the connection to the network. The network tariff will include fixed and variable components, and if the customer's demand is met entirely by the micro-generator, then the levied charge will only be the fixed connection component.

Energex's compliance with clause 6.18.4(a)(3) of the *Rules* is demonstrated by the proposed network pricing arrangements for customers participating in the Queensland Solar Bonus Scheme, which ensure that these customers are treated no less favourably than other customers as the billed consumption of these customers will be unaffected by their participation in the Scheme.

Figure 3.1 - Assignment of customers to SCS tariff classes (Flowchart A)

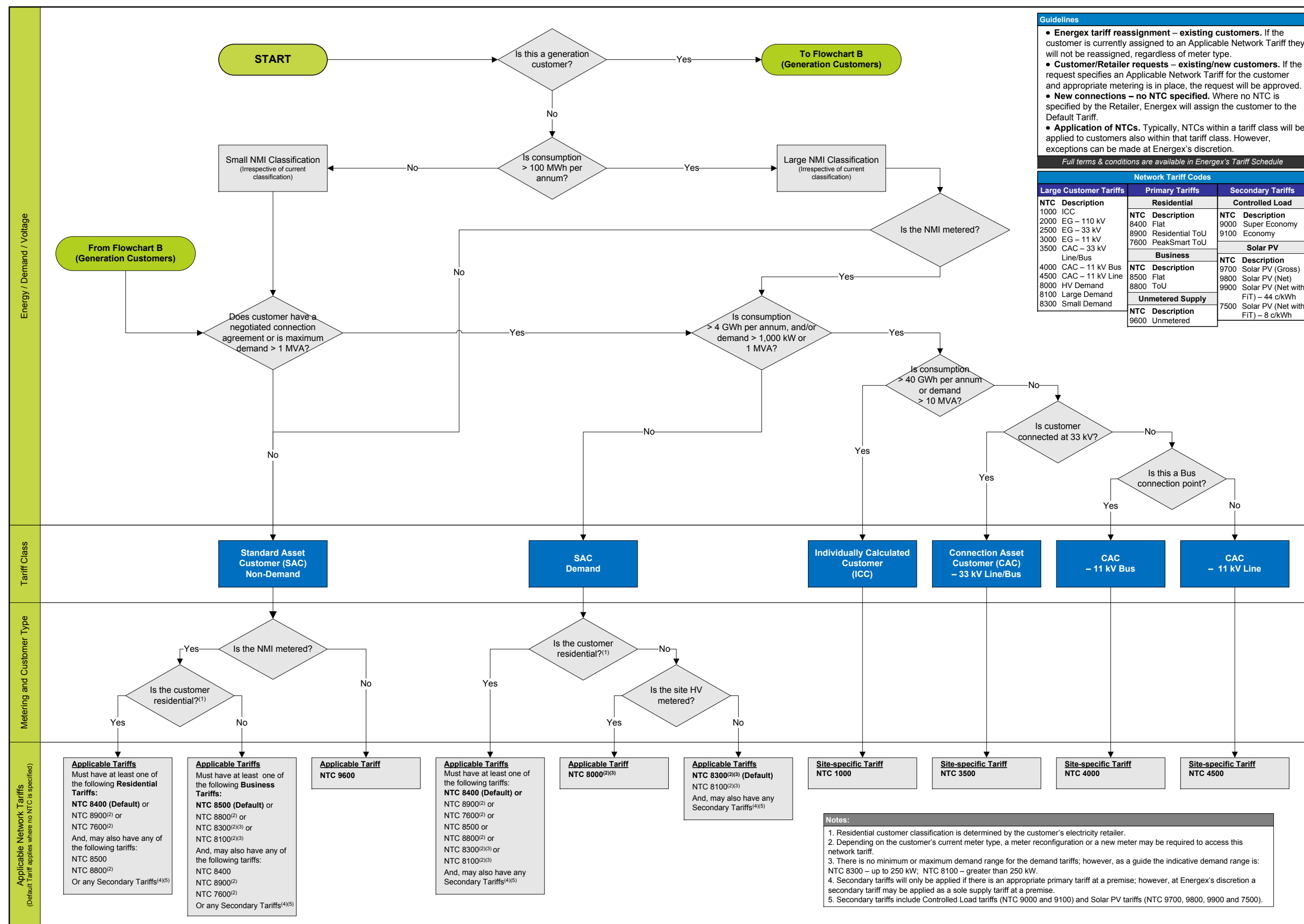
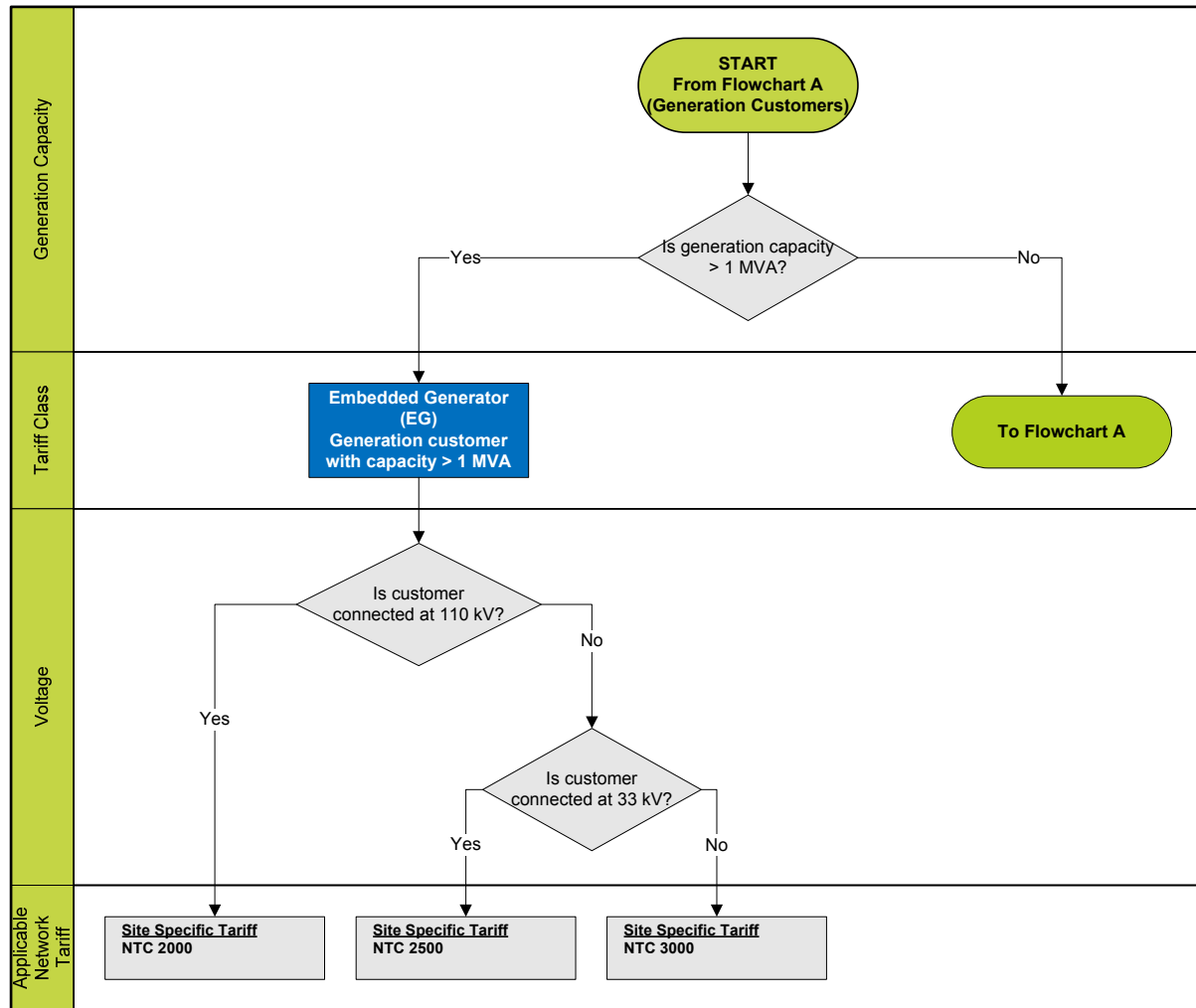


Figure 3.2 - Assignment of customers to SCS tariff classes (Flowchart B)



3.2.3 Customer notification process for tariff class changes

In accordance with Appendix B of the Final Determination, customers and their retailer will be notified of the tariff class to which they have been assigned or re-assigned. The process for notifying customers of tariff class changes is outlined in Table 3.2.

Table 3.2 - Customer notification process for tariff class changes

Input to tariff class assignment process	Notification process
Energex driven re-assignment based on a change in usage or connection	Based on NMI classification, Energex identifies customers who are assigned to an incorrect tariff class. The correct tariff class is determined based on the process outlined in Figure 3.1. The customer is notified in writing of the tariff class re-assignment, and is given the opportunity to request a review¹ of the decision.
Customer driven re-assignment (through Energex Form 1634 - QESI)	Energex receives a completed Form 1634 – QESI from the retailer for tariff re-assignment. If the request is approved the customer and their retailer are notified in writing of the tariff re-assignment and subsequent tariff class re-assignment. If the request is not approved the customer and their retailer are notified in writing that the tariff re-assignment and subsequent tariff class re-assignment has not been approved. The customer is given the opportunity to request a review¹ of the decision.
New connection	Energex receives notification of a new customer connection. For CAC, EG and ICC customers: ▪ The correct tariff class is determined based on the process outlined in Figure 3.1. ▪ The customer is notified of the tariff classification as part of the Connection Agreement, and is given the opportunity to request a review¹ of the decision. For SAC Demand and SAC Non-Demand customers: ▪ Where a tariff code is nominated on the connection request thus informing tariff class assignment, Energex will confirm if this is appropriate. ▪ If a tariff code is not nominated on the connection request, the correct tariff class and tariff code is determined based on the process outlined in Figure 3.1, and customers will be assigned to the Default Tariff. ▪ Notification to the retailer will occur electronically by way of a Change Request notice through Market Settlement and Transfer Solution and the customer is given an opportunity to request a review¹ of the decision.

Note 1. The process for tariff class review is outlined in Section 3.2.4.

Under market rules, it is the responsibility of the retailer to provide Energex with the correct customer details. Where the customer contact details are known, the customer will be notified directly; their retailer will also be notified. If the customer details are unknown or if the letter is returned to Energex, Energex will notify the customer's retailer.

3.2.4 Tariff class assignment review process

The notice given to advise of a tariff class assignment or re-assignment will include advice that the customer may request further information from Energex and that they may request a review of the proposed assignment or re-assignment.

This notice will include:

- a copy of Energex's internal review procedures or the link to where such information is available on the Energex website
- advice that if the review is not resolved to the satisfaction of the customer then:
 - for small customers – to the extent that resolution of such disputes is within the jurisdiction of a state based energy ombudsman scheme the customer is entitled to escalate the matter to such a body
 - for large customers – the customer is entitled to escalate the matter to the Department of Energy and Water Supply for resolution
- advice that if the request for review is not resolved to the customer's satisfaction under Energex's internal review system, then the customer is entitled to seek resolution via the dispute resolution process available under Part 10 of the NEL.

If a customer requests a review of the assignment or re-assignment, Energex will reconsider this request taking into account clauses 6.18.4(a)(1)–(3) of the *Rules*, and process detailed in Figure 3.1 and Figure 3.2 of this Pricing Proposal. Energex will notify the customer and their retailer in writing of its decision and the reasons for that decision.

4. SCS tariffs and charges

4.1 Description of tariffs

The tariffs for SCS are described in Table 4.1. Typically, customers are restricted to accessing tariffs allocated to the tariff class to which they are assigned. However, in some circumstances, and at Energex's discretion, customers may be able to access tariffs from a tariff class different to the one to which they are assigned.

4.2 Proposed charges

RULE REQUIREMENT

Clause 6.18.2 Pricing proposals

- (b) A pricing proposal must:
 - (2) set out the proposed tariffs for each tariff class.

The proposed charges for SCS, including DUOS, DPPC and total Network Use of System (NUOS), are provided in Table 4.2. Site-specific tariffs for ICC, CAC and EG customers are provided in Appendix 1.

The tariffs for SCS are set at the beginning of the regulatory year; however, within a regulatory year there may sometimes be a requirement to include either a tariff for a new ICC, CAC or EG customer or revise the site-specific tariff for an existing customer.

Revision of a site-specific tariff may result from the requirements of a signed connection agreement with the customer, a change in connection assets, or for an ICC, a change in their specific usage of the upstream shared network. This ensures that customers with similar characteristic are treated equitably and is consistent with clause 6.18.4 of the *Rules* by specifically taking into account the nature and extent of their usage and the nature of their connection to the network. If new or revised charges are required, they will be calculated in accordance with the approved Pricing Proposal and the customer will be notified in accordance with the process outlined in Table 3.2.

Table 4.1 - Descriptions of SCS tariffs

Tariff Class	NTCs	Tariff Description			Tariff Details
ICC	1000	Tariff class typically applies to customers with electricity consumption greater than 40 GWh per year at a single connection point; and/or where the customers demand is greater than or equal to 10 MVA; or where a customer's circumstances mean that the average shared network charge becomes meaningless or distorted. Where there is a network on private property and there are site-specific Energex costs associated with operating, maintaining and accessing the network, these costs should be applied directly to the users of those assets when it is economically efficient to do so.			The tariffs for ICCs are calculated on a site-specific basis and are confidential – they are provided in Appendix 1.1 (Confidential). Energex will provide site-specific tariffs directly to the customer and their retailer. DPPC charges for ICC tariffs are based on the relevant transmission connection point, plus charges associated with the customer’s shared distribution network used for the electricity supply, plus connection charges based on the customer’s connection assets. This provides the greatest cost-reflectivity for these customers and is a feasible method for calculating charges since the number of such customers is relatively small.
CAC	3500 4000 4500	Tariff class typically applies to customers with electricity consumption greater than 4 GWh, but less than 40 GWh per year at a single connection point; and/or where demand is greater than or equal to 1 MVA at a single connection point; and/or where a customer has a dedicated supply system with connection assets; or where the customer has contributed to their dedicated connection assets; or where the uniqueness of the connection assets would result in distortion of the SAC pricing. Where there is a network on private property and there are site-specific Energex costs associated with operating, maintaining and accessing the network, these costs should be applied directly to the users of those assets when it is economically efficient to do so. CAC tariffs are: • 33 kV Line/Bus (3500) • 11 kV Line (4500) • 11 kV Bus (4000)			Tariffs for CACs include a mix of site-specific charging parameters (fixed charge) and general tariff class charging parameters (capacity, demand and volume charges). Tariffs for the site-specific charging parameter (fixed charge) are provided in Appendix 1.2 (Confidential). Energex will provide these site-specific charges directly to the customer and their retailer. DPPC charges for CAC tariffs are based on average DPPC charges, plus average shared network charges plus site-specific connection charges based on the customer’s connection assets. This provides a significant degree of cost-reflectivity for this group of customers while recognising the practical difficulties of calculating individual shared network charges for each customer. The other charging parameters for CACs are outlined in Table 4.2.
EG	2000 2500 3000	Tariff class typically applies to generators with an installed capacity greater than 1 MVA in accordance with the Energy Networks Association definition: Medium: 1 – 5 MVA (Low Voltage – LV or High Voltage – HV) or less than 1 MVA (HV); and Large: greater than 5 MVA. Tariffs for connection and access services for medium and large EGs will be developed on a similar basis to site-specific customers. This is due to the nature of connections, which are typically non-standard and may require additional embedded generator protection system upgrades. In accordance with the <i>Rules</i>, all EGs will receive a charge for connection services regardless of whether they are a net importer or exporter of electricity. However, DUOS charges will not be incurred for the export of electricity generated by the user into the distribution network. EGs who are net importers of electricity will receive appropriate network charges. Micro and small generators with an installed capacity less than or equal to (≤) 1 MVA (LV) will be connected under an applicable SAC tariff based on the consumption level and metering type of the customer. However, a small generator may be treated as a CAC if there is a supply arrangement which is different to that which would be provided for a load customer of similar size or as per CAC requirements. EG tariffs are: • 110 kV (2000) • 33 kV (2500) • 11 kV (3000)			Tariffs for EGs include a mix of site-specific charging parameters (fixed charge) and general tariff class charging parameters (capacity, demand and volume charges). Tariffs for the site-specific charging parameter (fixed charge) are provided in Appendix 1.2 (Confidential). Energex will provide these site-specific charges directly to the customer and their retailer. The other charging parameters for EGs are outlined in Table 4.2.
SAC Demand	8000 8100 8300	Tariff class typically applies to customers with consumption less than 4 GWh per year, but greater than 100 MWh per year. Customers must have a meter installed that is capable of measuring energy consumption (kWh) and demand (kW), and records total energy consumption and demand over 30 minute periods. Each tariff has been built to include a demand parameter so that the customer’s actual demand is reflected in the price that they pay for their electricity. The highest demand reading for that month is used to calculate the customer’s electricity bill. Tariffs with a demand element are generally more cost-reflective than those with only a volume (energy) charge. SAC Demand tariffs are:			Tariffs are based on an average shared network prices and average connection price. The tariff charges for SACs are outlined in Table 4.2. Capital contributions may apply to newly connecting SACs and are sought as prepayment for a revenue shortfall in the case of an uneconomic connection. Energex’s capital contributions policy is available on the Energex website.
		• HV Demand	8000	This tariff is available to customers connected at HV with demand up to 1,000 kW.	
		• Large Demand	8100	This tariff generally applies to customers with maximum demand between 250 and 1,000 kW.	
		• Small Demand	8300	This is the Default Tariff for customers between 100 megawatt hours (MWh) and 4 GWh per year where no previous demand history exists. Generally, the small demand tariff applies for customers with maximum demand up to 250 kW. Customers with consumption less than 100 MWh can choose to access this tariff on a voluntary basis.	

Tariff Class	NTCs	Tariff Description			Tariff Details
SAC Non-Demand	7500	Tariff class typically applies to customers with consumption below 100 MWh per year. A Non-Demand tariff may also apply when the customer's connection point has a meter installed that is capable of measuring total energy consumption (kWh) only. SAC Non-Demand tariffs are:			Tariffs are based on an average shared network prices and average connection price. The tariffs for SACs are outlined in Table 4.2. Capital contributions may apply to newly connecting SACs and are sought as prepayment for a revenue shortfall in the case of an uneconomic connection. Energex's capital contributions policy is available on the Energex website. For Super Economy (NTC 9000) and Economy (NTC 9100) tariffs, specified connected appliances are detailed in Energex's Tariff Schedule.
	7600				
	8400				
	8500	• Business Flat 8500	This tariff is the Default Tariff for business customers with consumption less than 100 MWh per annum.		
	8800	• Business ToU 8800	This tariff is available to business customers with consumption less than 100 MWh per annum. This ToU tariff accounts for when, as well as how much, electricity is used by each customer. With ToU, electricity is priced at multiple levels, depending on the time of day. Tariffs are lower during off-peak hours and higher during peak hours.		
	8900				
	9000				
	9100	• Residential Flat 8400	This tariff is the Default Tariff for residential customers, regardless of their size and cannot be used in conjunction with Residential ToU (NTC 8900) or PeakSmart ToU (NTC 7600).		
	9600				
	9700				
	9800	• Residential ToU 8900	These tariffs are available to residential customers, regardless of their size and cannot be used in conjunction with Residential Flat (NTC 8400). Depending on the time of day, the tariff is priced differently with highest rates during peak hours and lower rates the rest of the day. Customers must have a ToU capable meter to access both tariffs, and for PeakSmart ToU additional eligibility requirements must be met.		
	9900	• PeakSmart ToU 7600			
		• Solar PV (net with FiT) 9900	These tariffs are part of the Queensland Solar Bonus Scheme.		
		• Solar PV (net with FiT) 7500	These tariffs are available to eligible customers participating in the Scheme. The Queensland Government sets the FiT rate (cents per kWh – c/kWh) to be paid for the excess energy generated and fed back into the electricity grid: NTC 9900 – FiT rate of 44 c/kWh, paid by Energex NTC 7500 – FiT rate of 8 c/KWh, paid by Energex The 44 c/kWh FiT rate (NTC 9900) was available to eligible customers until 9 July 2012 and, for existing customers, is available until 2028 where they continue to meet eligibility requirements. The 8 c/kWh FiT rate (NTC 7500) is the current rate available to eligible customers.		
		• Solar PV (net) 9800	These tariffs are not part of the Queensland Solar Bonus Scheme.		
		• Solar PV (gross) 9700	Under a Net arrangement (NTC 9800), energy generated is used in the premises first and any excess energy exported to the grid is metered. Under a Gross agreement (NTC 9700), all energy produced is exported to the grid and metered. No FiT is payable by Energex for either of these tariffs. For NTC 9800, the electricity retailer purchases excess energy generated and for NTC 9700, all energy generated. The rate that is paid is negotiated between the customer and the retailer or other third party. If a customer's eligibility changes, they can apply for the Queensland Solar Bonus Scheme tariff available at the time of application.		
		• Super Economy 9000	Specified connected appliances are controlled by network equipment so supply will be permanently available for a minimum period of 8 hours at the absolute discretion of Energex, but usually between the hours of 10:00 pm and 7:00 am.		
		• Economy 9100	Specified connected appliances are controlled by network equipment so supply will be available for a minimum period of 18 hours per day during time periods set at the absolute discretion of Energex.		
		• Unmetered 9600	This tariff is applicable to unmetered supplies. This includes facilities such as street lighting, public telephones, traffic signals, and public barbecues and watchman lights. Energex only provides connection to the network for these services. The unmetered supply tariff therefore seeks to only recover a contribution towards the shared network (use of system charge). For the provision of street lighting services, additional levies may be incurred; these will be recovered as an ACS.		

Detailed terms and conditions relating to each tariff are included in Energex's Tariff Schedule.

Table 4.2 - 2013/14 SCS tariff charges (proposed DUOS, DPPC and NUOS charges)

Tariff Class	Tariff Description	NTC	DUOS Charges								DPPC (TUOS) Charges						NUOS							
			Fixed (\$/day)	Capacity (\$/kVA/ month)	Demand (\$/kVA/ month)	Demand (\$/kW/ month)	Volume Flat (c/kWh)	Off Peak Volume (c/kWh)	Shoulder Volume (c/kWh)	Peak Volume (c/kWh)	Fixed (\$/day)	Demand (\$/kW/ month)	Volume Flat (c/kWh)	Off Peak Volume (c/kWh)	Shoulder Volume (c/kWh)	Peak Volume (c/kWh)	Fixed (\$/day)	Capacity (\$/kVA/ month)	Demand (\$/kVA/ month)	Demand (\$/kW/ month)	Volume Flat (c/kWh)	Off Peak Volume (c/kWh)	Shoulder Volume (c/kWh)	Peak Volume (c/kWh)
ICC	ICC	1000	Tariffs for ICC customers are confidential (see Appendix 1.1).																					
CAC	CAC - 33kV Line/Bus	3500	Site Specific (see Appendix 1.2)	1.091	2.664			0.065		0.089	Site Specific (see Appendix 1.2)	1.159		0.140		0.140	Site Specific (see Appendix 1.2)	1.091	2.664	1.159		0.205		0.229
	CAC - 11kV Bus	4000		2.111	4.303			0.065		0.089		1.159		0.140		0.140		2.111	4.303	1.159		0.205		0.229
	CAC - 11kV Line	4500		2.896	6.289			0.065		0.089		1.159		0.140		0.140		2.896	6.289	1.159		0.205		0.229
EG	EG - 110kV	2000	Site Specific (see Appendix 1.2)	0.034	0.094			0.065		0.089	Site Specific (see Appendix 1.2)	1.157		0.140		0.140	Site Specific (see Appendix 1.2)	0.034	0.094	1.157		0.205		0.229
	EG - 33kV	2500		1.091	2.664			0.065		0.089		1.159		0.140		0.140		1.091	2.664	1.159		0.205		0.229
	EG - 11kV	3000		2.896	6.289			0.065		0.089		1.159		0.140		0.140		2.896	6.289	1.159		0.205		0.229
SAC Demand	HV Demand	8000	41.471			12.404	0.089				10.119	2.251	0.947				51.590			14.655	1.036			
	Large Demand	8100	22.595			16.783	0.089				10.119	2.251	0.947				32.714			19.034	1.036			
	Small Demand	8300	4.811			18.636	0.089				1.961	2.251	1.350				6.772			20.887	1.439			
SAC Non-Demand	Business Flat	8500	0.470				10.303				0.255		1.878				0.725				12.181			
	Business ToU	8800	0.470					6.422		12.187	0.255			1.673		2.040	0.725					8.095		14.227
	Residential Flat	8400	0.360				10.247				0.079		1.710				0.439				11.957			
	Residential ToU	8900	0.500					7.484	9.747	15.364	0.079			1.295	1.710	3.777	0.579					8.779	11.457	19.141
	PeakSmart ToU	7600	0.500					5.484	9.747	15.364	0.079			1.295	1.710	3.777	0.579					6.779	11.457	19.141
	Super Economy	9000					4.130						0.708								4.838			
	Economy	9100					7.484						1.295								8.779			
	Unmetered	9600					8.041						1.407								9.448			
	Solar PV (net with FIT)	7500	FIT rate legislated by State Government.																					
	Solar PV (net with FIT)	9900	FIT rate legislated by State Government.																					
	Solar PV (net)	9700	FIT rate negotiated with third party.																					
	Solar PV (gross)	9800	FIT rate negotiated with third party.																					

5. Charging parameters

RULE REQUIREMENT

Clause 6.18.2 Pricing proposals

- (b) A pricing proposal must:
- (3) set out, for each proposed tariff, the charging parameters and the elements of service to which each charging parameter relates.

5.1 Application of charging parameters

Consistent with the *Rules*, the tariffs proposed by Energex comprise a number of charging parameters to recover revenue associated with the elements of service, that is DUOS or DPPC. The charging parameters relating to each element of service are included in Table 5.1.

Charging parameters are structured to provide signals to customers about the efficient use of the network and their impact on future network capacity and costs. The tariff structure and the proportioning of charge parameters have been developed to achieve the Pricing Principles in the *Rules* and Energex's Pricing Objectives, as discussed in Chapter 2. Energex has sought to select charging parameters for each tariff that signal the impact end users will have on the network while managing demand and volume variance risk, minimising boundary issues within and between tariff classes, and avoiding any signals that may result in perverse outcomes.

Table 5.1 - Charging parameters relating to each element of service

Element of service	Fixed charges	Capacity charges	Demand charges	Volume charges
DUOS	✓	✓	✓	✓
DPPC	✓		✓ ¹	✓

Note 1. Monthly maximum demand charge for ICCs is the locational charge as published by Powerlink and consists of the nominated demand plus average demand multiplied by rate.

5.2 Recovery of DUOS

Network tariffs and charging parameters are designed to recover Energex's approved revenue, consistent with the calculation of the MAR as outlined in Chapter 2. The network charging parameters adopted by Energex for the recovery of DUOS for SCS tariffs are detailed in Table 5.2. As demonstrated in Table 5.2, Energex does not recover DUOS on electricity generated by customers that is exported by them into the distribution network, as required by clause 6.1.4(a) of the *Rules*.

Table 5.2 - Tariff charging parameters for DUOS charges

Tariff Class	Tariff	Network Tariff Code (NTC)	Tariff Charging Parameter				
			Fixed Charge (\$/day)	Capacity Charge ^{1,2} (\$/kVA/month)	Monthly Maximum Demand Charge ¹ (\$/kVA month or \$/kW/month)	Volume Charge Flat (c/kWh)	Volume Charge ToU (c/kWh)
ICC	ICC	1000	✓	✓	✓		✓
CAC	CAC – 33kV Line/Bus	3500					
	CAC – 11kV Line	4500	✓	✓	✓		✓
	CAC – 11kV Bus	4000					
EG	EG	2000, 2500, 3000	✓	✓	✓		✓ ³
SAC Demand	HV Demand	8000					
	Small Demand	8100	✓		✓	✓	
	Large Demand	8300					
SAC Non-Demand	Business Flat	8500	✓			✓	
	Business ToU	8800	✓				✓
	Residential Flat	8400	✓			✓	
	Residential ToU	8900	✓				✓
	PeakSmart ToU	7600	✓				✓
	Solar PV	7500, 9700, 9800, 9900	Not Applicable				
	Super Economy	9000				✓	
	Economy	9100				✓	
	Unmetered	9600				✓	

Notes:

1. The application of demand based charges (authorised), is limited by the type of metering installed. Demand charges are not appropriate for those customers with metering equipment only capable of measuring and recording delivered electricity volume.
2. Typically, the capacity charge is levied on the basis of either contracted demand or maximum demand.
3. These rates apply to EGs only when they are importing electricity to their site (i.e. as a consumer).

5.2.1 Fixed charges

For large customers, where network usage signals are provided by other charging parameters, fixed charges reflect the incremental costs that arise from the connection and management of the customer. For small customers, due to metering limitations, the fixed charges can be used instead of capacity charges to reflect the average capacity set aside on the shared network for a typical customer using the tariff.

5.2.2 Demand and capacity charges

Demand charges are reflective of augmentation costs associated with customer demand activity. There are two demand charge parameter types:

1. A monthly maximum demand charge. This charge is levied on the basis that network users who place greater pressure on the system should incur higher charges. Network expansion becomes necessary where there is a likelihood of demand exceeding available capacity.
2. A capacity charge. This charge is similar to a monthly maximum demand charge, but more effectively assigns an adequate share of costs associated with system augmentations to network users. The capacity charge reflects the amount of network which is set aside for the customer which could be used by the customer at any time.

Demand charges signal to customers that they can reduce their electricity costs by reducing their peak demand. Capacity charges account for augmentation costs at the customer connection level and all associated upstream augmentation costs to provide sufficient network capacity to accommodate peak demand.

5.2.3 Flat volume charges

Flat volume charge parameters provide a mechanism to recover those costs that are related to the volume of the customer's consumption but not specifically the demand the customer places on the network. For SAC Non-Demand customers, the volume charge parameter recovers costs that would have been recovered from a demand-based or capacity charge that are not recovered from the fixed charge.

5.2.4 ToU volume charges

ToU tariffs offer lower charges during off-peak periods and higher charges during peak periods and can be used instead of a demand charge. The objective of a ToU volume charge is to reduce the demand on the network during peak times by encouraging customers to switch non-essential electricity use to off-peak and/or shoulder periods. This can reduce the infrastructure expenditure required to meet increasing peak demand and ensure resources are used more efficiently to potentially benefit all customers through reduced network costs over the long term.

Due to metering limitations for small customers, prior to 1 July 2012, ToU tariffs had not previously been offered. However, since 2007 all new meters installed in Energex's area are capable (once programmed) to record energy consumption at different periods of the day resulting in numerous residential customers now having meters capable of supporting ToU pricing.

To provide an incentive for users to decrease consumption during periods of peak demand, the Residential ToU tariff (NTC 8900) was introduced in 2012/13. This tariff allows price signals to be sent to customers via the variable charging structure; it also brings Energex's pricing structure in line with other Distribution Network Service Providers (DNSPs) across Australia. A second voluntary ToU tariff, PeakSmart ToU (NTC 7600), is being introduced in 2013/14 for residential customers. More information about the PeakSmart ToU tariff is in Section 10.3.

The charging timeframes for Energex's ToU energy tariffs are included in Table 5.3.

Table 5.3 - ToU charging timeframes

Tariff	NTC	Charging timeframes	Weekdays	Weekends
Residential ToU	8900	Off-Peak	10pm – 7am	10pm – 7am
		Shoulder	7am – 4pm, 8pm – 10pm	7am – 10pm
PeakSmart ToU	7600	Peak	4pm – 8pm	No Peak
Business ToU	8800	Off-Peak	9pm – 7am	Anytime
		Peak	7am – 9pm	No Peak
Business ToU – ICC, CAC, EG	All NTCs	Off-Peak	11pm – 7am	Anytime
		Peak	7am – 11pm	No Peak

5.3 Recovery of DPPC

Most electricity is delivered from generators to Energex's network via Powerlink's transmission network. Energex pays DPPC to Powerlink on behalf of its customers and recovers these costs through network tariffs. Energex's transmission cost recovery tariffs are based on a forecast of DPPC for each year, adjusted for over or under recoveries.

The DPPC from Powerlink comprises both fixed and variable charges. Where administratively efficient, the forecast DPPC will be passed on to customers in the same form of price structure as it is received.

A forecast of the DPPC is provided to Energex by Powerlink in March each year, allowing Energex to develop tariff components for recovery of the anticipated costs. The network charging parameters applied to each tariff for DPPC services are detailed in Table 5.4.

Table 5.4 - Tariff charging parameters for DPPC

Tariff Class	Tariff	NTC	Tariff Charging Parameter			
			Fixed Charge (\$/day)	Monthly Maximum Demand Charge ¹ (\$/kW/month)	Volume Charge Flat (c/kWh)	Volume Charge ToU (c/kWh)
ICC	ICC	1000	✓	✓ ²	✓ ³	
CAC	CAC – 33kV Line/Bus	3500	✓	✓		✓
	CAC – 11kV Line	4500				
	CAC – 11kV Bus	4000				
EG	EG	2000, 2500, 3000	✓	✓ ⁴		✓ ⁴
SAC Demand	HV Demand	8000	✓	✓	✓	
	Small Demand	8100				
	Large Demand	8300				
SAC Non-Demand	Business Flat	8500	✓		✓	
	Business ToU	8800	✓			✓
	Residential Flat	8400	✓		✓	
	Residential ToU	8900	✓			✓
	PeakSmart ToU	7600	✓			✓
	Solar PV	7500, 9700, 9800, 9900	Not Applicable			
	Super Economy	9000			✓	
	Economy	9100			✓	
	Unmetered	9600			✓	

Notes:

1. The application of demand based charges is limited by the type of metering installed. Demand charges are not appropriate for those customers with metering equipment only capable of measuring and recording delivered electricity volume.
2. Monthly maximum demand charge for ICCs is the locational charge as published by Powerlink and consists of the nominated demand plus average demand multiplied by rate.
3. Volume charge for ICCs is a combination of General and Common Charge as published by Powerlink.
4. These rates apply to EGs only when they are importing electricity to their site (i.e. as a consumer).

6. Weighted average revenue

RULE REQUIREMENT

Clause 6.18.2 Pricing proposals

- (b) A pricing proposal must:
- (4) set out, for each tariff class related to standard control services, the expected weighted average revenue for the relevant regulatory year and also for the current regulatory year.

The Weighted Average Revenue (WAR) for SCS tariffs for 2012/13 and 2013/14 is outlined Table 6.1.

Table 6.1 - Expected Weighted Average Revenue by tariff class

Tariff Class	Current regulatory year 2012/13 ¹ (\$/m)	Relevant regulatory year 2013/14 ¹ (\$/m)
ICC	32.4	35.7
CAC	105.4	127.6
EG	1.4	1.6
SAC Demand	335.5	404.1
SAC Non-Demand	944.9	1,131.3
Total²	1,419.8	1,700.4

Notes:

1. All prices exclude GST.
2. Due to rounding, individual components may not sum to the total.

7. Application of pricing principles

RULE REQUIREMENT

Clause 6.18.5 Pricing principles

- (a) For each tariff class, the revenue expected to be recovered should lie on or between:
 - (1) an upper bound representing the stand alone cost of serving the retail customers who belong to that class; and
 - (2) a lower bound representing the avoidable cost of not serving those retail customers.
- (b) A tariff, and if it consists of 2 or more charging parameters, each charging parameter for a tariff class:
 - (1) must take into account the long run marginal cost for the service or, in the case of a charging parameter, for the element of the service to which the charging parameter relates; and
 - (2) must be determined having regard to:
 - (i) transaction costs associated with the tariff or each charging parameter;
 - (ii) whether retail customers of the relevant tariff class are able or likely to respond to price signals.
- (c) If, however, as a result of the operation of paragraph (b), the Distribution Network Service Provider may not recover the expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.

7.1 Estimating avoidable and stand alone costs

In accordance with clause 6.18.5(a) of the *Rules*, the revenue expected to be recovered from each tariff class should lie on or between the bounds of stand alone and avoidable costs. This section describes how this expectation is met to ensure that the revenue recovered from each tariff class reflects costs and is free from cross subsidies.

Table 7.1 provides estimates which indicate that the expected revenue to be recovered from tariff classes for 2013/14 is between the two bounds created by avoidable and stand alone costs. These costs are described as:

- *Avoidable costs* – the costs that would be avoided if all customers in a tariff class were no longer connected to the network, assuming that all other customers remain connected
- *Stand alone costs* – the costs that would be incurred to service all customers in a tariff class on an individual (i.e. stand alone) basis.

Stand alone costs can be estimated using two methods:

1. By assuming that a new hypothetical network is constructed specifically to service the individual tariff class
2. By calculating the cost of using the existing network to supply the individual tariff class.

For the purposes of this document, Energex will use the second method for estimating stand alone costs. Due to economies of scale, the total stand alone cost based on the first method is greater than

the figure calculated using the second method. However, as the estimated stand alone cost represents the upper bound when calculating expected revenue (and, subsequently, tariffs), this method is considered suitable.

7.1.1 The DCOS model

The DCOS model, that Energex uses to calculate tariffs (as described in Chapter 2) generates DUOS tariffs based on the full distribution of the building block costs approved by the AER. Appendix 3 outlines the tariff cost allocation process used by Energex.

The DCOS model has also been used to estimate the stand alone and avoidable costs for each tariff class. Table 7.1 outlines which of the DCOS cost categories are used in the calculation of the two price boundaries and converts the DCOS cost categories into tariff charging parameters.

Table 7.1 - DCOS categories used in price boundary calculations and the conversion of DCOS categories into tariffs

DCOS Cost Category		Pricing Boundaries		Tariff Charging Parameter		
		Avoidable cost	Stand alone cost	Fixed Charge (\$/period)	Volume Charge ¹ (c/kWh)	Capacity / Demand Charge (\$/kW or \$/kVA)
Operating and Maintenance (O&M)²	Non-contributed Connection Assets	✓	✓	✓		
	Contributed Connection Assets	✓	✓	✓		
	Network Assets		✓	✓	✓ ⁵	✓
Regulatory Depreciation	Non-contributed Connection Assets	✓	✓	✓ ⁴		
	Contributed Connection Assets ³					
	Network Assets		✓	✓ ⁴	✓ ⁵	✓
Return on Capital	Non-contributed Connection Assets	✓	✓	✓		
	Contributed Connection Assets ⁴					
	Network Assets		✓	✓ ⁴	✓ ⁵	✓
Common Services			✓		✓	
Non-System			✓	✓	✓	✓

Notes:

1. Volume charges can be structured as a flat rate or a ToU rate.
2. O&M represents the application of the AER Building Block, 'Operating Expenditure.'
3. There is no regulatory depreciation or return on capital for contributed connection assets.
4. For SACs, the fixed charge comprises a small component of the shared network costs.
5. Applicable to customers on volume (energy) based network tariffs.

7.1.2 Lower bound test (avoidable cost)

As shown in Table 7.1, the avoidable costs for a tariff class include the cost of connection and the costs of operation and maintenance (O&M) for that connection. The fixed charge for the customer includes these costs, making it the floor price. Any use of the shared network will incur additional charges (in the form of the volume and/or capacity / demand charge parameters), taking the charge paid by any customer above the avoidable cost of supply (the economic cost floor).

7.1.3 Upper bound test (stand alone cost)

In the DCOS model, the infrastructure and O&M costs for upstream assets are shared across multiple customers and tariff classes. For this reason, the allocated cost of supply for each tariff class will be equal to or below the stand alone costs of supply.

In the case of smaller network customers connected at the distribution level (11 kV and below), the allocated cost will be well below stand alone costs as the costs for high voltage assets are shared with larger customers. For larger network customers connected at the sub-transmission level (33 kV and above), the allocated cost model includes a site-specific parameter for supply network costs.

The cost-based tariffs for CACs take into account the specific connection costs as well as an allocation of the upstream shared network costs. ICC cost-based tariffs are determined by mapping the actual supply network and allocating the relevant proportion of costs to the customers on the basis of their use of that network. Therefore, as there is an allocation of costs and/or the full network costs are allocated in the case of a single user asset, the revenue recovered from tariffs must be equal to or below the stand alone cost of supply (economic cost ceiling).

7.1.4 Cost estimates

The avoidable cost estimate for each tariff class has been developed based on:

- Avoidable capital – the return on capital and depreciation allocations for the non-contributed connection assets
- Avoidable O&M – those costs allocated to all connection assets for the tariff class in the DCOS model (i.e. annual O&M). Avoidable O&M does not include common and non-system assets as they are incurred irrespective of whether one particular tariff class is no longer connected
- Total avoidable costs – the sum of avoidable capital and avoidable O&M is represented as an annual charge in Table 7.2.

The stand alone cost estimate for each tariff class has been developed based on:

- The existing network is used to supply the particular tariff class on a standalone basis
- Stand alone capital – Avoidable capital plus all network (shared) assets required to service the tariff class
- Stand alone O&M – All annual O&M costs allocated for each tariff class in the DCOS model, including costs for connection and network assets, common services and non-system assets

- Total stand alone costs – The sum of stand alone capital and stand alone O&M is represented as an annual charge in Table 7.2.

Table 7.2 - 2013/14 stand alone and avoidable cost boundaries

Tariff Class	Avoidable cost (\$m)	Expected revenue¹ (\$m)	Stand alone cost (\$m)
ICC	11.0	35.7	41.9
CAC - 33kV	2.3	4.5	5.3
CAC - 11kV Bus	6.0	23.3	24.8
CAC - 11kV Line	18.5	99.8	114.5
EG	0.8	1.6	2.5
SAC Demand	19.8	404.1	427.6
SAC Non-Demand	95.5	1,131.3	1,148.4
Total²	153.7	1,700.4	1,765.0

Notes:

1. Excluding side constraint adjustment.
2. Due to rounding, individual components may not sum to the total.

7.2 Long run marginal cost

Marginal cost is the change in total cost that arises when the quantity produced changes by one unit. In the case of an electricity network, the marginal cost could be the cost incurred from one additional customer connecting to the network or an additional megawatt of demand or electricity consumed.

Marginal costs can be calculated as either Short Run Marginal Costs (SRMC) or Long Run Marginal Costs (LRMC). In the short run, investment in capacity is fixed; therefore, the SRMC refers to the cost of a customer connecting to the network but using only the existing network capacity. In the long run, investment in capacity is variable; hence, LRMC indicates an estimate of the cost of connecting the customer when an augmentation to the capacity of the network is necessary.

Pricing on the basis of LRMC assumes that prices should be based on the cost of meeting an increase in demand over an extended period of time. As demand on the electricity network increases, network capacity needs to be expanded to accommodate the additional demand. In order to provide correct signals for customers, prices must account for LRMC as it can provide appropriate signalling to customers and will assist in achieving a stable tariff structure over the forecast period.

Energex has estimated LRMC values at the voltage level using the Average Incremental Cost (AIC) method, described in Equation 7.1. The LRMC values are used as a test to ensure capacity / demand charges and energy charges incorporated into each tariff are reasonable.

Equation 7.1 - LRMC: average incremental cost method

$$\text{LRMC (AIC)} = \frac{\text{PV(Capex)} + \text{PV(Opex)}}{\text{PV(Incremental Demand)}}$$

where:

'PV (Capex)' and 'PV (Opex)' represent the Present Value (PV) costs associated with meeting the additional demand over the Regulatory Control Period.

This calculation method provides an estimate that allows irregular capital expenditure to be smoothed over time, while providing an indication of the costs associated with the increased demand. Thus, the LRMC indicates the level at which future increments of output must be priced to ensure total cost recovery given forecast demand.

The incremental capital and operating costs associated with increased demand are included in the Energex forecast Capex and Opex programs for the 2010–15 Regulatory Control Period. Energex receives a return on these (and other) costs in the form of a return on capital and depreciation charges, as approved by the AER. These costs are then allocated to each tariff class in the DCOS model.

In accordance with clause 6.18.5(b)(1) of the *Rules*, Energex designs tariffs to include a combination of charging parameters to reflect LRMC and recover the total allowable revenue. The charging parameters include:

- Capacity charges – where appropriate, takes into account the long term demand peak and can provide effective pricing signals to customers of the next increment of load
- Demand charges – used to take into account short term peaks in demand
- Volume charges – used, where appropriate, due to limitations with current metering
- Fixed charges – used to ensure any remaining costs, including costs associated with connection assets, are recovered.

7.3 Transaction costs

For each tariff, Energex has selected a number of charging parameters, identified in Chapter 5 for SCS and Chapter 12 for ACS. Each combination of tariff charging parameters has been selected to reflect the need for both fixed and variable components, as required by clause 6.18.5(b)(2)(i).

A combination of various parameters is required to achieve economic functionality and to ensure that appropriate pricing signals are provided to customers. However, the number and design of these parameters has been selected with regard to minimising the associated transaction and pricing administration costs.

As explained in Section 3.1, Energex groups individual tariffs in the same way as tariff classes. This approach ensures that customers are grouped on an economically efficient basis and there are not an excessive number of tariffs available. Ultimately, this minimises transaction costs that may be incurred

by customers through switching between tariffs and by Energex in managing the provision of an excessive number tariffs.

Additionally, for customers with demand levels that fluctuate annually, Energex may apply a reasonable tolerance limit up to 20 per cent on tariff thresholds to mitigate frequent tariff re-assignment and reduce transaction costs for customers and Energex.

7.4 Response to price signals

Consistent with clause 6.18.5(b)(2)(ii) of the *Rules*, the tariffs proposed by Energex provide signals to customers about the efficient use of the network and the impact of future network capacity and costs. The charging parameters used to signal customers include capacity, demand and volume charges, which been priced to allow customers to respond to the signal provided. These parameters are discussed in Section 5.2.

Energex is aware of customers putting systems in place to manage their network charges in response to price signals. For example, commercial and industrial customers (who have these elements as part of the network tariff) may stagger the starting time of equipment such as motors, air-conditioning (A/C) units and large lighting installations, thereby reducing their maximum demand on the network as all the start-up loads are not simultaneous. The short term benefit to the customer is a lower monthly demand charge and, if they effectively manage their longer term maximum demand, they will benefit from reduced capacity charges.

Energy usage costs are affected by the overall electrical efficiency of installations. If customers improve the efficiency of their usage they will reduce their energy-related charges. For example, this can be achieved by installing more efficient lighting and A/C, and improving thermal insulation of cold rooms. Some outcomes of Energex's current demand management programs are highlighted in Section 11.2.

7.5 Tariff adjustment to address revenue shortfalls

When setting network tariffs, Energex uses a combination of charging parameters. These are developed taking into account the LRMC, transaction costs and customer response to pricing signals, as required under Clause 6.18.5(b) of the *Rules*. However, the expected revenue will not be recovered if charging parameters considered only those items identified. For example, building block revenue is greater than LRMC since it allows for recovery of sunk costs (i.e. it allows for recovery of long run average cost).

Accordingly, the charging parameters outlined in Table 5.2 are applied to allow for any shortfall in expected revenue. These parameters are selected in a manner which complements the chosen pricing signals and minimises price distortion, as required by clause 6.18.5(c) of the *Rules*.

8. Side constraints for SCS tariffs

RULE REQUIREMENT

Clause 6.18.6 Side constraints on tariffs for standard control services

- (a) This clause applies only to tariff classes related to the provision of standard control services.
- (b) The expected weighted average revenue to be raised from a tariff class for a particular regulatory year of a Regulatory Control Period must not exceed the corresponding expected weighted average revenue for the preceding regulatory year in that Regulatory Control Period by more than the permissible percentage.
- (c) The permissible percentage is the greater of the following:
 - (1) the CPI-X limitation on any increase in the Distribution Network Service Provider's expected weighted average revenue between the two regulatory years plus 2%;
Note: The calculation is of the form $(1 + \text{CPI})(1 - X)(1 + 2\%)$
 - (2) CPI plus 2%.
Note: The calculation is of the form $(1 + \text{CPI})(1 + 2\%)$
- (d) In deciding whether the permissible percentage has been exceeded in a particular regulatory year, the following are to be disregarded:
 - (1) the recovery of revenue to accommodate a variation to the distribution determination under rule 6.6 or 6.13;
 - (2) the recovery of revenue to accommodate pass through of designated pricing proposal charges to retail customers;
 - (3) the recovery of revenue to accommodate pass through of jurisdictional scheme amounts for approved jurisdictional schemes; and
 - (4) the recovery of revenue to accommodate any increase in the Distribution Network Service Provider's annual revenue requirement by virtue of an application of a formula referred to in clause 6.5.2(1).
- (e) This clause does not, however, limit the extent a tariff for retail customers with remotely-read interval metering or other similar metering technology may vary according to the time or other circumstances of a customer's usage.

In this document Energex is required to demonstrate that the proposed DUOS tariffs for the next year (t) will, for each tariff class, meet the side constraint formula in Equation 8.1. This formula and requirement is set by the AER in Section 4.5.2 of the Final Determination.

Equation 8.1 - Side constraint formula

$$\frac{\sum_{j=1}^m d_t^j \times q_t^j}{\sum_{j=1}^m d_{t-1}^j \times q_t^j} \leq (1 + \Delta CPI_t) \times (1 - X_t) \times (1 + 2\%) \pm S_t \pm C_t \pm \text{transitional}_t \pm \text{passthrough}_t \pm \text{unders and overs}_t$$

where each tariff class 'j' has up to 'm' components, and where:

d_t^j is the proposed price for component j of the tariff class for year t

d_{t-1}^j is the price charged by Energex for component j of the tariff class in year $t-1$

q_t^j is the forecast quantity of component j of the tariff class in year t

ΔCPI_t is the annual percentage change in the ABS Consumer Price Index All Groups, Weighted Average of Eight Capital Cities from March in regulatory year $t-2$ to March in regulatory year $t-1$

X_t is the X Factor for each year of the Regulatory Control Period. If $X > 0$, then X will be set equal to zero for the purposes of the side constraint formula

S_t is the Service Target Performance Incentive Scheme (STPIS) factor to be applied in regulatory year t

C_t is the annual adjustment factor for the difference between actual and forecast capital contributions in year $t-2$

transitional_t is a transitional factor for matters such as over/under in tax paid during the current Regulatory Control Period and over/under adjustments related to shared assets used for purposes other than SCS

pass-through_t is an annual adjustment factor that reflects the pass through amounts approved by the AER with respect to regulatory year t

$\text{unders and overs}_t$ is an annual adjustment factor related to the balance of the DUOS under and over account with respect to regulatory year t .

For 2013/14, the permissible percentage change is 22.32% for all tariff classes.

The CPI used to calculate the permissible percentage is 2.50%. More information about the annual variation to CPI is available in Section 10.1.1.

In accordance with clause 6.18.6(d) of the *Rules*, and as per the side constraint formula, the permissible percentage change excludes all DPPC charges and an adjustment is made for any approved pass throughs. Energex has applied the side constraint to all tariff classes in 2013/14. The side constraint results in a net impact on revenue of \$0.0 million (as shown in Figure 2.1).

Table 8.1 demonstrates Energex's compliance with the side constraint formula for each tariff class in 2013/14. Specifically, in accordance with clause 6.18.6(b) of the *Rules*, the table illustrates that the expected WAR to be raised from each tariff class in 2013/14, does not exceed the WAR for each tariff class in 2012/13 by more than the permissible percentage.

Table 8.1 - Compliance with side constraint formula

Tariff Class	Weighted Average Revenue 2012/13 ¹ (\$m)	Expected Revenue 2013/14 ² (\$m)	Calculated percentage change (%)	Permissible percentage change (%)
ICC	32.4	35.7	10.2	22.3
CAC - 33kV	3.9	4.5	16.2	22.3
CAC - 11kV Bus	19.2	23.3	21.0	22.3
CAC - 11kV Line	82.3	99.8	21.3	22.3
EG	1.4	1.6	14.0	22.3
SAC Demand	335.5	404.1	20.4	22.3
SAC Non-Demand	944.9	1,131.3	19.7	22.3
Total³	1,419.8	1,700.4	-	-

Notes:

1. Regulatory year t-1.
2. Regulatory year t, including side constraint adjustment.
3. Due to rounding, individual components may not sum to the total.

In order to calculate the percentage change for each tariff class, the WAR for 2013/14 is based on 2012/13 rates and 2013/14 quantities¹¹, as stipulated in the side constraint formula.

¹¹ Quantities include forecast demand, capacity, customer numbers and energy consumption.

9. Transmission cost recovery

RULE REQUIREMENT

Clause 6.18.2 Pricing proposals

- (b) A pricing proposal must:
- (6) set out how designated pricing proposal charges are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous regulatory year.

Clause 6.18.7 Recovery of designated pricing proposal charges

- (a) A pricing proposal must provide for tariffs designed to pass on to retail customers the designated pricing proposal charges to be incurred by the Distribution Network Service Provider for transmission use of system services.
- (b) The amount to be passed on to retail customers for a particular regulatory year must not exceed the estimated amount of the designated pricing proposal charges adjusted for over or under recovery in accordance with paragraph (c).
- (c) The over and under recovery amount must be calculated in a way that:
 - (1) subject to subparagraphs (2) and (3) below, is consistent with the method determined by the AER in the relevant distribution determination for the Distribution Network Service Provider;
 - (2) ensures a Distribution Network Service Provider is able to recover from retail customers no more and no less than the designated pricing proposal charges it incurs; and
 - (3) adjusts for an appropriate cost of capital that is consistent with the allowed rate of return used in the relevant distribution determination for the relevant regulatory year.
- (d) Notwithstanding anything else in this clause 6.18.7, a Distribution Network Service Provider may not recover charges under this clause to the extent these are:
 - (1) recovered through the Distribution Network Service Provider's annual revenue requirement;
 - (2) recovered under clause 6.18.7A; or
 - (3) recovered from another Distribution Network Service Provider.

DETERMINATION REQUIREMENT

Energex, in accordance with clause 6.12.1(19) of the *Rules*, is to maintain a record of the amount of revenues recovered from TUOS charges and associated payments in accordance with Appendix E of the Final Determination.

In accordance with clauses 6.18.2(b)(6) and 6.18.7 of the *Rules*, tariffs outlined in this Pricing Proposal will allow for the pass through of DPPC, including any adjustments for over or under recovery.

To comply with the *Rules*, information reported as part of this Pricing Proposal includes:

- Expenses –
 - regulated DPPC paid to Transmission Network Service Providers (TNSPs)
 - avoided charges for the locational component of prescribed TUOS Services (to be referred to as avoided TUOS)

- payments made to other DNSPs for use of their network.
- Revenue –
 - payments received from network users
 - payments received from other DNSPs.
- Adjustments for over or under recovery – difference between revenue and expenses.

9.1 Expenses

9.1.1 DPPC paid to TNSPs

Energex connects to the Powerlink network at multiple transmission network connection points (TNCPs). Powerlink, as a regulated TNSP, recovers its revenue from directly connected customers and DNSPs connected to its network.

In accordance with the connection agreement with Powerlink, Energex is required to pay DPPC to Powerlink on a monthly basis.

9.1.2 Avoided customer TUOS charges

RULE REQUIREMENT

Clause 5.5 Access arrangements relating to Distribution Networks

- (h) A Distribution Network Service Provider must pass through to a Connection Applicant the amount calculated in accordance with paragraph (i) for the locational component of prescribed TUOS services that would have been payable by the Distribution Network Service Provider to a Transmission Network Service Provider had the Connection Applicant not been connected to its distribution network ('avoided charges for the locational component of prescribed TUOS services').

In accordance with the *Rules*, for EGs where prices for the locational component of prescribed DPPC services were applicable at the relevant TNCP during the relevant financial year, Energex will:

- determine the charges for the locational component of prescribed DPPC services that would have been payable by Energex had the EG not injected any energy at its connection point during that financial year
- determine the amount by which the charges calculated in (a) exceed the amount for the locational component of prescribed DPPC services actually payable by Energex
- credit the value from (b) to the EG account.

In 2013/14, avoided TUOS payments will generally be remitted in the form of a lump sum payment after 30 June 2014, similar to previous years.

Avoided TUOS to EGs by Energex, reflect the avoided costs of upstream transmission network reinforcement to SEQ – that is, avoided TUOS does not solely impact on the TNCP to which the EG is connected. Consequently, the benefits of avoided TUOS relate to all customers and have been assigned to all tariff classes.

9.1.3 Payments to other DNSPs

In contingency circumstances, Essential Energy (the DNSP in northern NSW) provides supply from its Terranora Substation to Energex's Kirra Zone Substation. Under this arrangement, Essential Energy requires Energex to pay for the use of its assets.

The charges established by Essential Energy in respect of this arrangement are based on approved rates for each month in which the alternate supply is utilised. These costs have been incorporated into the TUOS costs for the Mudgeeraba TNCP and are consequently passed through to users.

9.2 Revenue

9.2.1 Recovery of DPPC through tariffs

A description of how DPPC is recovered through standard control tariffs is given in Section 5.3. The total revenue received is indicated in Table 9.1.

9.2.2 Payments received from other DNSPs

Energex does not currently receive any transmission related payments from other DNSPs.

9.3 DPPC overs and unders accounts

As part of the requirements of the Final Determination, Energex is required to provide amounts for the following entries in its DPPC overs and unders account for the most recently completed regulatory year (t-2) and the next regulatory year (t):

1. The opening balance for each year. The opening balance for year t-2 should be zero
2. The amount of revenue recovered from DPPC applied in respect of that year, less the amounts of all transmission related payments made by Energex in respect of that year
3. An interest charge for two years related to the net amount in item 2 for year t-2. This adjustment should be calculated using the approved nominal weighted average cost of capital (WACC)
4. The total of items 1 to 3 to derive the closing balance for each year.

Table 9.1 outlines the DPPC overs and unders calculation and provides separate identification of any over or under recovery relating to prior years included in the current year revenue. This distinction allows correct calculation of the over or under recovery for the respective year.

Table 9.1 - DPPC overs and unders account

Over/under account element	2011/12 actual (\$'000)	2013/14 forecast (\$'000)
Revenue from DPPC charges	345,921	419,420
Less over/under adjustment approved for year t-2	32,672	N/A
Total transmission related payments	389,972	405,721
• Transmission charges to be paid to TNSPs	388,975	404,109
• Avoided customer TUOS payments	517	564
• Inter-DNSP payments	481	1,048
Over/Under recovery for regulatory year	-11,379	13,699
DPPC overs and unders Account		
Nominal WACC 2011-12	9.72%	N/A
Opening Balance	0	-13,699
Over/Under recovery for regulatory year	-11,379	13,699
Interest on over/under recovery for regulatory year	-2,320	N/A
Closing balance¹	-13,699	0
Note 1. Due to rounding, individual components may not sum to total.		

The AER requires that the amounts used in the table above for the most recently completed regulatory year (t-2) are audited.

Energex believes this requirement has been met as the amounts are either:

- (a) audited as part of the statutory or regulatory account audits undertaken by Queensland Audit Office (QAO);
- (b) provided by the AER in the Final Determination; or
- (c) calculated based on underlying figures which are included under points (a) and/or (b) above.

Amounts for the next regulatory year (t) are forecast amounts.

In proposing variations to the amount and structure of DPPC for a given regulatory year (t), Energex will achieve a zero expected balance on the DPPC overs and unders account at the end of each regulatory year in the Regulatory Control Period.

10. Changes from previous regulatory year

RULE REQUIREMENT

Clause 6.18.2 Pricing proposals

(b) A pricing proposal must:

- (8) describe the nature and extent of change from the previous regulatory year and demonstrate that the changes comply with the *Rules* and any applicable distribution determination.

This section outlines changes between 2012/13 and 2013/14, including adjustments to revenue cap components, and changes to tariffs and the approach to price setting. It also details potential changes within the regulatory year. An analysis of the customer impacts of the various changes is included in Chapter 13.

10.1 Summary of annual adjustments

As shown in Table 2.2 and Table 2.3, various adjustments are made to the annual revenue requirement for the relevant regulatory year to calculate the MAR. Some adjustments are required by the Final Determination; these are outlined in Section 10.1.1. A summary of the annual adjustments is included in Table 10.1.

These adjustments ensure the MAR accounts for changes in the value of various revenue cap components between regulatory years. These adjustments also impact the calculation of the annual applicable side constraint, as outlined in Chapter 8.

Table 10.1 - Summary of annual adjustments

Component / Adjustment	2012/13 value (\$m)	2013/14 value (\$m)	Reason for change
Revenue Reduction (ENCAP)	-16.1	-59.9	Reduction in Capex due to ENCAP Review, subject to SCI ¹
CPI	1.58%	2.50%	Annual adjustment per ABS
Capital Contributions under recovery	0.8	9.3	2011/12 under recovery
DUOS under recovery	20.5	0.2	2011/12 under recovery, subject to SCI ¹
STPIS	0	0.3	2011/12 STPIS recovery, subject to SCI ¹
Solar PV FiT pass through	17.1	78.6	2011/12 FiT pass through
X Factor	-11.04%	-11.04%	N/A

Note 1. Adjustments detailed in Energex's 2013/14 SCI. Further information is included in Section 10.2.

10.1.1 Adjustments required by the Final Determination

The Final Determination sets out a number of adjustments to be made annually for calculating the MAR, including:

- CPI (Section 4) – Energex is to use a CPI factor based on the year to March t-1. This CPI factor is the annual percentage change in the ABS CPI for All Groups, Weighted Average of Eight Capital Cities from March in year t-2 to March in year t-1. For 2013/14, that is March 2012 to March 2013.
- Capital Contributions (Section 4) – An annual adjustment to the MAR is to be made for over or under recovery of capital contributions in year t-2, i.e. 2011/12.
- DUOS over or under recovery (Section 4) – An annually adjusted amount reflect the actual overs or unders in year t-2, i.e. 2011/12.
- Solar PV FiT pass through (Section 15) – Annually adjusted figure to account for FiT event in year t-2, i.e. 2011/12, as approved by the AER on 10 January 2013.¹²
- STPIS (Section 12) – Annual adjustment to reflect the STPIS reward/penalty from year t-2, i.e. 2011/12. As per Energex's 2012/13 Pricing Proposal, Energex elected to forego the entire STPIS reward to which it was eligible in 2012/13. In 2013/14, Energex's STPIS adjustment is subject to Energex's 2013/14 SCI.

10.1.2 Other adjustments

The Revenue Reduction (ENCAP) is the result of a Queensland Government direction¹³ and is detailed in Energex's 2013/14 SCI. Further adjustments to the recovery of DUOS and STPIS are also detailed in the SCI. These changes are outlined in Section 10.2.

The X Factor is pre-determined¹⁴ and based on the Australian Competition Tribunal's order and revised determination¹⁵. Under this revised determination, the X Factor was set at -11.04% for 2012/13 and 2013/14, hence there has been no change.

10.2 Adjustments detailed in Energex's 2013/14 SCI

When compared to 2012/13, the impact of the increased annual revenue requirement, the diminishing volume of electricity sales, increasing revenue under recoveries and the pass through of significant Solar PV FiT payments will result in a significant increase in 2013/14 network prices.

Based on Energex's forecasts of 2013/14 and 2014/15 network prices, and anticipated easing of pricing pressures in the next regulatory period, Energex is employing the following initiatives to mitigate the impact of the significant 2013/14 electricity price increase on customers and to smooth volatility in prices over the longer term:

- As the 2011/12 (t-2) DUOS under recovery is in excess of 2%, but less than 5%, of the MAR, Energex is proposing to spread recovery over 2013/14 and 2014/15
- Energex is deferring 99% of its STPIS payment to 2014/15

¹² Details of this approval are included in Section 2.2. Refer to Footnote 8 for full citation.

¹³ Details of the Queensland Government direction are included in Section 2.2.

¹⁴ The X Factor was set at -11.04% for first four years of the Regulatory Control Period; the final year (2014/15) is set at -1.42%.

¹⁵ Details of the Tribunal's Decision are included in Section 2.2. Refer to Footnote 3 for full citation.

- Reducing Revenue by \$59.9 m, due to ENCAP.

Energex expects that price increases will moderate over the next regulatory period; incorporating these adjustment mechanisms will reduce the potential volatility of the current and longer term price impacts on customers. Accordingly, Energex will be spreading the recovery of the 2011/12 DUOS under recovery across two years.

As approved by the AER¹⁶, Energex will defer 99% of the STPIS payment for which it is eligible in 2013/14 (accrued during 2011/12) for one regulatory year, to 2014/15. Additionally, as per the Queensland Government direction¹⁷ and as shown in Table 2.2 and Table 2.3, Energex has excluded its expected Capex reduction for 2013/14 resulting from the ENCAP review (\$59.9 m) from its MAR.

10.3 Changes to tariffs

Energex has an ongoing program for reviewing network tariffs for its customers. In 2012/13, the focus was on the SAC Non-Demand residential tariffs; however, in 2013/14, the focus broadened to review all SCS tariffs. The changes proposed for 2013/14 are designed to give customers more choice by providing a more robust suite of network tariffs, specifically:

- improving cost-reflectivity in tariffs and encouraging maximum asset by replacing the power factor adjustment (PFA) methodology with straight kVA pricing for ICC, CAC and EG DUOS charges. This approach is reflective of current practices by DNSPs in other Australian jurisdictions, and aligns the Final Determination
- supporting more specific pricing signals for residential customers and providing an incentive to facilitate demand management of smart appliances through the introduction of a second, voluntary ToU tariff, PeakSmart ToU, which reflects the cost of network augmentation and reflects the value of demand management to the network.

10.3.1 Straight kVA pricing for large customers

From 1 July 2013, it is proposed that straight kVA pricing will be introduced for ICC, CAC and EG customers replacing the existing power factor adjustment (PFA) methodology and reflective of current practices in other Australian jurisdictions. The calculation methodology Energex will use to calculate kVA pricing is included in Appendix 1.3.

A tariff based on kVA is a more accurate measure of a customer's impact on the network, relative to a tariff based on kW, as it better reflects the costs imposed on the network by the customer. Accordingly, a kVA tariff better meets Energex's Pricing Objectives as it is more cost-reflective and encourages improved network efficiency. Providing the correct pricing signals will assist in achieving maximum asset utilisation – an outcome that benefits all customers by reducing the need for Energex to spend additional funds on increasing the capacity of the network long term, thereby minimising tariff increases. The introduction of straight kVA pricing also provides an additional incentive for customers to improve their power factors which should, ideally, be as close as possible to unity (i.e. one (1)).

¹⁶ Approval letter from AER to Energex, dated 22 April 2013. AER Ref: 50729.

¹⁷ Details of the Queensland Government direction are included in Section 2.2.

The PFA methodology, which is being replaced by straight kVA pricing, has been used for calculating charges since the introduction of kVA tariffs on 1 July 2010. It was introduced as a transitional measure to assist customers with understanding the concept of kVA and the impact of kVA on the network and network charges, prior to the planned introduction of the straight kVA pricing. In 2012/13, PFA factors were set as close as reasonably possible to achieve the equivalent of straight kVA prices, effectively smoothing the transition to straight kVA prices for 2013/14 tariffs. This change only impacts on DUOS charges, as DPPC is passed through in kW, preserving the economic signals present in the DPPC structure.

Energex has written letters to all customers within the impacted tariff classes specifically advising of the change to charging methodology and has responded directly to any enquiries received. In addition, since 2010/11 when the PFA methodology was introduced, customers have received advice of this future transition to straight kVA charging in their Annual Statement of Charges.

10.3.2 PeakSmart ToU tariff for residential customers

In addition to the residential default flat rate tariff and existing voluntary ToU tariff, from 1 July 2013, the tariff structure will include a second, voluntary ToU tariff, PeakSmart ToU.

The PeakSmart tariff has been designed to provide a financial incentive for customers to activate Demand Response Ready (DRR) appliances, such as PeakSmart A/C units, to allow demand management of these appliances by Energex. Like the existing ToU tariff, the PeakSmart tariff will be a three-part tariff with different rates for peak, shoulder and off-peak periods, as outlined in Section 5.2.4.

The ToU structure of the PeakSmart tariff allows Energex to send price signals to customers to encourage them to reduce their demand on the network during peak times by shifting non-time-critical electricity use to the cheaper shoulder and off-peak periods. However, this tariff also enables essential appliance usage, such as A/C to be demand managed during critical network peaks.

Residential customers with at least 4 kW cooling capacity (or equivalent aggregated and activated demand response ready appliance rated input load), including at least one activated PeakSmart A/C unit under demand management by Energex will be eligible for the tariff. The customer benefit proposed through building an incentive into the tariff reflects the cost of deferred capital expenditure on the network.

Residential A/C use is a primary driver of residential peak demand that is increasing the need for network infrastructure and, consequently, driving upward pressure on electricity prices. By activating demand management of DRR appliances by Energex, consumers are allowing Energex to cap the energy use of these appliances during peak times and are helping to reduce peak loads. Successful Energex A/C demand management trials have demonstrated that this method of controlled demand reduction does not impact customer comfort and allows the A/C unit to work to maintain comfort levels. Unlike controlling A/C or other appliances under the existing load control tariffs that provide regular on/off switching, demand response ready appliances do not require a separately metered circuit nor do they cause the A/C to be completely switched off during the peak period.

Currently, only PeakSmart A/C units are available as DRR appliances; however, A/C is the first of a likely growing smart appliance market for which this tariff establishes a solid foundation to provide future price signals to assist in managing these growing loads.

PeakSmart ToU tariff is an any time supply tariff and will supply electricity to the premises at all time; however, there will be a number of specific 'controllable' appliances within the premise. Depending on the type of appliance, Energex's control signal will either put the appliance in economy mode or put the appliance in sleep mode. Appliances, such as A/C, would maintain comfort levels by operating in economy mode, whereas pool pumps may operate in sleep mode, effectively turning it off for a period of time. It should be noted that a large number of appliances, including televisions and computers, would not be affected.

10.4 Changes to pricing approach

10.4.1 Smoothing transition between tariff classes

In response to queries around the tariff structure of SAC Demand tariffs and the transition between the SAC Non-Demand and SAC Demand classes in 2012/13, Energex has reviewed the structure in 2013/14 and smoothed the transition accordingly to ensure that customer impact is minimised.

The customer impact of the 2012/13 network demand tariffs was largely caused by Energex's response to the change in the methodology used by Powerlink to calculate DPPC charges. Energex passes these charges through to customers as network charges. In 2012/13 the AER required Powerlink to move to demand-based pricing. This resulted in approximately 70% of DPPC charges being fixed with the remainder being demand-based.

Where practical, to minimise pricing distortions, Energex network tariffs preserve the economic signals present in the structure of DPPC. In some cases, this is limited by the available tariff charge elements duplicated for each network tariff. Consequently, the step change in 2012/13 transmission charging methodology reflected in the DPPC was incorporated into Energex tariffs. Accordingly, there was an increase in the fixed DPPC charge elements for demand-based tariffs, bringing these tariffs into line with the large customer network tariffs for CACs and ICCS where DPPC closely reflects the Powerlink charging structure. The DPPC charges incorporated into the 2012/13 network tariff charges were approved in the Energex's 2012/13 Pricing Proposal.

In response to customer feedback and considering customer impacts, in 2013/14 the DPPC charges for NTC 8300 will be recovered in a different manner to NTC 8100, specifically a greater percentage of revenue will be recovered from demand charges and less from fixed charges. By moving more into demand and softening the fixed charge significantly the new structure will more precisely accommodate the relative size difference of customers who have access to the demand tariffs. This will reduce the price impact on small customers and smooth the transition between the SAC Non-Demand and SAC Demand tariff classes, i.e. from NTC 8500 and NTC 8800 to NTC 8300.

10.4.2 Allocation of Solar PV FiT across all tariff classes

Energex has changed the mechanism for passing through Solar PV FiT payments to ensure that it is shared across all tariff classes. This is consistent with the treatment of similar payments by other DNSPs.

10.5 Proposed removal of tariffs

Energex does not anticipate the removal of any tariffs during 2013/14.

10.6 Changes within the 2013/14 regulatory year

RULE REQUIREMENT

Clause 6.18.2 Pricing proposals

(b) A pricing proposal must:

- (5) set out the variation or adjustment to the tariff that could occur during the course of the regulatory year and the basis on which it could occur.

SAC customer tariffs typically remain unchanged within the regulatory year; however, some SAC customers may participate in voluntary tariff trials. These tariffs will be specifically designed to determine customer behaviour and may have special tariff elements for this purpose. More details on these tariffs are contained in Chapter 11.

The tariffs for larger customers are most likely to change where there are changes specific to an individual customer's connection arrangements and/or there is a material change in use of the site. There is also the possibility of Government-mandated tariff changes. An example of such a change is the setting of prices for some ACS via Schedule 8 of the *Queensland Electricity Regulation 2006*. Another example is the Solar PV FiT, where a new tariff was introduced with pricing legislated by the Queensland Government.

Any tariff changes for either SCS or ACS during the 2013/14 regulatory year would be developed taking into account the principles and objectives outlined in Chapter 2.

11. Pricing strategy and tariff trials

Energex undertakes an ongoing program of tariff development that includes introducing new tariffs and improving the structure of existing tariffs. The strategic outlook in Section 11.1 identifies a number of potential tariff changes currently being considered by Energex for future implementation. There is also a number of tariff trials and programs currently underway being run to support decision-making around tariff changes, particularly in the longer term. Refer to Section 11.2 for more information.

11.1 Strategic outlook

11.1.1 ToU demand charges for large customers

Energex's network tariffs include a demand component for all large customers that is currently based on the maximum demand recorded in the month by the customer. Demand tariffs signal to customers that they can reduce their network charges by reducing the variability of their consumption.

In the medium term, Energex is considering the introduction of a ToU signal in combination with the demand signal for SAC Demand customers, providing greater cost-reflectivity and encourage more efficient use of the network. This tariff structure would reward large customers who are able to shift demand to off-peak periods as a lower peak demand reduces the pressure that they place on the network.

11.1.2 Shift to straight fixed charges for unmetered supply

Typically, unmetered connections are small loads such as street lighting, barbeques and security lighting. Due to the nature of these loads, no physical meter is installed; instead, monthly consumption is calculated based on pre-determined usage tables. As these are stable loads, they are grouped into one tariff class for charging efficiency.

Moving from a c/kWh variable charge to a \$/day fixed charge will more accurately reflect the cost of providing network for unmetered connections by providing a flat rate rather than a volume charge. This also provides greater price stability for customers with these loads.

11.1.3 Capacity and/or demand charges for small customers

Demand charges are well established and accepted as components of network tariffs for large customers. In the medium to long term, Energex will look at progressively rolling out demand and/or capacity based tariffs for small business and, eventually, residential customers.

Additionally, in the long term, Energex is considering locational tariffs to assist in managing localised peak demand on the network and various tariffs (including ToU and capacity-based tariffs) with Dynamic Peak Pricing (DPP) elements. Further consideration of the inclusion of DPP elements in tariffs will be informed by the outcomes of the recently completed Reward Based Tariff (RBT) Trial which is discussed in Section 11.2.2.

11.1.4 Widening the application of kVA pricing for DUOS charges

As kVA charging more accurately measures a customer's impact on the network, and following the introduction of straight kVA pricing for DUOS for ICC, CAC and EG customers in 2013/14 (as detailed in Section 10.3.1) Energex is considering widening the application of this approach across tariff classes and rolling out a kVA-based charging structure for SAC Demand tariffs in the medium-term.

11.1.5 Move towards fixed charges comprising greater percentage of tariffs

Large customers (such as ICC and CAC) already have fixed capacity charges that reflect their long term network capacity requirements. In the absence of demand and capacity charge elements, based on typical usage profiles, a fixed charge can be used to reflect the typical capacity requirement of a SAC Non-Demand customer. Moving towards an increase in the proportion of tariffs comprised of fixed charges will improve the cost-reflectivity of providing network for small customers.

11.1.6 Australian Energy Market Commission power of choice review

In preparation for possible future requirements that may result from the Australian Energy Market Commission (AEMC) Power of choice review¹⁸, Energex is assessing options to separate the costs of metering for small customers (i.e. with annual consumption <100 MWh) into a separate tariff element. This change would impact the tariff structure for SAC Demand and is anticipated to take effect from 1 July 2014.

11.2 Tariff trials and demand management programs

Energex is undertaking a number of tariff trials and residential programs as part of a strategy to encourage more efficient network use. These programs and associated customer research provide Energex with information on potential customer behaviour with respect to new tariffs.

11.2.1 Controlled load tariffs

Controlled load is load that can be managed at Energex's discretion within specific periods as agreed with the customer. Controlled load provides a means to manage peak demand during times of maximum network utilisation, reducing network capacity requirements. Load may be interrupted without loss of utility benefits to customers or, where customers volunteer, some utility benefit may be forgone for a financial and/or environmental reward.

Air-conditioning demand management

Domestic and commercial A/C load continues to be a significant contributor to peak demand. During the 2010–15 Regulatory Control Period Energex is continuing to build on the early success of its 'time for a cool change' (Cool Change) trials, by rolling out PeakSmart A/C across Energex's distribution area from September 2012. This work supports the new demand management technology being applied by several A/C manufacturers. As part of this program, a signal receiver is supplied for customers who purchase and activate a new PeakSmart capable A/C unit and a cash 'reward' is provided for signing up to the program. To date, approximately 5,000 customers have applied for the

¹⁸ AEMC 2012. Power of choice review - giving consumers options in the way they use electricity, Final Report, 30 November 2012, Sydney. <http://www.aemc.gov.au/Media/docs/Final-report-1b158644-c634-48bf-bb3a-e3f204beda30-0.pdf> (Date accessed: 12/04/2013)

PeakSmart program. Sustainable price signals are critical to the ongoing success of the PeakSmart program and will be supported by the introduction of the new PeakSmart ToU tariff.

Pool pump direct load control

The Cool Change and Energy Conservation Communities programs also included work focused on improving load control solutions for pool pumps. Estimates indicate pool pumps contribute significantly to peak demand and initial analysis has indicated that benefits exist in shifting pool pump usage from peak use times. Accordingly, in addition to offering and promoting a better rate for electricity used by pool pumps at non-peak times through a controlled load tariff, Energex will continue to offer alternative customer choice in relation to pool pump load control where tariff conversion may not be practical or feasible.

In September 2011, Energex launched a broad based pool program (with significant pool industry and electrician support) rebating customers who either connect their pool pump to a controlled load tariff (such as NTC 9100) or convert a fixed speed pool pump to a minimum five star rated energy efficient pool pump. To date, over 10,000 customers have applied for this offer with an ongoing program proposed for 2013/14.

Hot water load control

Energex is currently conducting a campaign across SEQ offering incentives to convert water heating from a continuous supply tariff (NTC 8400 or NTC 8900) to a controlled load tariff that provides supply for a minimum of eight hours (NTC 9000) or 18 hours (NTC 9100) per day. Energex reviews switch times on an ongoing basis to optimise load managed during peak times. A rewards campaign is scheduled to run through the 2013/14 winter period to incentivise customers to take advantage of tariff conversions.

11.2.2 Reward Based Tariff Trial

The RBT Trial was a cooperative project between Energex and Ergon Energy which trialled how electricity prices may be used to encourage customers to alter their electricity consumption. The trial, which ran for just over 2 years, was completed on 31 March 2013 and involved approximately 3,700 customers in selected suburbs throughout Cairns, Toowoomba and Brisbane.

Two types of tariffs were tested throughout the trial, including a time varying tariff (incorporating a DPP and ToU price) forming the consumption trial; and a maximum demand (capacity based) tariff, specifically targeting high demand days. The trial did not affect the way customers were billed by their electricity retailers, rather they received statements that provided feedback on their performance based on the trial's theoretical electricity prices. Customers were encouraged to respond to these theoretical trial prices through the use of financial incentives.

The RBT trial has provided Energex and Ergon Energy with valuable insights into customer acceptance and behaviour towards tariffs of this nature and the effectiveness of such tariffs in reducing peak demand. Preliminary results from the first year of the trial indicate that customers can and will take action to reduce peak demand in response to pricing signals. The final analysis report for the trial will be released in July 2013. Information obtained through the trial will be analysed and feed into future tariff policy development.

12. Alternative control services

12.1 ACS tariff classes

RULE REQUIREMENT

Clause 6.18.2 Pricing proposals

(b) A pricing proposal must:

- (1) set out the tariff classes that are to apply for the relevant regulatory year.

For the 2010–15 Regulatory Control Period, the AER has classified fee-based services, quoted services and street lighting as ACS. These three services form the basis of tariff classes for ACS which are described in Table 12.1.

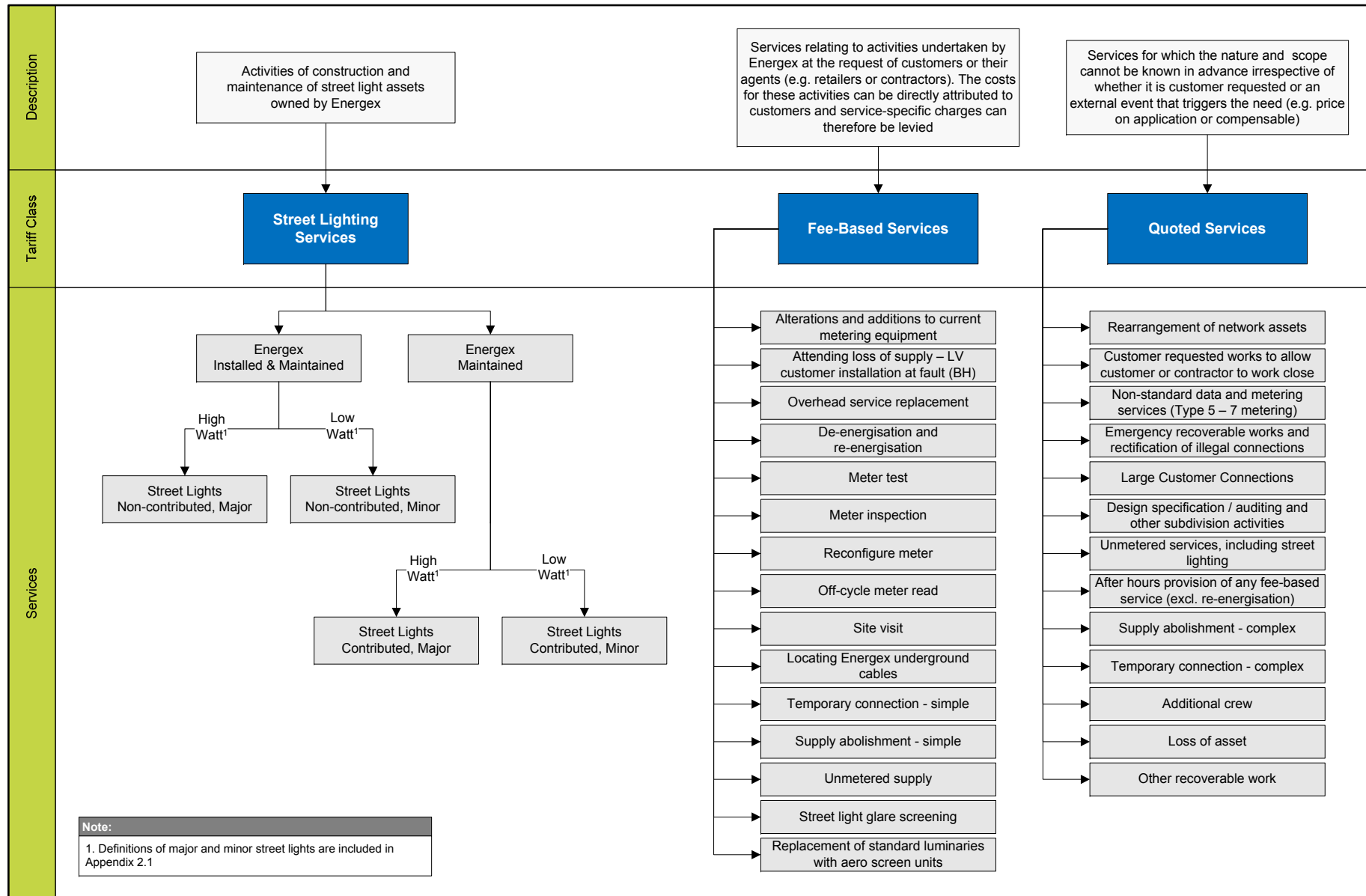
Table 12.1 - 2013/14 ACS tariff classes

Tariff Class	Applicable Customer
Street-lighting	Street lighting services relate to activities of provision, construction and maintenance of street light assets owned by Energex (conveyance of electricity for street lights remains a SCS).
Fee-based	Activities undertaken by Energex at the request of customers or their agents (e.g. retailers or contractors). The costs for these activities can be directly attributed to customers and service-specific charges can therefore be levied.
Quoted services	The nature and scope of services cannot be known in advance, irrespective of whether it is customer requested or an external event that triggers the need (e.g. price on application).

Prior to receiving supply for an ACS, a customer will be assigned to the relevant tariff class based on the type of ACS required. Similar to tariff class membership requirement for SCS, described in Section 3.1.1, an ACS customer will not receive the service prior to being allocated to the appropriate tariff class. The procedure for assigning customers to the appropriate ACS tariff class, is illustrated in Figure 12.1.

Prices for ACS are calculated based on the current costs to provide a particular service. The pricing signals now differentiate between services performed during business hours, those performed after hours and those requested on a priority basis. Customers are able to minimise the costs incurred for these services by choosing to have them delivered during business hours and within standard timeframes, where possible.

Figure 12.1 - Assignment of customers to ACS tariff classes



12.2 Street lighting

This section describes Energex's approach to the provision of the ACS of street lighting for the 2010–15 Regulatory Control Period.

12.2.1 Overview

Since 1 July 2010, the bundled prices that customers are charged for street lighting consist of a SCS and an ACS. The conveyance of electricity to street lights remains a SCS, while services relating to the provision, construction and maintenance of street lighting assets have been classified by the AER as ACS. The ACS element of street lighting services is addressed in this chapter.

In the 2010–15 Regulatory Control Period, Energex will continue to provide for non-contributed and contributed street lighting services. Since 1 July 2010, the non-contributed service only applies where Energex has constructed standard street lights. Customers will still receive an ongoing charge for the standard level of service provided under the non-contributed service. Non-standard street lights will be available as a contributed service or as a quoted service, charged as an incremental cost to a standard street light. Charges associated with these services will need to be paid upfront by the customer. In the instances where work is required outside of business hours due to maintenance access restrictions or customer requirements, these services will be provided as a quoted service.

The AER approved a price cap form of control for street lighting services over the 2010–15 Regulatory Control Period, comprising:

- a schedule of fixed prices for street lighting services for 2010/11
- a price path for the remaining years of the 2010–15 Regulatory Control Period.

12.2.2 Framework

The approach for the treatment of street lighting assets, contributed and non-contributed, satisfies the requirements of the *Rules* and delivers network charges which directly correlate with the level of service provided.

Street lighting services where Energex constructs the asset

In the 2010–15 Regulatory Control Period, where a non-standard street light is requested by the customer, a separate charge for the incremental cost difference will be levied as a quoted service. The customer will still receive an ongoing charge for the non-contributed service.

Street lighting services where customer or agent constructs the asset

In the 2010–15 Regulatory Control Period, Energex will recognise gifted assets as contributed assets and will not seek to recover any asset related costs from the customer. The customer will receive an ongoing charge for the maintenance of the street light under the contributed service.

Existing contributed street lighting services

Contributed street lights existing pre-1 July 2010, will be recognised within SCS. This aligns with the historical capital contributions treatment where benefits from the street light asset contributions were accrued by all SACs. This reflected a view that all SACs are the end beneficiary of the contributed street lighting service.

12.2.3 Proposed charges

RULE REQUIREMENT

Clause 6.18.2 Pricing proposals

- (b) A pricing proposal must:
- (2) set out the proposed tariffs for each tariff class.

DETERMINATION REQUIREMENT

Energex to provide for street lighting, the proposed price for each service in the regulatory year and the revenues collected from the provision of each service in the preceding regulatory year.

In Appendix 2.1, Energex has provided a price schedule for the provision, construction and maintenance of street lights for 2013/14. The prices are based on the methodology approved in the AER Final Determination and charges are tailored to enable the customer to be charged according to the level of service requested. The prices reflect standardised lights and no restriction on access for operation, maintenance and repair. In the case of restricted access, an additional charge may apply.

In line with the AER requirements, the revenues collected from the provision of each service in the preceding regulatory year are provided in Appendix 2.5. The applicable terms and conditions for each street light service are outlined in Energex's published Tariff Schedule.

12.2.4 Charging parameters

RULE REQUIREMENT

Clause 6.18.2 Pricing proposals

- (b) A pricing proposal must:
- (3) set out, for each proposed tariff, the charging parameters and the elements of service to which each charging parameter relates.

Energex's revenue requirements for street lighting services have been determined based on the revenue building block components consistent with the approach used for its SCS. Energex developed prices for the first year of the Regulatory Control Period using a methodology that applies a revenue allocation based on the relative installation costs for major and minor street lights and the applicable asset funding arrangement (non-contributed and contributed).

Street lights are allocated into two categories, major and minor, according to luminaire type and size, and to non-contributed and contributed based on the funding arrangement. Definitions of major and minor street lights are included in Appendix 2.1.

12.3 Fee-based and quoted services

This section describes Energex's approach to the provision of fee-based and quoted services for the 2010–15 Regulatory Control Period.

12.3.1 Overview

Energex's fee-based services and quoted services are usually provided at the explicit request of third parties. These services are defined as:

- Fee-based services – Services relating to activities undertaken by Energex at the request of customers or their agents (e.g. retailers or contractors). The costs for these activities can be directly attributed to customers and service-specific prices can be charged.
- Quoted services – Services for which the nature and scope cannot be known in advance irrespective of whether it is requested by the customer or triggered by an external event.

For the 2010–15 Regulatory Control Period, the prices for fee-based and quoted services will be subject to an annual escalation process which will take into account a weighted average price increase for each of the components of the price formula.

12.3.2 Framework

ACS are provided on an individual fee-for-service basis to retailers and end use customers. Energex will provide a fixed-fee or quoted price, depending on the service.

The AER has approved a formula based price cap form of control mechanism for fee-based services and quoted services over the 2010–15 Regulatory Control Period. This consists of:

- a schedule of fixed prices for fee-based services and a schedule of rates for quoted services for the first year of the 2010–15 Regulatory Control Period
- an annual escalation process for the remaining years of the 2010–15 Regulatory Control Period.

This formula outlined below for fee-based services and quoted services has been designed to ensure prices will be representative of the efficient costs of providing and delivering the service.

From 1 July 2010, large customer connections were classified as a quoted service under ACS. Additional detail on the framework and approach for large customer connections is provided in Section 12.4.

12.3.3 Proposed charges

RULE REQUIREMENT

Clause 6.18.2 Pricing proposals

- (b) A pricing proposal must:
- (2) set out the proposed tariffs for each tariff class.

DETERMINATION REQUIREMENT

Energex to provide for fee-based and quotes services, the proposed price for each service in the regulatory year, in addition to the volume of services provided and revenues collected from the provision of each service in the preceding regulatory year.

Prices for fee-based services have been calculated using the formula in Equation 12.1 (Section 12.3.4) and forecast costs for labour, fleet on-costs, overheads and capital allowance. The pricing signals now differentiate between services performed during business hours, those performed after hours and those requested on a priority basis. Customers are able to minimise the costs incurred for these services by choosing to have them delivered during business hours and within standard timeframes, where possible.

The proposed price schedule for fee-based services in 2013/14 is provided in Appendix 2.2. The prices were determined using the AER's approved formula based price cap control mechanisms. Revised cost escalators and on-cost and overhead rates were calculated in accordance with the Final Determination. Additional information on the calculations and rates used is included in Appendix 2.4 (Confidential).

The proposed price schedule for quoted services (illustrative configuration) in 2013/14 is provided in Appendix 2.3. These prices were determined using the AER's approved formula based price cap control mechanisms. The AER's approved prices do not represent a binding capped price for an individual quoted service.

Energex has retained its current policy of not establishing a fixed price where variations in the precise nature of the services being sought mean that averaging would result in significant inequity for customers. The prices for quoted services will be calculated using the formula above to reflect the actual cost of service provision based on the specific requirements of the customer.

In line with the AER requirements, the volume of services and the revenues collected from the provision of each fee-based and quoted service in the most recently completed regulatory year are provided in Appendix 2.5.

12.3.4 Charging parameters

RULE REQUIREMENT

Clause 6.18.2 Pricing proposals

- (b) A pricing proposal must:
- (3) set out, for each proposed tariff, the charging parameters and the elements of service to which each charging parameter relates.

Energex uses the formula in Equation 12.1 to calculate the price for fee-based and quoted services. This formula includes cost parameters for different services to determine the price.

Equation 12.1 - Formula for pricing fee-based and quoted services

$$\text{Price} = \text{Labour} + \text{Contractor Services} + \text{Materials} + \text{Capital Allowance} + \text{GST}$$

where:

Labour is all labour costs directly incurred in the provision of the service, labour on-costs, fleet on-costs and overheads. The labour cost for each service is dependent on the skill level, travel time, number of hours and crew size required to perform the service

Contractor services is all costs associated with the use of external labour in the provision of the service, including overheads and any direct costs incurred as part of performing the service (e.g. traffic control, road closure permits)

Materials is the cost of materials directly incurred in the provision of the service, material storage and logistics on-costs and overheads

Capital allowance is a return on, and return of, capital for non-system assets used in the delivery of the service

GST is Goods and Services Tax, where applicable.

12.3.5 Long run marginal costs

RULE REQUIREMENT

Clause 6.18.5 Pricing principles

- (b) A tariff, and if it consists of 2 or more charging parameters, each charging parameter for a tariff class:
- (1) must take into account the long run marginal cost for the service, or in the case of a charging parameter, for the element of the service to which the charging parameter relates.

The LRMC is taken into consideration when determining prices for ACS, as per clause 6.18.5(b)(1) of the *Rules*. Given that there is only a single charge parameter for each service, prices have been set at average cost ensuring total costs are recovered for these services, as allowed for under clause 6.18.5(c) of the *Rules*.

12.3.6 Estimating avoidable and stand alone costs

The formula used for calculating prices for fee-based services and quoted services has been designed to ensure prices will represent the efficient costs of providing and delivering the service, and signal the economic costs of service provision by being subsidy free.

Prices based on this formula will be cost-reflective, representing costs derived through the same allocation method as that used to determine costs for SCS, in accordance with the AER approved Cost Allocation Method (CAM). The prices for each tariff class within ACS will be between the bounds of avoidable and stand alone costs due to the economies of scale in providing each service.

The avoidable cost for a particular service is equivalent to the direct labour and materials cost. Overhead costs will be incurred regardless of whether the service is provided. Consequently, the price calculated using the formula in Equation 12.1 (Section 12.3.4) (which includes an allowance for overhead costs) will be greater than the avoidable cost.

The stand alone cost is equal to the costs of serving each tariff class within ACS on a stand alone basis. For example, the stand alone cost would require the use of dedicated resources and assets (including costs associated with corporate support, capital allowance for non-system assets, etc.). As these costs can be shared among all of the tariff classes within ACS, the cost calculated for each individual service will be less than the stand alone cost.

12.4 Large customer connections

12.4.1 Overview

From 1 July 2010, for new or upgraded connections (requested by a customer) of greater than 1 MVA or 4 GWh per annum (that is a tariff class of EG, CAC or ICC), the design and construction of the connection assets is a quoted service and will be priced as detailed in Section 12.3.

As part of the new service classification, customers may choose to have the service provided by other suppliers subject to those suppliers complying with Energex's Quality Assurance and Materials Specification requirements.

The commissioning, operation and maintenance of all connection assets, including large connections, was determined by the AER to be a SCS which is consistent with the previous regulatory approach. In addition, the existing pricing methodology arrangements for the use of the shared non-system assets are unchanged.

12.4.2 Framework

The framework for new and existing large customer connections (LCC) is provided in Figure 12.2 and Table 12.2.

The design and construction of LCC will be classified as one of the following:

- ACS – All new connections or upgrades to existing connections, which are paid for by the customer and gifted to Energex. This may include an upfront payment for the design and construction of the connection asset, calculated in accordance Equation 12.1 (Section 12.3.4). These assets will form part of the Contributed Asset Base (CAB). Items in

the CAB will have no return on capital or regulatory depreciation cost allocated to them. However there will be an allocation for O&M costs recovered through DUOS as per the tariff cost allocation process.

- SCS – LCC assets, existing prior to 1 July 2010, which are owned and maintained by Energex, or for Energex driven asset replacement. These services will continue to form part of the Regulatory Asset Base (RAB). The methodology for calculating network charges for these customers is consistent with the previous Regulatory Control Period. Accordingly, these connection assets have costs allocated to them for return on capital, regulatory depreciation and O&M as per the tariff cost allocation process detailed in Appendix 3.
- Unregulated Services – Connection assets that are funded, owned and operated by the customer. These services will attract no specific connection asset charges.

12.4.3 Proposed charges

RULE REQUIREMENT

Clause 6.18.2 Pricing proposals

- (b) A pricing proposal must:
- (2) set out the proposed tariffs for each tariff class.

Due to the variability in the scope, the design and construction of LCC is a quoted service for which no specific prices are published. The price for the illustrative service configuration for large customer connections is shown in Appendix 2.3.

12.4.4 Charging parameters

RULE REQUIREMENT

Clause 6.18.2 Pricing proposals

- (b) A pricing proposal must:
- (3) set out, for each proposed tariff, the charging parameters and the elements of service to which each charging parameter relates.

Energex will calculate the price for the design and construction of LCC assets in accordance with the formula in Equation 12.1 (Section 12.3.4).

Figure 12.2 - Service classification for LCC

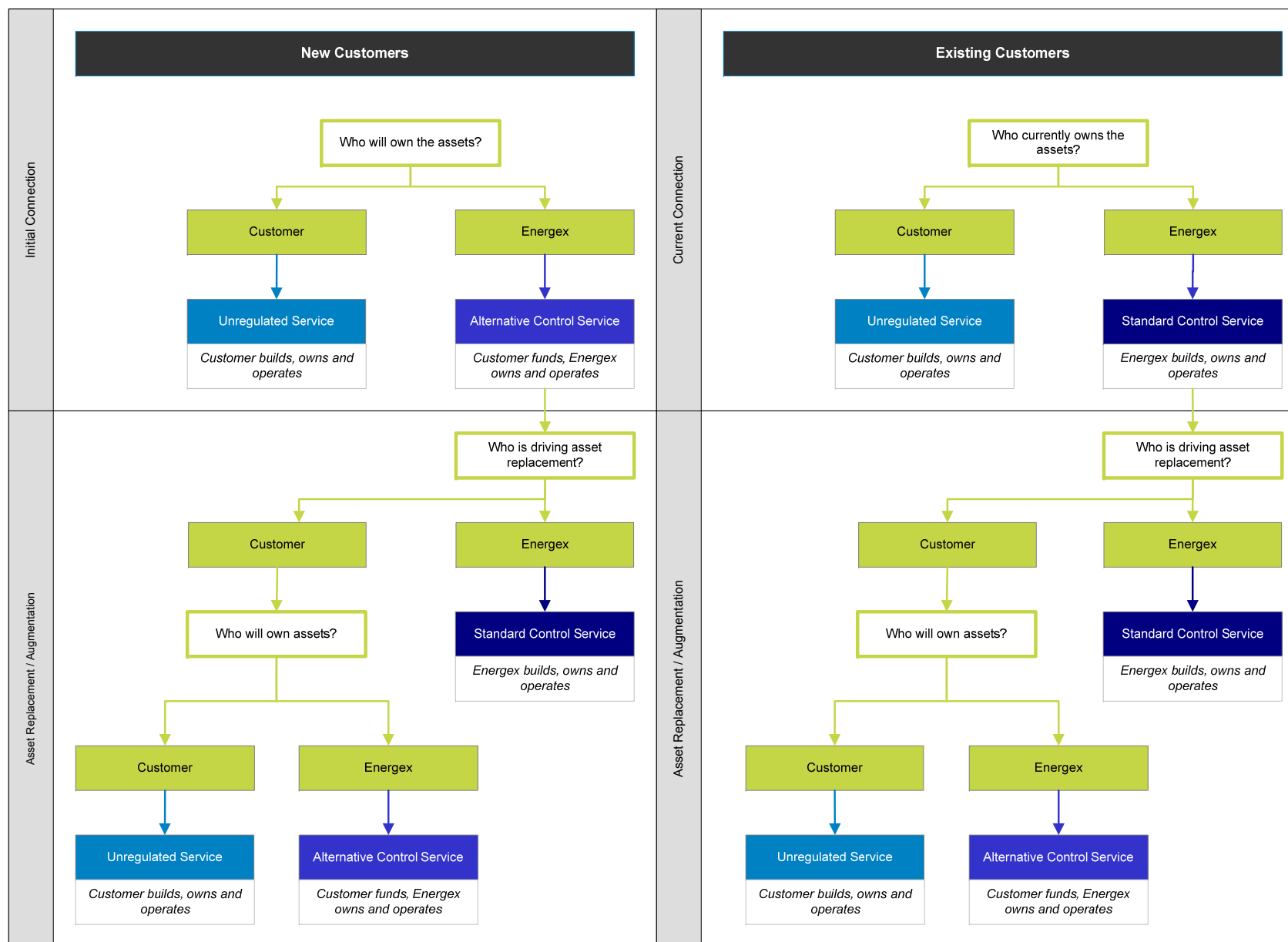


Table 12.2 - LCC pricing framework

Initial Connection Date	Description	Ownership ¹	Asset Classification	Service Classification	Asset Base	Upfront Customer payment (quoted price, relating to design and construction costs)	Tariff Charging Parameter (site-specific fixed charge)	
							Recovery of depreciation and return on capital (through DUOS)	Recovery of operating expenditure (through DUOS)
Before 1 July 2010 (or part of transitional arrangement) Asset constructed under previous framework	Existing connection	Energex	Non-contributed	SCS	RAB	N/A	✓	✓
	Upgrade - Energex driven ²	Energex	Non-contributed	SCS	RAB	N/A	✓	✓
	Upgrade - Customer request	Energex (gifted)	Contributed	ACS	CAB	✓	N/A	✓
After 1 July 2010 Asset constructed under new framework	New Connection	Energex (gifted)	Contributed	ACS	CAB	✓	N/A	✓
	Upgrade - Customer request	Energex (gifted)	Contributed	ACS	CAB	✓	N/A	✓
	Upgrade - Energex driven ²	Energex	Non-contributed	SCS	RAB	N/A	✓	✓
Asset constructed under either framework	Replacement - During warranty period for gifted assets	Energex (gifted)	Contributed	N/A	CAB	N/A (covered under warranty)	N/A	✓
	Replacement - Outside manufacturer's warranty period	Energex (gifted)	Contributed	ACS	CAB	✓	N/A	✓
		Energex	Non-contributed	SCS	RAB	N/A	✓	✓
	Any service	Customer	N/A	Unregulated	N/A	No specific connection asset charges		

Notes:

1. If the customer chooses to retain ownership of the asset, the service is unregulated and there are no specific connection asset charges.
2. An Energex driven upgrade to a customer's connection assets could occur, when for network reasons, the connection arrangement needs to be altered.

13. Customer impacts

13.1 SCS

Energex is aware of the changing expectations of customers and the current upward pressure being exerted on energy prices and has considered this when developing its network tariffs. Energex is committed to achieving a balanced commercial outcome while meeting its obligations to customers and managing sustainability and risk.

13.1.1 Customer numbers and consumption

Energex's revenue stream is derived from its customer base, in the proportions shown in Figures Figure 13.1 and Figure 13.2. Based on the most recently completed regulatory year, 2011/12 (t-2), Figure 13.1 breaks down Energex's actual customer numbers into tariff classes, and Figure 13.2 breaks down actual total consumption into tariff classes. 2013/14 expected revenue by tariff class is available in Table 2.4 and Table 8.1.

The relationship between consumption and revenue in 2013/14

Reflective of the assets required to service customers, the ratio of revenue consumption for ICC and CAC tariff classes is lower than the ratio for the SAC Non-Demand tariff class. The ICC and CAC customers, though they use large volumes, are connected high up in the distribution network, whereas the SAC Non-Demand customers' smaller volumes but use the electricity network at the LV connection, which is the lowest part of the network. As such, the cost-reflective price is higher for SAC Non-Demand customers connected at the LV level, compared to the cost-reflective price for the ICC and CAC customers who are connected at the HV or sub-transmission voltage level.

Figure 13.1 - 2011/12 - Customer numbers (count) by tariff class

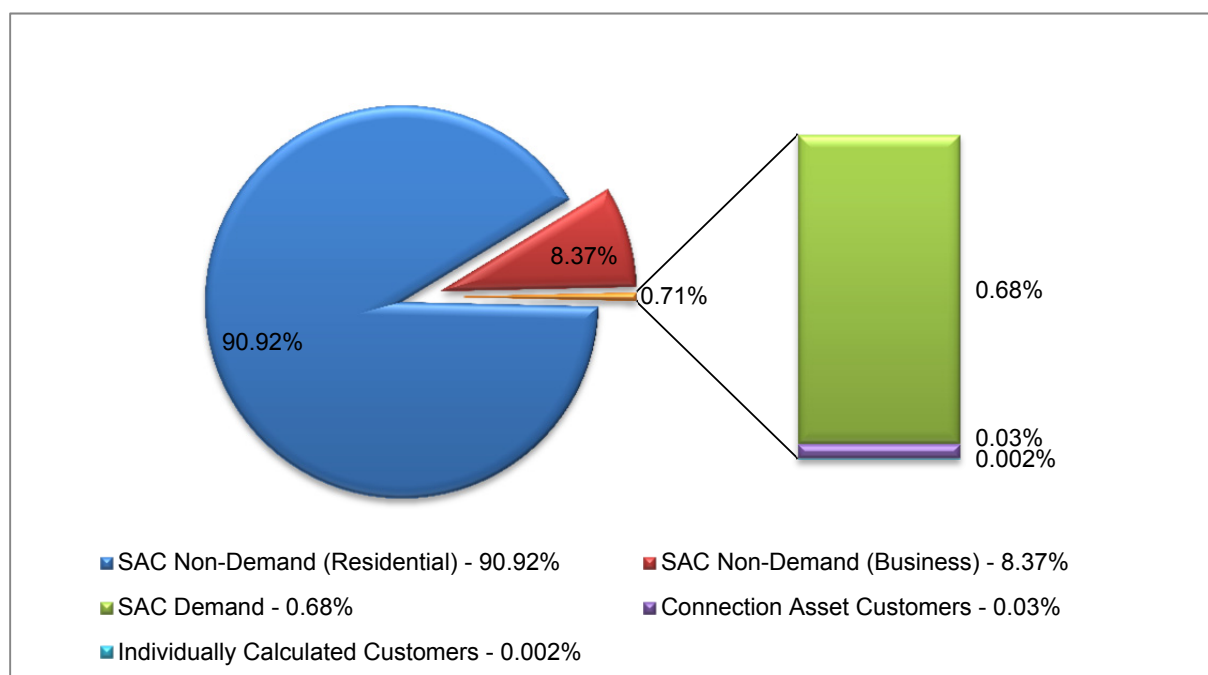
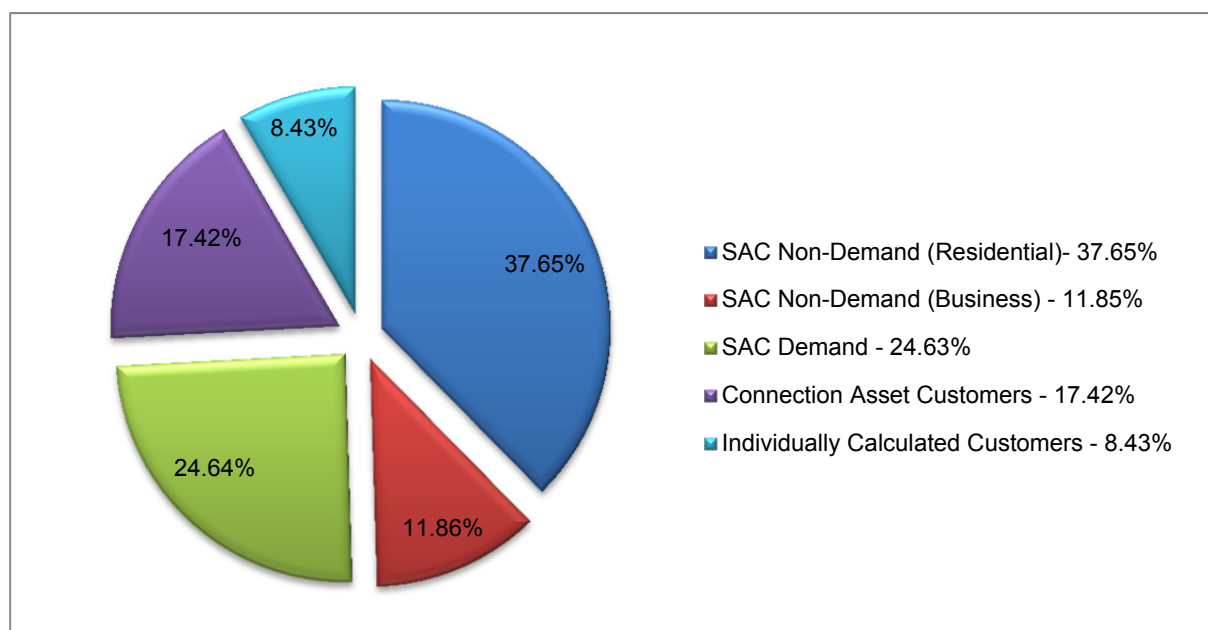


Figure 13.2 - 2011/12 Consumption by tariff class



13.1.2 2013/14 Price impacts

The ongoing average price increases across the Regulatory Control Period are necessary for Energex to deliver the capital and operating expenditure programs which support development and growth in SEQ, as well as improving reliability and security of supply.

Table 13.1 provides an estimate of the charges in 2013/14 for the average consumption level in each tariff class. This table provides an estimate of the percentage change for the average customer. The impact on each customer will be dependent on the individual customer's demand and consumption patterns.

Table 13.1 - Outline of estimated average percentage price change by tariff class from 2012/13 to 2013/14

Tariff Class	Approved DUOS charge 2012/13 (c/kWh)	Estimated DUOS charge 2013/14 (c/kWh)	Average percentage change (%)
ICC	1.81	1.99	10.17
CAC – 33kV Line/Bus	2.89	3.35	16.16
CAC – 11kV Bus	2.39	2.89	20.99
CAC – 11kV Line	2.82	3.42	21.32
EG	1.99	2.26	14.01
SAC Demand	6.05	7.28	20.42
SAC Non-Demand	9.54	11.43	19.73

Analysis undertaken by Energex on the network price increase that may be experienced by low, typical and high usage customers on tariffs within the SAC Demand and SAC Non-Demand tariff classes are included in Table 13.2. The prices quoted in the table are indicative only. All analysis is based on data from Energex's most recently completed regulatory year (t-2), i.e. 2011/12.

The network prices used for the analysis comprise total annual NUOS including GST, and represent the typical 'N' (Network) component of a customer's bill. These NUOS prices are the AER approved prices for 2012/13 and the proposed 2013/14 prices included in this document for AER approval.

For SAC Non-Demand tariffs, the low, typical and high usage scenarios considered for each tariff are based on the 10th, 50th and 90th percentiles of annual energy consumption. For SAC Demand, an average load factor has been applied to the 10th, 50th and 90th percentiles of monthly demand for the purposes of analysis. ICC, CAC and EG tariffs comprise various site-specific charges, as such, have been excluded from this analysis. Additionally, due to its complexity, the Unmetered Supply tariff has also been excluded.

Table 13.2 - Indicative NUOS price change from 2012/13 to 2013/14 for varying usage profiles for SAC Demand and SAC Non-Demand tariffs

SAC Demand tariffs	Usage type ¹	Monthly demand ² (kW)	2012/13 NUOS (\$)	2013/14 NUOS ³ (\$)	Typical annual NUOS increase ⁴ (\$)	Typical annual NUOS increase ⁴ (%)
HV Demand – 8000	Not included due to insufficient customer numbers					
Demand Large – 8100	Low usage	260	77,153.00	89,359.08	12,206.08	15.82
	Typical usage	365	103,612.00	120,142.66	16,530.66	15.95
	High usage	623	168,625.54	195,782.31	27,156.77	16.10
Demand Small – 8300	Low usage	24	12,493.51	10,633.03	-1,860.48	-14.89
	Typical usage	70	24,988.62	25,803.22	814.60	3.26
	High usage	180	54,868.23	62,079.74	7,211.51	13.14
SAC Non-Demand tariffs	Usage type ¹	Annual consumption ² (kWh)	2012/13 NUOS ⁵ (\$)	2013/14 NUOS ³ (\$)	Typical Annual NUOS Increase ⁴ (\$)	Typical Annual NUOS Increase ⁴ (%)
Business Flat – 8500	Low usage	1,010	356.99	426.42	69.43	19.45
	Typical usage	8,970	1,240.30	1,492.99	252.69	20.37
	High usage	47,220	5,484.82	6,618.14	1,133.32	20.66
Business ToU – 8800	Low usage	2,290	478.77	569.13	90.37	18.87
	Typical usage	27,890	3,092.99	3,677.40	584.40	18.89
	High usage	105,720	11,040.86	13,127.26	2,086.40	18.90
Residential Flat – 8400⁵	Low usage	1,700	331.27	399.84	68.57	20.70
	Typical usage	4,300	622.99	741.78	118.79	19.07
	High usage	9,390	1,194.08	1,411.19	217.11	18.18
Residential ToU – 8900	Not included due to insufficient customer numbers					
PeakSmart ToU – 7600	Not included as new tariff					
Super Economy – 9000	Low usage	890	72.99	85.95	12.95	17.74
	Typical usage	2,130	174.69	205.69	31.00	17.74
	High usage	3,890	319.04	375.65	56.61	17.74
Economy – 9100	Low usage	1,110	50.21	59.06	8.85	17.63
	Typical usage	2,210	99.96	117.59	17.62	17.63
	High usage	3,620	163.74	192.61	28.87	17.63

Notes:

1. Low, typical and high usage scenarios are considered for each tariff based on the 10th, 50th and 90th percentiles of annual energy consumption for SAC Non-Demand and Mean Maximum Monthly Demand for SAC Demand.
2. Energy usage profiles are based on actual data from the 2011/12 regulatory year and may differ from future forecast data.
3. Total Annual NUOS including GST represents the typical 'N' component of a customer's bill.
4. Price increases shown in this table are indicative only. Individual customers should consider their specific circumstances to determine their likely network tariff impact.
5. In 2012/13, the Queensland Government capped the Residential Flat (NTC 8400) DUOS fixed charge. Consequently, the observed customer impact calculated from the capped price will be greater than the observed customer impact calculated from the uncapped price (i.e. the approved price in Energex's 2012/13 Annual Pricing Proposal).

13.2 ACS

The price changes between 2012/13 and 2013/14 for ACS (street lighting and fee-based services) are provided in Table 13.3 and Table 13.4, respectively.

Table 13.3 - Street light services percentage increase from 2012/13 to 2013/14

Street lighting service ¹	Percentage change (%)
Major non-contributed	8.91
Major contributed	11.11
Minor non-contributed	7.32
Minor contributed	9.09
Note 1. Definitions of major and minor lamps are included in Appendix 2.1.	

Table 13.4 - Fee-based services percentage increase from 2012/13 to 2013/14

Fee-based service	Percentage change (%)
Alterations and additions to current metering equipment	3.78
Attending loss of supply – LV customer installation at fault (BH)	3.78
Overhead service replacement – single phase	3.78
Overhead service replacement – multiple phase	3.78
De-energisation	3.78
Meter test	3.78
Meter inspection	3.78
Reconfigure meter	3.78
Off-cycle meter read	3.78
Site visit	3.78
Locating Energex underground cables	N/A
Temporary connection	3.78
Re-energisation (BH)	3.78
Re-energisation (AH)	3.78
Re-energisation (visual) (BH)	3.78
Re-energisation (visual) (AH)	3.78
Re-energisation non-payment (visual) (BH)	3.78
Re-energisation non-payment (visual) (AH)	3.78
Supply abolishment	3.78
Unmetered supply	3.78
Street light glare screening	3.78
Replacement of standard luminaries with aero screen units (per street light)	3.78

14. Publication of information about tariffs and tariff classes

RULE REQUIREMENT

Clause 6.18.9 Publication of information about tariffs and tariff classes

- (a) A Distribution Network Service Provider must maintain on its website:
- (1) a statement of the provider's tariff classes and the tariffs applicable to each class; and
 - (2) for each tariff – the charging parameters and the elements of the service to which each charging parameter relates; and
 - (3) a statement of expected price trends (to be updated for each regulatory year) giving an indication of how the Distribution Network Service Provider expects prices to change over the Regulatory Control Period and the reasons for the expected price changes.
- (b) The information for a particular regulatory year must, if practicable, be posted on the website 20 business days before the commencement of the relevant regulatory year and, if that is not practicable, as soon as practicable thereafter.

Once approved, Energex's Annual Pricing Proposal (this document) and Energex's Annual Tariff Schedule will be published on the Energex website. As required by clause 6.18.9(a)(1) and (2), these documents describe Energex's tariff classes, each of the tariffs applicable to each class, and the charging parameters and the elements of the service to which each charging parameter relates.

The Statement of Expected Price Trends is also published annually on the Energex website. This document details that:

- for SCS, any change in prices will be subject to the approved X Factor from the Final Determination, the change in CPI and any applicable yearly adjustments
- in each regulatory year (as part of the requirements of a pricing proposal under the *Rules*), Energex will demonstrate that the proposed tariffs for DUOS will meet the side constraints formula for each tariff class
- for ACS including street lighting and fee-based services, the prices will be subject to an annual escalation process.

In accordance with clause 6.18.9(b) of the *Rules*, this document, the Tariff Schedule, and the Statement of Expected Price Trends will be published on the Energex website on 3 June 2013, or as soon as practicable thereafter.

15. Confidential information

Energex claims confidentiality over the sections of this document specified in Table 15.1 on the grounds that the information contained therein is either commercial-in-confidence or contains intellectual property.

Energex requests that the AER does not disclose the information contained in these sections to any person outside the AER.

Table 15.1 - Confidential information

Chapter	Section	Title
Appendix 1	Appendix 1.1	ICC site-specific tariffs
	Appendix 1.2	CAC and EG site-specific tariffs
Appendix 2	Appendix 2.4	Price development for ACS
	Appendix 2.5	Additional reporting requirements for ACS

16. Appendices

16.1 Table of contents (Appendices)

APPENDIX 1. 2013/14 TARIFFS – SCS	A-1
Appendix 1.1 – ICC site-specific tariffs (CONFIDENTIAL)	A-2
Appendix 1.2 – CAC and EG site-specific tariffs (CONFIDENTIAL).....	A-5
Appendix 1.3 – kVA calculation methodology	A-28
APPENDIX 2. 2013/14 TARIFFS – ACS	A-29
Appendix 2.1 – Street lighting services	A-30
Appendix 2.2 – Fee-based services	A-31
Appendix 2.3 – Quoted services	A-33
Appendix 2.4 – Price development for ACS (CONFIDENTIAL).....	A-34
Appendix 2.5 – Additional reporting for ACS (CONFIDENTIAL).....	A-39
APPENDIX 3. TARIFF COST ALLOCATION PROCESS	A-42
APPENDIX 4. SUMMARY OF COMPLIANCE	A-55
Appendix 4.1 – Compliance with the National Electricity Rules	A-56
Appendix 4.2 – Compliance with the Final Determination	A-60
APPENDIX 5. GLOSSARY	A-62
Appendix 5.1 – Acronyms and abbreviations	A-63
Appendix 5.2 – Measurements.....	A-65
Appendix 5.3 – Definitions.....	A-66

16.2 List of tables (Appendices)

Table A-1.1 - 2013/14 ICC site-specific prices	Confidential
Table A-1.2 - 2013/14 CAC and EG site-specific prices	Confidential
Table A-2.1 - 2013/14 Prices for street lighting services	A-30
Table A-2.2 - 2013/14 Prices for fee-based services	A-31
Table A-2.3 - 2013/14 Prices for quoted services	A-33
Table A-2.4 - 2013/14 Proposed cost escalators	A-33
Table A-2.5 - 2013/14 Proposed on-cost and overhead rates	A-33
Table A-2.6 - Service level assumptions and forecast volume used for 2013/14 price calculations	Confidential
Table A-2.7 - 2011/12 Revenue collected from street light services.....	Confidential
Table A-2.8 - 2011/12 Volume of fee-based services and collected revenue	Confidential
Table A-2.9 - 2011/12 Volume of quoted services and collected revenue	Confidential
Table A-4.1 - Demonstrated compliance with the <i>Rules</i>	A-56
Table A-4.2 - Demonstrated compliance with the Final Determination.....	A-60
Table A-5.1 - Acronyms and abbreviations used through this document	A-63
Table A-5.2 - Units of measurement used through this document.....	A-65
Table A-5.3 - Multiples of prefixes (units) used through this document.....	A-65
Table A-5.4 - Definitions of terminology used through this document	A-66

16.3 List of figures and equations (Appendices)

Equation A-1.1 - kVA calculation methodology.....	A-28
Figure A-3.1 - Cost allocations to tariff classes.....	A-44
Figure A-3.2 - Time based cost allocation method.....	A-46
Figure A-3.3 - Demand profiles by voltage level	A-47
Figure A-3.4 - Daily cost allocation by voltage level (\$/MW).....	A-47
Figure A-3.5 - Demand profiles by tariff class	A-49
Figure A-3.6 - Daily cost allocation by tariff class (\$/MW)	A-49
Figure A-3.7 - Cost allocation method: ICC, CAC and EG tariff classes	A-51
Figure A-3.8 - Cost allocation method: SAC Demand tariff class	A-52
Figure A-3.9 - Cost allocation method: SAC Non-Demand tariff class	A-53

APPENDIX 1.

2013/14 Tariffs – Standard Control Services

CONFIDENTIAL

Appendix 1.1 – ICC site-specific tariffs

The ICC tariffs are site-specific and are not published due to the confidentiality requirements of the customer. Energex will provide these site-specific tariffs directly to the customer and their retailer.

CONFIDENTIAL

Appendix 1.2 – CAC and EG site-specific tariffs

The CAC tariffs are site-specific and are not published due to the confidentiality requirements of the customer. Energex will provide these site-specific tariffs directly to the customer and their retailer.

Appendix 1.3 – kVA calculation methodology

From 1 July 2013, it is proposed that straight kVA pricing will be introduced for ICC, CAC and EG customers replacing the existing power factor adjustment (PFA) methodology and reflective of current practices in other Australian jurisdictions.

Energex will apply the formula in Equation A-1.1 to calculate the kVA value used to determine demand and capacity charges for each customer. Aligning with industry standards, kVA will be calculated for 30 minute intervals.

Equation A-1.1 - kVA calculation methodology

$$kVA = \sqrt{kW^2_{PeakLoad} + kvar^2_{PeakLag}} \text{ for each 30 minute interval}$$

where,

kVA (kilovolt amperes) is used to measure demand; it measures the apparent power flow which is a measure of the total capacity required to supply a customer's load

kW (kilowatt) is a measure of the real component of power being consumed by a customer's load

PeakLoad is the maximum kW per 30 minute interval.

kvar (kilovolt amperes reactive) is a measure of reactive power which exists when the current and voltage are out of phase, i.e. not changing simultaneously.

PeakLag is the maximum kvar per 30 minute interval.

To calculate kVA using the formula:

- The maximum kW per 30 minute interval (kW PeakLoad) will be determined by the real time summation of the individual meters within a premise. For meters that store the interval data in 15 minute intervals, the real time summation will also include the summation to 30 min intervals. The maximum kW for a 30 minute interval will not be netted with any subsequent generation data that may occur within the 30 minute interval
- The maximum kvar per 30 minute interval (kvar PeakLag) will be determined using the same methodology with no netting of leading kvar data within the 30 min interval
- Once the maximum kW and maximum kvar are determined for each 30 minute interval in the billing period, then the maximum kVA for billing purposes will be calculated as the greatest kVA for any 30 min interval in the billing period, in the same 30 minute interval as the maximum kW and kvar.

APPENDIX 2.

2013/14 Tariffs – Alternative Control Services

Appendix 2.1 – Street lighting services

This section provides the prices for the provision, construction and maintenance of street light services. The prices, included in Table A-2.1 are approved in the Final Determination.

Table A-2.1 - 2013/14 Prices for street lighting services

Street light Service	Price ¹ (\$/light/day)
Major non-contributed	1.10
Major contributed	0.30
Minor non-contributed	0.44
Minor contributed	0.12
Note 1. All prices exclude GST.	

In accordance with the Energex Public Lighting Manual, the following definitions apply for major and minor lamps:

- Major Lamps – in common use of Major Road lighting and includes the following:
 - High Pressure Sodium 100 watt (S100) and above
 - Metal Halide 150 watt (H150) and above
 - Mercury Vapour 250 watt (M250) and above.
- Minor Lamps – all lamps in common use for Minor Road lighting other than Major Lamps and include High Pressure Sodium, Mercury Vapour and Fluorescent.

Appendix 2.2 – Fee-based services

This section provides the proposed prices for individual fee-based services provided by Energex. The prices are included in Table A-2.2. These prices were determined using the AER's approved formula-based price cap control mechanisms.

Table A-2.2 - 2013/14 Prices for fee-based services

Fee-based service	Price ¹ (\$/service)
Alterations and additions to current metering equipment	105.45
Attending loss of supply – low voltage customer installation at fault (BH)	117.79
Overhead service replacement – single phase	322.47
Overhead service replacement – multiple phase	379.88
De-energisation ²	52.81
Meter Test	122.04
Meter Inspection	94.26
Re-configure meter	78.14
Off-cycle meter read	8.80
Site visit	67.68
Locating Energex underground cables	N/A
Temporary connection	937.65
Re-energisation (BH) ²	46.02
Re-energisation (AH) ²	131.32
Re-energisation (visual) (BH) ²	78.24
Re-energisation (visual) (AH) ²	171.56
Re-energisation non-payment (visual) (BH) ²	78.24
Re-energisation non-payment (visual) (AH) ²	171.56

Fee-based service	Price ¹ (\$/service)
Supply abolishment	365.23
Unmetered supply	161.19
Street light glare screening	155.78
Replacement of standard luminaries with aero screen units (per street light)	360.14

Notes:

1. All prices exclude GST.
2. Prices for these services are subject to Schedule 8 of the Queensland *Electricity Regulation 2006*. As Schedule 8 prices for 2013/14 are yet to be published, the rates in this table represent the proposed Energex costs using the ACS formula. The Schedule 8 prices will be included in Energex 2013/14 Tariff Schedule which will be published on the Energex website prior to 1 July 2013.

Appendix 2.3 – Quoted services

The prices for the illustrative configuration of Energex's quoted services are outlined in Table A-2.3. These prices were determined using the AER's approved formula based price cap control mechanism. The prices below do not represent a binding capped price for an individual quoted service.

Table A-2.3 - 2013/14 Prices for quoted services

Quoted service ¹	Price ² (\$/service)
Rearrangement of network assets	4,529.38
Customer requested works to allow customer or contractor to work close	6,701.17
Non-standard data and metering services (Type 5 – 7 metering)	122.00
Emergency recoverable works and rectification of illegal connections	10,061.82
Large customer connections	377,227.69
Design specification / auditing and other subdivision activities	1,463.99
Unmetered services, including street lighting	1,963.73
AH provision of any fee-based service (excluding re-energisation)	1,743.55
Supply abolishment – complex	485.54
Additional crew	128.67
Temporary connection – complex	49,412.54
Loss of asset	9,946.26
Other recoverable work ³	0.00

Notes:

1. Illustrative configuration only.
2. All prices exclude GST.
3. There is no common configuration of the 'other recoverable work' service. The service is applied only in those circumstances where the service requested is not covered by any of the other service categories or would not otherwise have been requested for the efficient management of the network.

CONFIDENTIAL

Appendix 2.4 – Price development for ACS

The information contained within this appendix is confidential. Energex will provide this information directly to the AER.

CONFIDENTIAL

Appendix 2.5 – Additional reporting for ACS

The information contained within this appendix is confidential. Energex will provide this information directly to the AER.

APPENDIX 3.

Tariff Cost Allocation Process

Appendix 3 – Tariff cost allocation process

Part 1. Tariff class cost allocation process

Energex's revenue cap (as determined by the AER) is based on a building block approach, which includes each of the following regulated cost components:

- Regulatory depreciation
- Return on capital
- Operating expenditure
- Tax allowance
- Capital contribution
- Revenue from shared assets.

The amount attributed to each component is provided in Energex's 2013/14 Pricing Proposal¹⁹. The purpose of the tariff cost allocation process is to allocate or assign the network costs to the tariff classes in the most economically efficient and cost-reflective way.

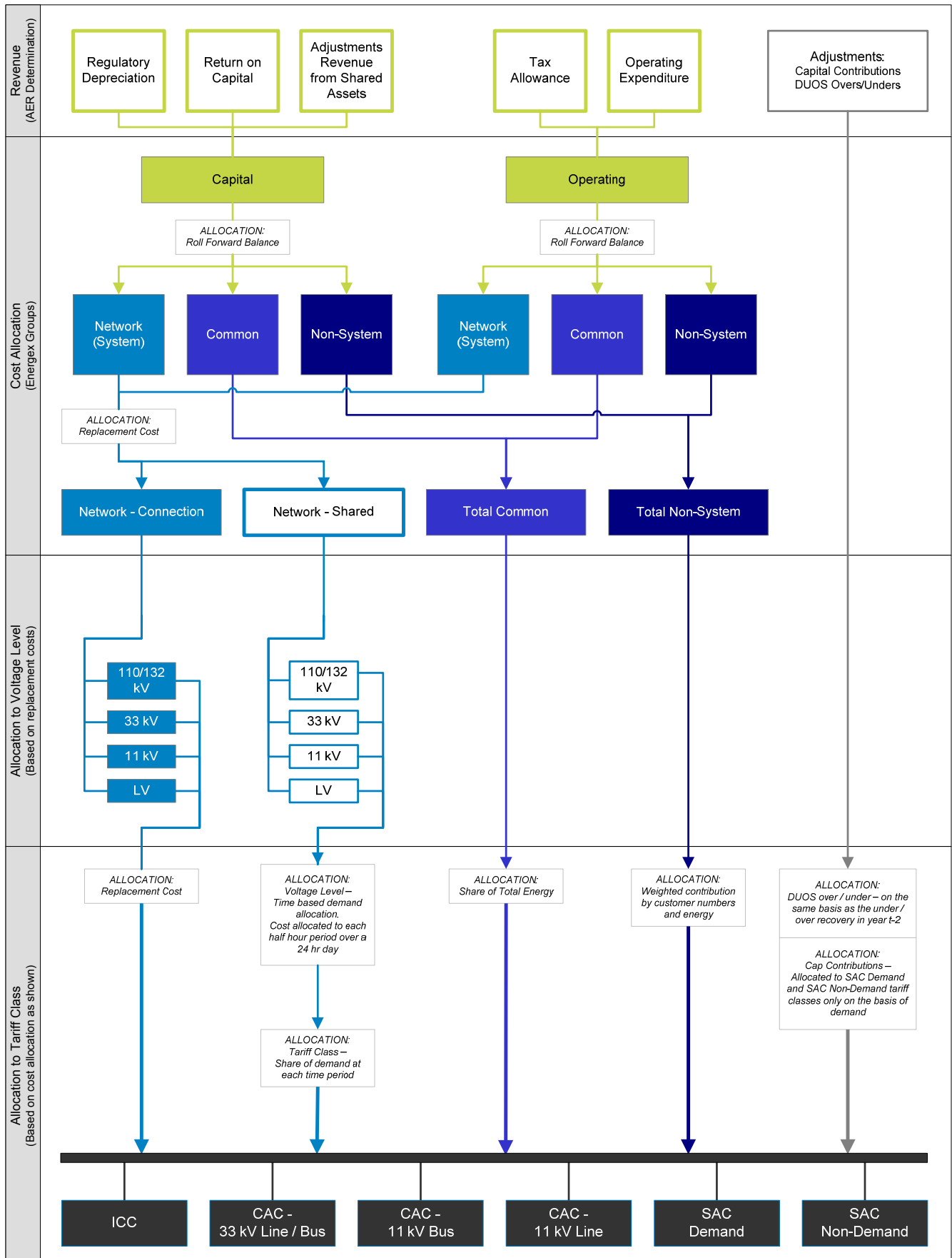
The major steps in the process are:

- Step 1 – Allocate AER building blocks to Energex DCOS cost groups
- Step 2 – Allocate Network (System) costs to voltage level
- Step 3 – Allocate costs to tariff classes.

These steps are illustrated in Figure A-3.1 and explained throughout the remainder of Part 1 of this appendix. Part 2 of this appendix addresses the allocation of costs to specific tariffs.

¹⁹ Refer to Table 2.2 in Energex's 2013/14 Pricing Proposal (Chapter 2, Section 2.2.1).

Figure A-3.1 - Cost allocations to tariff classes



Step 1 – Allocate AER building blocks to Energex DCOS cost groups

The regulated cost components specified by the AER are initially allocated into the Energex cost groups of:

- Network (system)
- Common services
- Non-system
- Adjustments.

Network (system)

Network (system) costs are the directly attributable costs associated with the provision of network connection and distribution services that are attributable to a single customer or group of customers. Network costs are allocated between connection assets and shared network assets based on the replacement cost of assets.

Network costs are further allocated to each of the following voltage cost groups based on the replacement costs of assets:

- 110/132 kV
- 33 kV
- 11 kV
- LV.

Common services

Common services costs are costs associated with those system assets that benefit the system as a whole and are not directly related to any single customer or group of customers. Assets included in this category are reactive plant, load control, control centres and communications.

Non-system

Non-system costs include corporate support (e.g. CEO, Finance, Human Resources and Legal), customer services, and occupancy costs that are not directly attributable to the operation and maintenance of the network but which are associated with network service delivery. These costs are treated consistently as a group as the cost drivers for this set of costs are consistent and it is impractical to manage a cost allocation stream for each of the specific components.

Step 2 – Allocate network (system) costs to voltage level

Individual calculated customers

The cost allocation to each ICC is performed on an individual basis. Connection assets are assigned to ICCs based on information obtained from Energex network panel diagrams. Each ICC is then allocated a share of upstream shared network. This is based on the ratio of the customer's individual demand to the total demand of the respective supply (substation). The share of network costs is based on a time based profile with the demand for the substation and the customer allocated across daily half hour periods (48 periods).

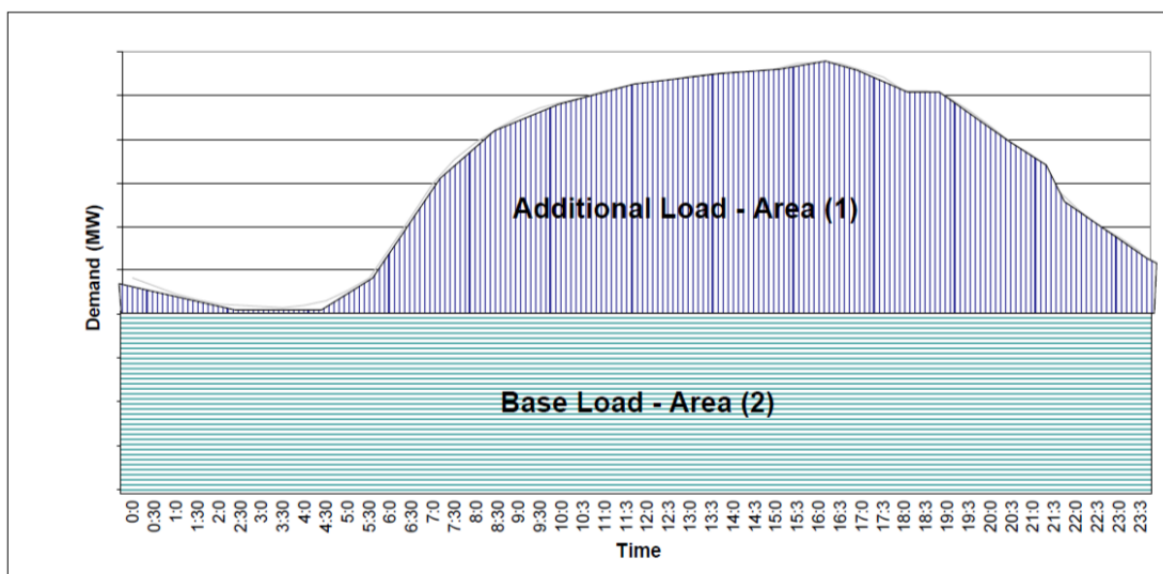
The ratio X in each period (customer demand / substation demand) is then applied to the replacement cost of system assets within the supply network to which the individual customer is connected. This is then aggregated to provide a total shared asset value for the customer.

Remaining allocation

Following the allocation of costs to ICCs, the remaining network costs are allocated to each of the voltage levels (110/132 kV, 33 kV, 11 kV and LV) on a time basis.

Network costs (capital and operating) are allocated to time periods based on the level of demand for that period. The basic principle is that the cost of providing the network (for system costs) should be allocated to the time periods when it is being utilised. Accordingly there is a “base load” that uses a proportion of the network all the time (Area 2 on Figure A-3.2). The cost of this base load of network can be recouped across all time periods. On the other hand, the extra network constructed for other periods, including the peak, should be recouped across the time periods when this is used (Area 1 on Figure A-3.2).

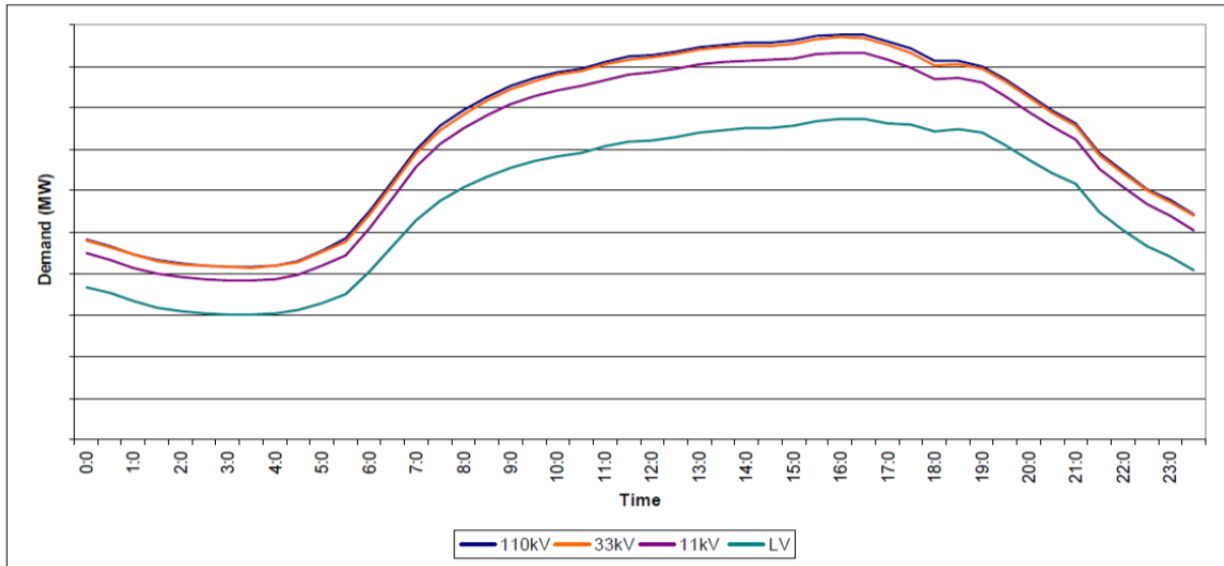
Figure A-3.2 - Time based cost allocation method



Note. Figure is indicative only.

Meter data is obtained for all interval metered customers (110/132 kV, 33 kV, 11 kV and LV Demand). From this meter data, an estimated peak daily load profile is obtained for each voltage level, as illustrated in Figure A-3.3.

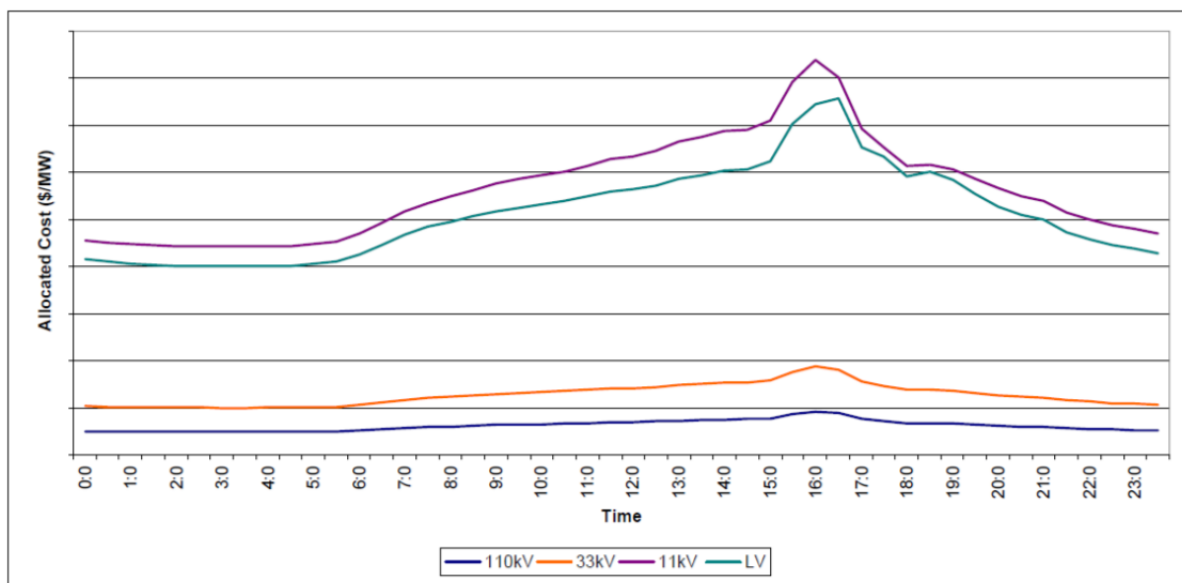
Figure A-3.3 - Demand profiles by voltage level



Note. Figure is indicative only.

Costs are allocated across daily half hour periods (48 periods) based on an estimate of the peak daily load profile for each voltage level. More costs are allocated to periods with a higher demand on the network, as shown in Figure A-3.4.

Figure A-3.4 - Daily cost allocation by voltage level (\$/MW)



Note. Figure is indicative only.

Step 3 – Allocated costs to tariff classes

There are seven SCS tariff classes²⁰ to which network costs are allocated:

1. ICC
2. CAC – 33 kV
3. CAC – 11 kV Bus
4. CAC – 11 kV Line
5. EG
6. SAC Demand
7. SAC Non-Demand.

Network

Connection assets

Connection assets are allocated to each tariff class based on their share of the replacement cost for each voltage level. Contributed connection assets are not used in the allocation of capital costs as these assets have already been paid upfront by the customer.

Shared network

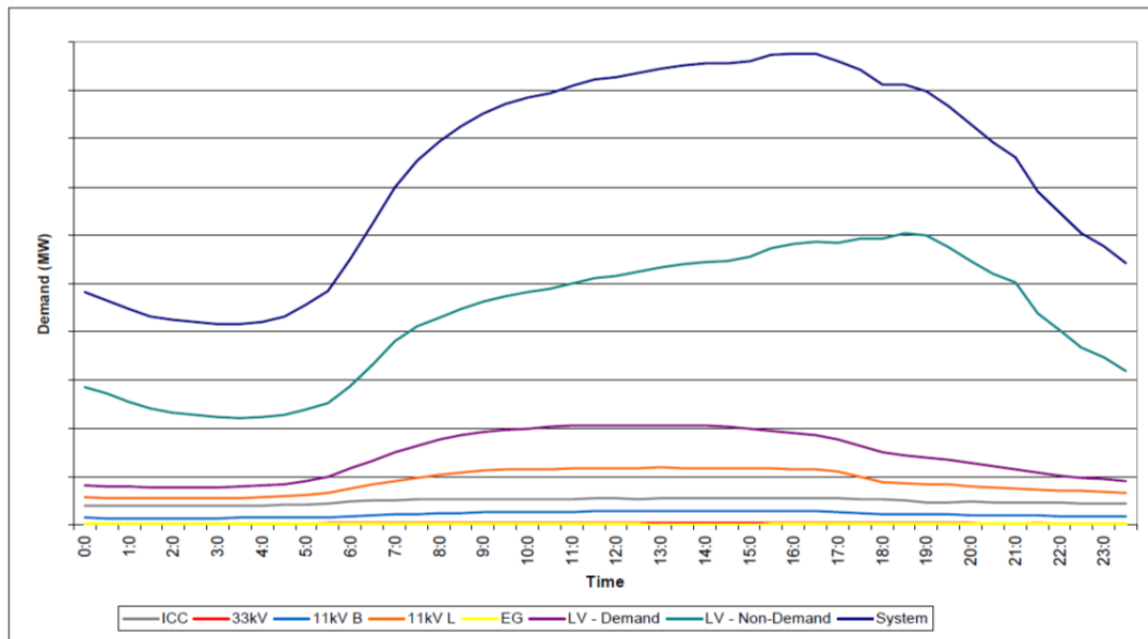
With the exception of ICCs, the cost for each voltage level during each time period is allocated to each tariff class based on their share of demand at that time. This results in a daily cost allocation for each tariff class for each time period. The total across all periods is then aggregated as the cost allocated to each tariff class as illustrated in Energex's 2013/14 Pricing Proposal.²¹

A period demand (ToU) allocation has been used for the shared system costs as it reflects the fact that demand is the primary driver of shared network costs. By using a period demand it allocates costs of the network to the time periods when it is being utilised. This method reflects the impact that demand (particularly peak demand) has on increasing network investment. Demand profiles for each of the SCS tariff class is shown in Figure A-3.5 and the daily cost allocation by tariff class is shown in Figure A-3.6.

²⁰ For more information about Energex's tariff classes, refer to Table 3.1 in Energex's 2013/14 Pricing Proposal (Chapter 3, Section 3.1).

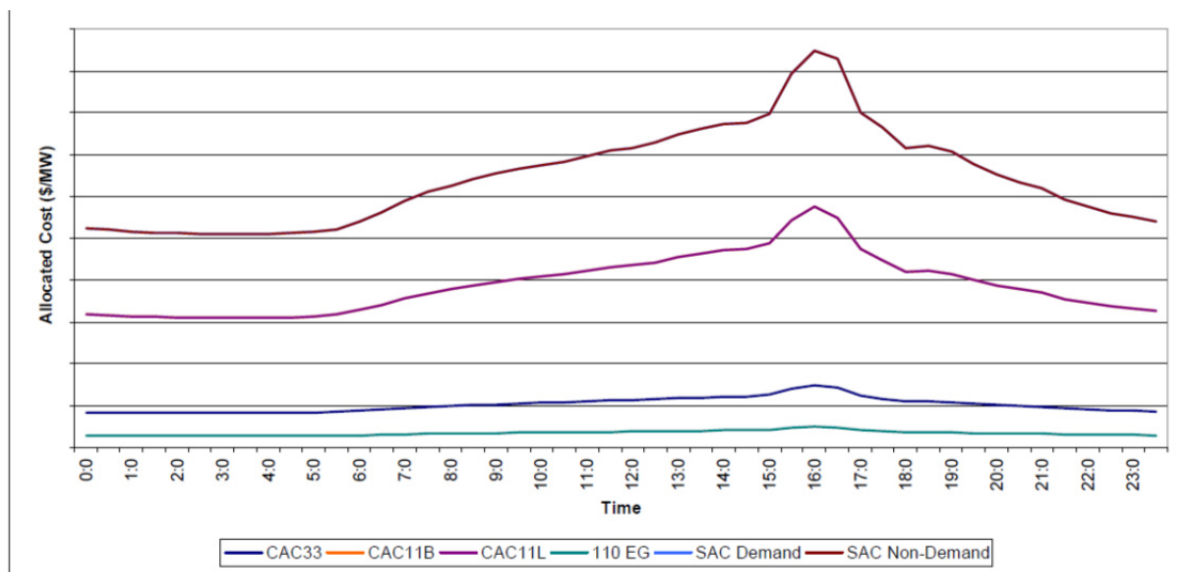
²¹ Refer to Figure 2.1 in Energex's 2013/14 Pricing Proposal (Chapter 2, Section 2.1).

Figure A-3.5 - Demand profiles by tariff class



Note. Figure is indicative only.

Figure A-3.6 - Daily cost allocation by tariff class (\$/MW)



Note. Figure is indicative only.

Common services

Common services costs are allocated to each tariff class using total energy as the allocation method. Anytime energy is used in cost allocation for costs which are related to the size of the customer but not specifically the demand the customer places on the network.

Non-system

Non-system costs are allocated to each tariff class using a hybrid allocation. Customer numbers (75%) and total energy (25%) is used. These weightings reflect that the number of customers is the primary driver of non-system costs.

Customer numbers and energy are used for the cost allocation method as these costs are associated with the number of connected customers and their expectations/service requirements. Energex has a number of costs that are customer number based. A significant proportion of the overhead costs of the business are driven by the number of staff and systems required to serve the customer base.

Part 2. Tariff (charging parameter) cost allocation process

Following the cost allocation to tariff classes, costs must be allocated to tariffs and ultimately to charging parameters (tariff elements), which may include any combination of the following:

- Fixed charges
- Capacity charges
- Demand charges
- Volume charges.

The purpose of the tariff (charging parameter) cost allocation process is to allocate or assign the costs to each parameter in the most efficient and cost-reflective way. Section 1 of this Appendix sets out the process utilised by Energex for the allocation of costs to charging parameters.

The process for cost allocation involves two major steps:

- Step 1 – Allocate tariff class costs to individual tariffs
- Step 2 – Allocate tariff costs to charging parameters.

These steps are undertaken for each tariff class and are illustrated in Figures A-3.7, A-3.8 and A-3.9.

Figure A-3.7 - Cost allocation method: ICC, CAC and EG tariff classes

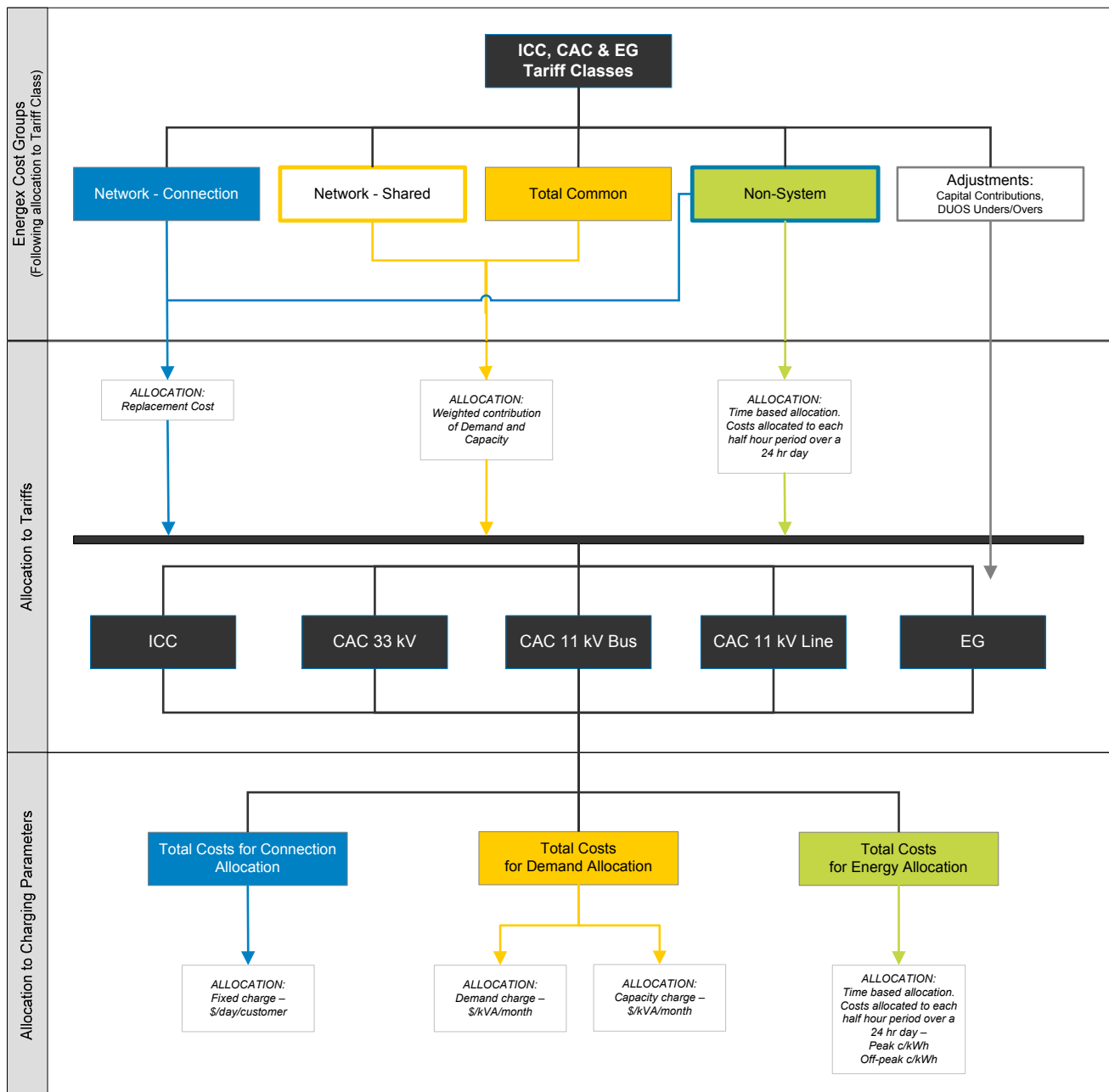


Figure A-3.8 - Cost allocation method: SAC Demand tariff class

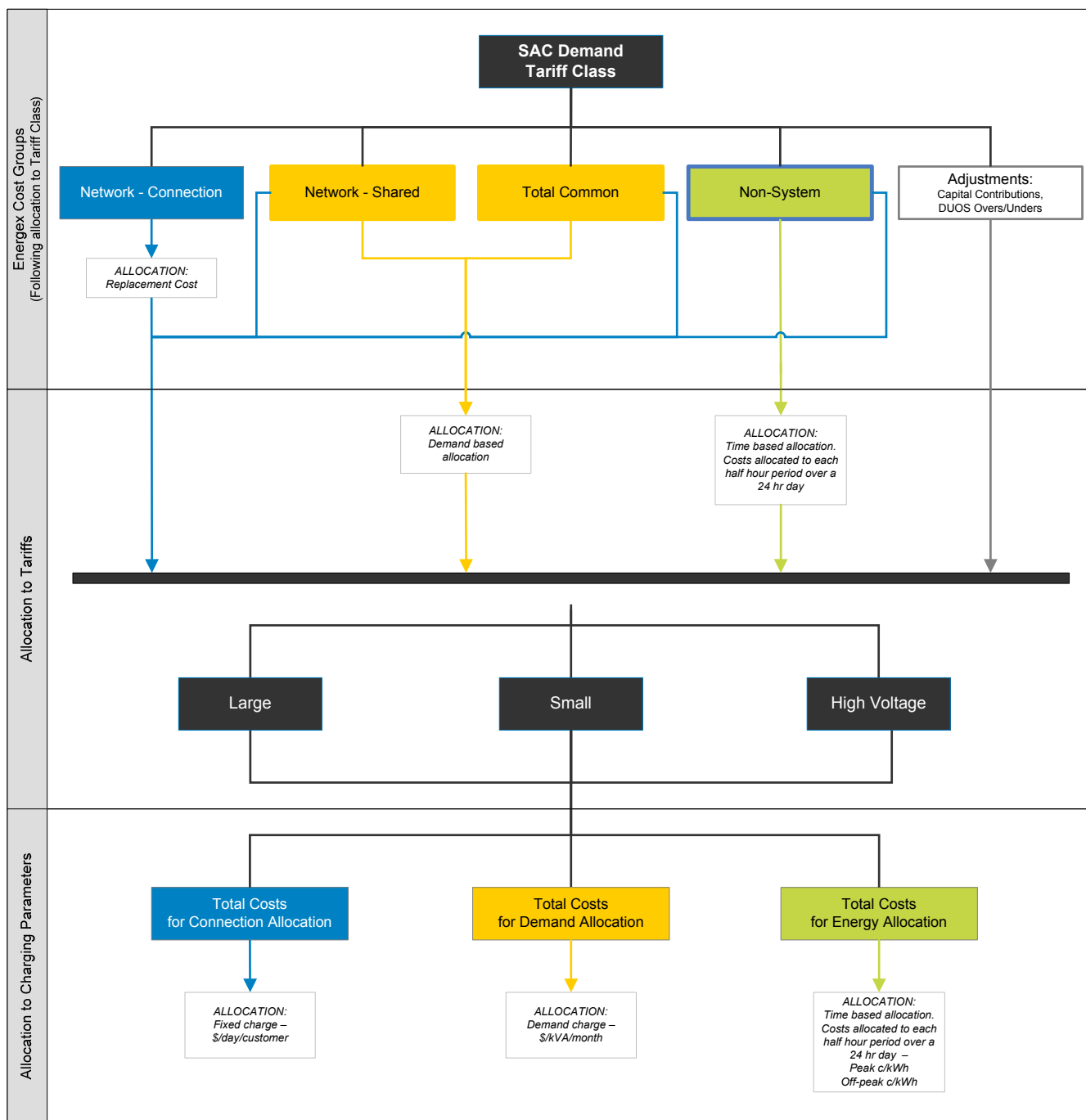
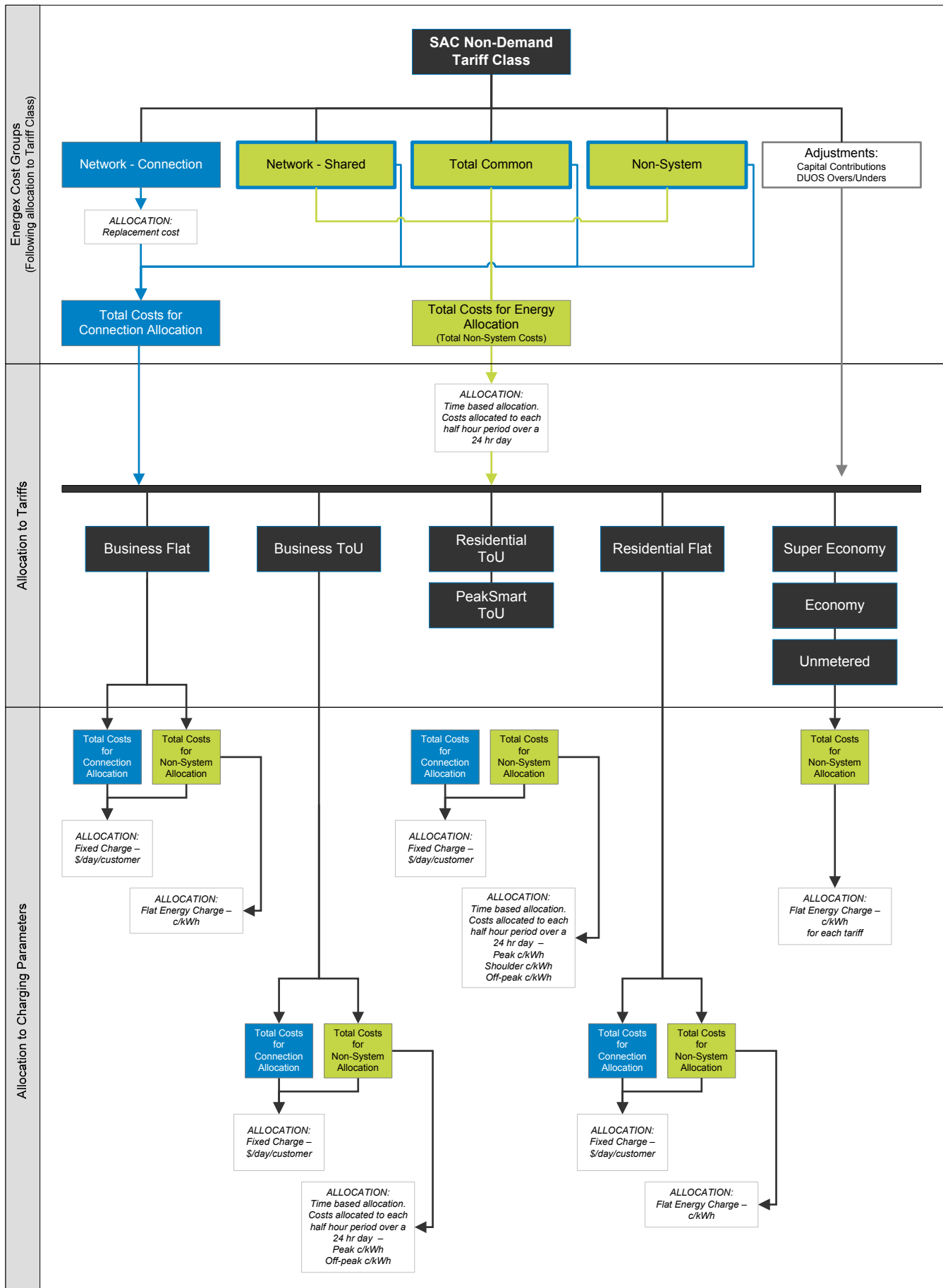


Figure A-3.9 - Cost allocation method: SAC Non-Demand tariff class



Variances from allocated cost based tariffs

Energex develops and applies network tariffs based on the DCOS model. For each tariff class, the costs outlined above are recovered through a combination of fixed charges, demand charges and/or volume charges. The network pricing methodology applied to each of those groups has precluded any possible bypass challenge on the basis that the network tariff is efficient and an alternative electricity service cannot be sourced at a lower economic value.

Providing it is consistent with the *Rules*, Energex may negotiate a tariff other than the tariff calculated using the CAM, if it can be demonstrated that:

- the cost-based network tariff is not efficient
- an economic bypass opportunity exists
- alternative electricity service could be utilised.

APPENDIX 4.

Summary of Compliance

Appendix 4.1 – Compliance with the National Electricity Rules

Table A-4.1 - Demonstrated compliance with the *Rules*

Clause	Pricing Proposal Requirement	Reference
6.1.4(a)	Demonstrates that Energex does not charge a Distribution Network User DUOS charges for the export of electricity generated by the user into the distribution network.	Chapter 4 s 4.2, Table 4.2 Chapter 5 s 5.2, Table 5.2
6.1.4(b)	Demonstrates that Energex charges for the provision of connection services as allowed in the <i>Rules</i> .	Chapter 4 s 4.2 Chapter 5 s 5.2
6.18.2(b)(1)	Sets out each tariff class (including the classes of alternative control services).	Chapter 3 s 3.1, Table 3.1 Chapter 12 s 12.1, Table 12.1
6.18.2(b)(2)	Sets out the proposed tariffs for each tariff class.	Chapter 3 s 3.1, Table 3.1 Chapter 4 s 4.2, Table 4.2 Chapter 12 s 12.1, Table 12.1 Appendix 1 A 1.1 and A 1.2 Appendix 2 A 2.1, A 2.2 and A 2.3
6.18.2(b)(3)	Set out for each proposed tariff, the charge parameters and the elements of service to which each charging parameter relates.	Chapter 5 s5.2, Table 5.1 s5.2, Table Table 5.2 s5.3, Table Table 5.4 Chapter 12 s12.2.4, s12.3.4, Equation 12.1, s12.4.4
6.18.2(b)(4)	Set out, for each tariff class related to standard control services, the expected weighted average revenue for the relevant regulatory year and also for the current regulatory year.	Chapter 6 Table 6.1

Clause	Pricing Proposal Requirement	Reference
6.18.2(b)(5)	Sets out the nature of any variation or adjustment to the tariff that could occur during the course of the regulatory year and the basis on which it could occur.	Chapter 10 s 10.6
6.18.2(b)(6)	Set out how DPPC are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous regulatory year.	Chapter 5 s5.3, Table Table 5.4 Chapter 9
6.18.2(b)(6A)	Set out how jurisdictional scheme amounts for each approved jurisdictional scheme are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those amounts.	Chapter 2 s 2.2.5
6.18.2(b)(6B)	Describe how each approved jurisdictional scheme that has been amended since the last jurisdictional scheme approval date meets the jurisdictional scheme eligibility criteria.	Chapter 2 s 2.2.5
6.18.2(b)(7)	Demonstrates compliance with the <i>Rules</i> and any applicable distribution determination.	Appendix 4 A 4.1 and A 4.2
6.18.2(b)(8)	Describes the nature and extent of change from the previous regulatory year and demonstrates that the changes comply with the <i>Rules</i> and any applicable distribution determination.	Chapter 10
6.18.3(a)	Define the tariff classes into which customers for direct control services are divided.	Chapter 3 s 3.1, Table 3.1 Chapter 4 s 4.1, Table 4.1. Chapter 12 s 12.1, Table 12.1
6.18.3(b)	Demonstrates that each customer for direct control services is a member of at least one tariff class.	Chapter 3 s 3.1.1 Chapter 12 s 12.1
6.18.3(c)	Sets out separate tariff classes for standard control and alternative control customers.	Chapter 3 s 3.1, Table 3.1 Chapter 12 s 12.1, Table 12.1
6.18.3(d)(1)	Demonstrates that tariff classes are formed based on groupings of customers on an economically efficient basis.	Chapter 3 s 3.1 Chapter 12 s 12.1
6.18.3(d)(2)	Demonstrates that customers and tariffs are grouped into tariff classes with regard to the need to avoid unnecessary transactions costs.	Chapter 3 s 3.1 Chapter 7 s 7.3 Chapter 12 s 12.1

Clause	Pricing Proposal Requirement	Reference
6.18.4(a)(1)(i), (ii) and (iii)	Demonstrates that customers are assigned (or re-assigned) to tariff classes on the basis of the nature and extent of their usage, the nature of their connection to the network, and the metering installed at the customer's premises.	Chapter 3 s 3.2.1, Figure 3.1
6.18.4(a)(2)	Demonstrates that customers with a similar usage and connection profile are treated on an equal basis.	Chapter 3 s 3.2.1
6.18.4(a)(3)	Demonstrates that customers with micro-generation facilities are treated no less favourably than customers without such facilities.	Chapter 3 s 3.2.2
6.18.4(a)(4), (b)	Demonstrates that customer assignment (or re-assignment) to a tariff class does not occur in the absence of an effective system of assessment and review.	Chapter 3 s 3.2.4
6.18.4(b)	Demonstrates that if the charging parameters for a particular tariff result in a basis of charge that varies according to the usage or load profile of the customer, an effective system of assessment and review should be provided.	Chapter 3 s 3.2.4
6.18.5(a)(1) and (2)	Demonstrates that revenue from a tariff class lies between the stand alone and avoidable cost.	Chapter 7 s 7.1, Table 7.2 Chapter 12 s 12.3.6
6.18.5(b)(1)	Demonstrates that tariffs and tariff components have regard for long run marginal cost.	Chapter 7 s 7.2, Equation 7.1 Chapter 12 s 12.3.5
6.18.5(b)(2)(i)	Demonstrates that tariffs and tariff components have regard for the transaction costs to customers.	Chapter 7 s 7.3
6.18.5(b)(2)(ii)	Demonstrates that tariffs and tariff components are set with regard to whether customers are able or likely to respond to price signals.	Chapter 7 s 7.4
6.18.5(c)	Demonstrates that if tariffs do not recover the required revenue as a result of the operation of clause 6.18.5(b), the tariffs have been adjusted with minimum distortion to efficient patterns of consumption.	Chapter 7 s 7.5
6.18.6(b)	Demonstrates that the weighted average revenue for a standard control tariff class does not exceed that for the previous year by more than the "permissible percentage" defined in clause 6.18.6(c) of the <i>Rules</i> .	Chapter 8 Table 8.1
6.18.6(c)(1) and (2)	Demonstrates the "permissible percentage" has been calculated in accordance with the definition set out in this clause of the <i>Rules</i> .	Chapter 8 Equation 8.1
6.18.6(d)(1) and (2)	Demonstrates that Designated Pricing Proposal Charges and pass-throughs were removed from the calculation of the side constraint.	Chapter 8 Equation 8.1

Clause	Pricing Proposal Requirement	Reference
6.18.7(a)	Demonstrates that tariffs passed on to customers include the charges to be incurred by Energex for transmission use of system services.	Chapter 5 s 5.3 Chapter 9
6.18.7(b)	Demonstrates that the DPPC charges passed on to customers do not exceed the estimated DPPC charges adjusted for over or under recovery in the previous regulatory year.	Chapter 9 s 9.3, Table 9.1
6.18.7(c)(1) , (2) and (3)	Demonstrates that any DPPC over or under recovery is the difference between what was paid to Powerlink and what was recovered from customers via DPPC charges and is adjusted for an appropriate cost of capital consistent with the allowed rate of return.	Chapter 9 s 9.3, Table 9.1
6.18.9(a)(1)	The tariff classes and the applicable tariffs (or prices) are published on Energex's website.	Chapter 14
6.18.9(a)(2)	The charging parameters within each tariff class (i.e. the fixed, demand and energy prices) and the elements of the service to which each charging parameter relates are published on Energex's website.	Chapter 14
6.18.9(a)(3)	The manner in which tariffs are expected to change over the remainder of the Regulatory Control Period, and the reasons for the changes, are published on Energex's website.	Chapter 14
6.18.9(b)	The information set out in clause 6.18.9(a) is published on Energex's website 20 days prior to the start of the relevant regulatory year or as soon as practicable thereafter.	Chapter 14

Appendix 4.2 – Compliance with the Final Determination

Table A-4.2 - Demonstrated compliance with the Final Determination

Section	Pricing Proposal Requirement	Reference
Section 2.5.2	Energex to follow the procedure for assigning customers to tariff classes or re-assigning customers from one tariff class to another as specified in Appendix B of the Final Determination.	Chapter 3 s 3.2, Figure 3.1, s 3.2.3 and s 3.2.4
Section 4.5.2	Demonstrates that the proposed DUOS prices for the next year (t) will meet the side constraints formula set out in Section 4.5.2 of the Final Determination.	Chapter 8 Equation 8.1, Table 8.1
Section 4.6	Include a record of the amount of revenues recovered from TUOS charges and associated payments in accordance with Appendix E of the Final Determination.	Chapter 9
Section 17.3.3	Include the revenues collected from the provision of each street light service in the preceding regulatory year.	Appendix 2 A-2.5
Section 18.3.3	Calculate the respective internal and external labour cost escalators for each regulatory year of the next Regulatory Control Period and provide both qualitative and quantitative supporting information to the AER as part of the annual pricing proposals.	Appendix 2 A-2.4
Section 18.3.4	Calculate the on cost and overhead rates in each regulatory year of the next Regulatory Control Period in accordance with the methodologies set out in the CAM and submit both qualitative and quantitative supporting information to the AER as part of the annual pricing proposal.	Appendix 2 A-2.4
Section 18.3.5	Include the volume of fee-based and quoted services provided and the revenues recovered from the provision of fee-based and quoted services in the preceding regulatory year.	Appendix 2 A-2.5
Section 18.3.5	Set out the nature and extent of any variation to an individual formula component, on cost or overhead rate from that applicable in the previous regulatory year that is above the indicative illustrative quoted service examples and fee-based services set out in Appendix I, J and K of the final decision.	Appendix 2 A-2.4

Section	Pricing Proposal Requirement	Reference
Section 18.3.5	Set out in its pricing proposal the nature and extent of any variation or adjustment to the methodology employed to derive a formula component escalator or on cost or overhead rate.	Appendix 2 A-2.4
Appendix D	Maintain a distribution use of system (DUOS) overs and unders account. Energex must provide information on this account to the AER as part of its annual pricing proposals.	Chapter 2 s 2.2.3, Table 2.4
Appendix E	Maintain a transmission use of system (TUOS) overs and unders account. Energex must provide information on this account to the AER as part of its annual pricing proposals.	Chapter 9 s 9.3, Table 9.1

APPENDIX 5.

Glossary

Appendix 5.1 – Acronyms and abbreviations

Table A-5.1 - Acronyms and abbreviations used throughout this document

Abbreviation	Description
ABS	Australian Bureau of Statistics
A/C	Air-conditioning
ACS	Alternative Control Service
AEMC	Australian Energy Market Commission
AER	Australian Energy Regulator
AH	After Hours
AIC	Average Incremental Cost
BH	Business Hours
CAB	Contributed Asset Base
CAC	Connection Asset Customers
CAM	Cost Allocation Method
Capex	Capital Expenditure
CPI	Consumer Price Index
DCOS	Distribution Cost of Supply
DLF	Distribution Loss Factor
DNSP	Distribution Network Service Provider
DPP	Dynamic Peak Pricing
DPPC	Designated Pricing Proposal Charges (previously known as TUOS)
DRR	Demand Response Ready
DUOS	Distribution Use of System
EG	Embedded Generators
ENA	Energy Network Australia
ENCAP	Electricity Network Capital Program
FIT	Feed-in-Tariff (Solar PV)
HV	High Voltage
ICC	Individually Calculated Customers
LCC	Large Customer Connection
LRMC	Long Run Marginal Cost
LV	Low Voltage

Abbreviation	Description
MAR	Maximum Allowable Revenue
NEL	National Electricity Law
NEM	National Electricity Market
NER	National Electricity Rules (or <i>Rules</i>)
NMI	National Meter Identifier
NSW	New South Wales
NTC	Network Tariff Code
NUOS	Network Use of System
O&M	Operating and Maintenance Allowance (Opex)
Opex	Operational Expenditure
PFA	Power Factor Adjustment
PV	Photovoltaic (Solar PV)
PV	Present Value
QAO	Queensland Audit Office
QCA	Queensland Competition Authority
QESI	Queensland Electricity Supply Industry
RAB	Regulatory Asset Base
RBT	Reward Based Tariff
<i>Rules</i>	National Electricity Rules (or NER)
SAC	Standard Asset Customers
SCI	Statement of Corporate Intent
SCS	Standard Control Service
SEQ	South East Queensland
SRMC	Short Run Marginal Cost
STPIS	Service Target Performance Incentive Scheme
TNCP	Transmission Network Connection Point
TNSP	Transmission Network Service Provider
TOU	Time of Use
TUOS	Transmission Use of System
WACC	Weighted Average Cost of Capital
WAR	Weighted Average Revenue

Appendix 5.2 – Measurements

Table A-5.2 - Units of measurement used throughout this document

Base Unit	Unit name	Multiples used in this document
A	ampere	kVA, MVA
h	hour	GWh, kWh, MWh
V	volt	kV, kVA, MVA
var	var	kvar
W	watt	W, kW, kWh, MW

Table A-5.3 - Multiples of prefixes (units) used throughout this document

Prefix symbol	Prefix name	Prefix multiples by unit	Prefixes used in this document
G	giga	10^9	GWh
M	mega	1 million or 10^6	MW, MWh, MVA
k	kilo	1 thousand or 10^3	kV, kVA, kvar, kW, kWh

Appendix 5.3 – Definitions

Table A-5.4 - Definitions of terminology used throughout this document

Term	Abbreviation / Acronym	Definition
After Hours	AH	Any time outside business hours.
Air-conditioning	A/C	An air-conditioning appliance; commonly used in the context of a unit, i.e. A/C unit, including in reference to the PeakSmart ToU tariff.
Alternative Control Service	ACS	This service class includes the provision, construction and maintenance of street lighting assets, and fee-based and quoted services.
Anytime		At the retailer's request, the work is performed within business hours to speed up the completion, and, where necessary after hour fees are charged.
Australian Bureau of Statistics	ABS	The ABS is Australia's national statistical agency that provides key statistics on a wide range of economic, environmental and social issues.
Australian Energy Market Commission	AEMC	A national, independent body that exists to make and amend the detailed rules for the NEM to ensure efficient, reliable and secure energy market frameworks which serve the long-term interests of consumers.
AEMC Power of choice review		Conducted by the AEMC, the Power of choice review sets out a substantial reform package for the NEM to provide consumers with more opportunities to make informed choices about the way they use electricity and manage expenditure.
Australian Energy Regulator	AER	The economic regulator of the national electricity market established under Section 44AE of the <i>Competition and Consumer Act 2010</i> (Commonwealth).
Business Hours	BH	8 am to 5 pm, Monday to Friday.
Capacity Charge		This part of the tariff seeks to reflect the costs associated with providing network capacity required by a customer on a long-term basis. It is levied on the basis of either contracted demand or the maximum demand in the previous calendar year. The charge is applied as a fixed dollar amount per kVA per month.
Capacity - Network		The maximum demand (kW) that the distribution network can provide for at any one time.
Capital Expenditure		Expenditure typically resulting in an asset (or the amount Energex has spent on assets).
Charging Parameter		The charges comprising a tariff. Parameters include demand, capacity, fixed and volume (flat or ToU) charges.

Term	Abbreviation / Acronym	Definition
Common Service		A service that ensures the integrity of a distribution system and benefits all Distribution Customers and cannot reasonably be allocated on a locational basis.
Connection Asset Customer	CAC	Typically, those customers with electricity consumption greater than 4 GWh, but less than 40 GWh, per year at a single connection point; or where demand is greater than or equal to 1 MVA; or where a customer has a dedicated supply system with significant connection assets or the customer has contributed to their dedicated connection assets.
Connection Asset (Contributed or Non-contributed)		Related to building connection assets at a customer's premises as well as the connection of these assets to the distribution network. Connection assets can be contributed (customer funded, then gifted to Energex) or non-contributed (Energex funded).
Connection Point		The point of electrical coupling between the electricity distribution network and a customer's electrical installation. The meter is installed as close as possible to this location.
Demand		The amount of electricity energy being consumed at a given time measured in either kilowatts (kW) or kilovolt amperes (kVA). The difference between the two is the power factor.
Demand Charge		This part of the tariff accounts for the actual demand a customer places on the electricity network. The actual demand levied for billing purposes is the metered monthly maximum demand. The charge is applied as a fixed dollar per kW per month or per kVA per month for EG and SAC Demand customers and a fixed dollar per kVA per month for ICC and CAC customers.
Demand Metered SAC		The customer's connection point has a meter installed that is capable of measuring energy consumption (kWh) and demand (kW). This meter records total energy consumption (kWh) and demand over 30 minute periods. A customer's demand is the average demand (kW) over the 30 minute period.
Demand Metered Tariff		The tariff has been structured to include a demand component so the customer's actual demand is reflected in the price they pay for their electricity. The highest demand reading for that month is used to calculate the customer's electricity bill.
Demand Response Ready	DRR	A DRR appliance, such as a PeakSmart A/C unit, is fitted with a signal receiver that allows demand management of these appliances by Energex.
Distribution Cost of Supply Model	DCOS	The Energex model used to allocate costs approved by the AER to the various tariff classes.
Distribution Loss Factor	DLFs	These represent the average electrical energy losses incurred when electricity is transmitted over a distribution network.
Distribution Use of System	DUOS	This refers to the network charges for the use of the distribution network.

Term	Abbreviation / Acronym	Definition
Designated Pricing Proposal Charge	DPPC	Refers to the charges incurred for use of the transmission network; previously referred to as Transmission Use of System (TUOS).
Economy		Specified connected appliances are controlled by network equipment so that supply will be available for a minimum period of 18 hours per day during time periods set at the absolute discretion of Energex.
Embedded Generator	EG	In line with the ENA classification, EGs are generally those generators with an installed capacity as follows: Medium: 1-5 MW (LV or HV) or < 1 MW (HV) Large: > 5 MW
Electricity Network Capital Program Review	ENCAP	The ENCAP Review was commissioned by the Queensland Government in late 2011. An independent panel was established to undertake a review of the capital infrastructure programs of Energex, Ergon Energy and Powerlink with the view to achieving efficiencies and cost savings while maintaining network security and reliability. The panel identified significant potential capital expenditure savings and Energex's program of capital work was reduced in line with these findings.
Energy		The amount of electricity consumed by a consumer (or all customers) over a period of time. Energy is measured in terms of watt hours (Wh), kilowatt hours (kWh), megawatt hours (MWh) or gigawatt hours (GWh).
Feed-in-Tariff	FiT	The rate that is to be paid for the excess energy generated by customers and fed back into the electricity grid under the Queensland Solar Bonus Scheme. The FiT rate is determined by the Queensland Government and is paid by the purchaser of the excess energy.
Final Determination		A distribution determination document published by the AER in its role as Energex's economic regulator and that provides for distribution charges to increase during Energex's Regulatory Control Period.
Fixed Charge		The fixed charge seeks to reflect the costs associated with customer's dedicated connection assets. The charge is applied as a fixed dollar amount per day.
Form 1634 – QESI		An Energex form, "QESI – Application for Review" which allows a customer's retailer to request a change to the customer's NTC or NMI Classification.
High Voltage	HV	Refers to the 11 kV or above network.
Individually Calculated Customer	ICC	Typically those customers with electricity consumption greater than 40 GWh per year at a single connection point; or where the customers demand is greater than or equal to 10 MVA; or where a customer's circumstances mean that the average shared network charge becomes meaningless or distorted.

Term	Abbreviation / Acronym	Definition
Long Run Marginal Cost	LRMC	An estimate of the cost (long term variable investment) of augmenting the existing network to provide sufficient capacity for one additional customer to connect to the network or an additional MW of demand.
Low Voltage	LV	Refers to the sub-11kV network
Maximum Allowable Revenue	MAR	The maximum revenue which can be recovered through tariffs for the regulatory year.
Maximum Demand		The maximum demand recorded at a customer's individual meter or the maximum demand placed on the electrical distribution network system at any time or at a specific time or within a specific time period, such as a month. Maximum demand is an indication of the capacity required for a customer's connection or the electrical distribution network.
Micro Generator		AS4777 compliant generators with an installation size of less than 10 kW (single phase) or 30 kW (three phase) connected to the LV network.
Market Settlement and Transfer Solution		The central repository for Standing Data for all NMIs in contestable markets.
National Electricity Law	NEL	The legislation that establishes the role of the AER as the economic regulator of the NEM and the regulatory framework under which the AER operates.
National Electricity Market	NEM	The interconnected electricity grid covering Queensland, New South Wales, Victoria, Tasmania, South Australia and the Australian Capital Territory.
National Electricity Rules	NER (the <i>Rules</i>)	The legal provisions (enforced by the AER) that regulate the operation of the NEM and the national electricity systems, the activities of market participants and the provision of connection services to retail customers.
National Metering Identifier	NMI	A unique number assigned to each metering installation.
Network Tariff Code	NTC	Energex's nominated code that represents the network tariff being charged to customers for network services.
Network Use of System	NUOS	The tariff for use of the distribution and transmission networks. It is the sum of both Distribution Use of System (DUOS) and Designated Pricing Proposal Charges (DPPC).
Non-demand Metered SAC		The customer's connection point has a meter installed that is capable of measuring the total energy consumption (kWh).
Non-demand Metered Tariff		The tariff is based around a fixed daily component and the actual energy (kWh) used by the customer.
Off-peak Period		All hours which are outside Peak and Shoulder periods.

Term	Abbreviation / Acronym	Definition
Operational Expenditure	Opex	Opex is the combined total of maintenance and operating costs. Maintenance Costs are those that are directly and specifically attributable to the repair and maintenance of network assets, while Operating Costs are those that relate to the day to day operations of Energex which are not maintenance costs.
Peak Period		Meter type 1–4 (ICC, CAC & SAC Demand): The hours between 7 am and 11 pm, Monday to Friday. Meter type 6 (SAC Non-Demand - Business): The hours between 7 am and 9 pm, Monday to Friday. Meter type 6 (SAC Non-Demand - Residential): The hours between 4 pm and 8 pm, Monday to Friday.
PeakSmart A/C		A PeakSmart A/C unit is a DRR A/C unit that, when its signal receiver is activated, allows it to be demand managed by Energex. In addition to other criteria, eligibility for the PeakSmart ToU tariff includes a customer activating at least one PeakSmart A/C at their premise.
Power Factor		Power factor, is the ratio of kW to kVA, and is a useful measure of the efficiency in the use of the network infrastructure. The closer the power factor is to one (1), the more efficiently the network assets are utilised. Power Factor = kW / kVA
Price Path		Outlines the escalation factors to be applied to the initial price over the Regulatory Control Period.
Pricing Objectives		Objectives established by Energex to complement (and ensure compliance with) the Pricing Principles and to provide clarity when formulating tariffs.
Pricing Principles		The Pricing Principles are established in Clause 6.18.5 of the NER and provide guidance to Energex for setting tariffs.
Pricing Proposal		This document. Prepared by Energex in accordance with Clause 6.18.2(a)(2) of the <i>Rules</i> , it is provided to the AER for approval and outlines how Energex will collect its approved revenue during the relevant regulatory year.
Queensland Government Solar Bonus Scheme		A program that pays residential and other small energy customers for the surplus electricity generated from roof-top solar photovoltaic (PV) systems that is exported to the Queensland electricity grid.
Regulatory Control Period		A standard Regulatory Control Period for DNSPs is a period of not less than 5 regulatory years; Energex's Regulatory Control Period is 2010-2015, as per the Final Determination.
Regulatory Depreciation		Also referred to as the return of capital – the sum of the (negative) straight-line depreciation and the (positive) annual inflation effect on the opening regulatory asset base (RAB).
Regulatory Year		A specific year within the Regulatory Control Period.
Return on Capital		The return necessary to achieve a fair and reasonable rate of return on the assets necessarily invested in the business.

Term	Abbreviation / Acronym	Definition
Reward Based Tariff	RBT	A reward based tariff provides a strong price signal to customers to drive changes in customer behaviour. If changes are achieved, the customer is rewarded by way of a financial incentive.
Revenue Cap		The amount of revenue Energex is approved by the AER to recover during the Regulatory Control Period and specific regulatory years.
Service Target Performance Incentive Scheme	STPIS	A scheme developed and published by the AER, in accordance with clause 6.6.2 of the <i>Rules</i> , that provides incentives (that may include targets) for DNSPs, (including Energex) to maintain and improve network performance.
Short Run Marginal Cost	SRMC	The cost (short term, fixed investment) of a customer connecting to the network but using only the existing network capacity.
Shoulder Period		Meter type 6 (SAC Non-Demand – Residential and Business): The hours between 7 am to 4 pm and 8 pm to 10 pm, Monday to Friday and 7 am to 10 pm weekends.
Side Constraint		A side constraint is an upper limit on price increases applied at the tariff class level and is calculated by taking into account volume forecasts, CPI, X Factor, STPIS and Capital Contributions. The purpose of a side constraint is to mitigate the impact of prices on customers from one year to the next.
Site-specific Charge		This charge is calculated specifically for a site and is specific to the individual connection point.
Solar Photovoltaic	Solar PV	A system that uses sunlight to generate electricity for residential use. The system provides power for the premises with any excess production feeding into the electricity grid.
Standard Asset Customer	SAC	Generally those customers with an annual electricity consumption below 4 GWh per year, whose supply arrangements are consistent across the customer group; and where there is no contribution for their dedicated connection assets. SAC Non-Demand are customers within SAC with consumption less than 100 MWh per year and who are on consumption (kWh) tariffs. SAC Demand are customers within SAC with consumption greater than 100 MWh per year and less than 4 GWh per year and who are on demand (kW) tariffs.
Standard Control Service	SCS	This service class includes network, connection and metering services.
Statement of Corporate Intent	SCI	The SCI is published at least annually and represents the agreement between Energex's Board of Directors and the shareholding Minister's on Energex's Performance in the relevant financial year in support of the state reform agenda. It is prepared in accordance with Section 7(2) of the <i>Government Owned Corporates Act 1993</i> (Queensland).

Term	Abbreviation / Acronym	Definition
Street Lights (Major)		Lamps in common use for Major Road lighting including: a) High Pressure Sodium 100 watt (S100) and above; b) Metal Halide 150 watt (H150) and above; and c) Mercury Vapour 250 watt (M250) and above.
Street Lights (Minor)		All lamps in common use for Minor Road lighting, including Mercury Vapour, High Pressure Sodium and Fluorescent.
Super Economy		Specified permanently connected appliances are controlled by network equipment so that supply will be permanently available for a minimum period of 8 hours at the absolute discretion of Energex but usually between the hours of 10:00 pm and 7:00 am.
Tariff		The set of charges applied to a customer in the respective billing period. A tariff consists of one or more charging parameters that comprise the total tariff rate.
Tariff class		A class of customers for one or more direct control services who are subject to a particular tariff or particular tariffs (as per the <i>Rules</i> definition).
Tariff Schedule		The Tariff Schedule is published by Energex annually at the beginning of the financial year and outlines its tariffs for SCS and ACS. It also provides information about how Energex assigns customers to tariff classes and the internal review process undertaken if a customer requests a review of a decision. The Tariff Schedule applies for the duration of the relevant financial year.
Time of Use	ToU	Refers to tariffs that vary according to the time of day at which the electricity is consumed.
Transmission Use of System	TUOS	Superseded terminology for Designated Pricing Proposal Charges (DPPC) which are charges incurred for use of the transmission network.
Unmetered Supply		A customer who takes supply where no meter is installed at the connection point.
Volume (Energy) Charge		This part of the tariff seeks to reflect costs not directly allocated to network drivers and costs that are proportional to the size of the customer. The energy consumption (kWh) for the period, as recorded by the customer's meter, is utilised to calculate this part of the tariff charge. This charge is applied as a fixed amount (cents) per kilowatt hour (kWh), i.e. c/kWh.
Volume (Energy) Charge (Off-peak)		This charge is applicable to those customers who are on a Residential and/or Business Time of Use tariff. The energy consumption (kWh) during off-peak periods (refer to Off-peak Period for times), as recorded by the customer's meter, is utilised to calculate this part of the tariff. This charge is applied as a fixed amount (cents) per kilowatt hour (kWh), i.e. c/kWh.

Term	Abbreviation / Acronym	Definition
Volume (Energy) Charge (Peak)		This charge is applicable to those customers who are on a Residential and/or Business Time of Use tariff. The energy consumption (kWh) during peak periods (refer to Peak Period for times), as recorded by the customer's meter, is utilised to calculate this part of the tariff. This charge is applied as a fixed amount (cents) per kilowatt hour (kWh) i.e. c/kWh.
Volume (Energy) Charge (Shoulder)		This charge is applicable to those customers who are on a Residential Time of Use tariff. The energy consumption (kWh) during shoulder periods (refer to Shoulder Period for times), as recorded by the customer's meter, is utilised to calculate this part of the tariff. This charge is applied as a fixed amount (cents) per kilowatt hour (kWh), i.e. c/kWh.
Weighted Average Cost of Capital	WACC	The minimum return a business must earn on an existing asset base. For Energex, the WACC is set by the AER in its Final Determination for a specific Regulatory Control Period.
Weighted Average Revenue	WAR	This is the average revenue that is expected to be recovered by tariff class during the relevant regulatory control year.
X Factor		Under the CPI – X form, prices or allowed revenues are adjusted annually for inflation (CPI) less an adjustment factor 'X'. The X Factor represents the change in real prices or revenues each year, so the DNSP can recover the costs that it expects to incur over the Regulatory Control Period.